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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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Objectives: The relationship between the duration from the emergency department (ED) to the intensive care units (ICUs) and in-hospital mortality among patients admitted directly to the ICUs from the ED remains controversial. The aim of the study was to utilize data from the Medical Information Mart for Intensive Care IV database to explore the relationship between the ED to ICUs time and patient outcomes.

Design: This retrospective observational cohort study utilized data from the Critical Care Medical Information Database IV. The relationship between the time from ED to ICUs and the primary outcome of in-hospital all-cause mortality was compared.

Setting: Admissions to the Beth Israel Deaconess Medical Center intensive care from 2008 to 2019.

Participants: This study included adult ICU patients aged 18 years and older who were admitted via the ED.

Primary and secondary outcome measures: The primary outcome was in-hospital all-cause mortality. Secondary outcomes included 28-day all-cause mortality, 1-year all-cause mortality and length of stay in ICU and hospital.

Results: During follow-up, 4,509 all-cause deaths (35.50%) occurred, including 1,185 (9.33%) in the ICU and 1,988 (15.65%) in-hospital. Mortality rates in the highest quartile of ED to ICU time decreased, with no differences in 1-year all-cause mortality. ICU stay shortened with each additional hour from ED to ICU. Restricted cubic spline analysis revealed a nearly linear relationship between ED to ICU time and both in-hospital (P for non-linearity = 0.669, P for trend < 0.001) and ICU mortality (P for non-linearity = 0.913, P for trend < 0.001). Inclusion of ED to ICU time enhanced the predictive accuracy of conventional severity scores for all-cause mortality, with consistent results in pre- and post-propensity score matching subgroup analyses.

Conclusions: Prolonged time from the emergency department to the ICUs, ranging from 5.65 to 34.55 hours, is associated with decreased in-hospital mortality in patients who went directly from ED to ICUs. However, this association does not extend to the 1-year mortality rate.

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Keywords: Emergency Department; Intensive Care Units; Mortality; Length of Stay; Patient Care

STRENGTHS AND LIMITATIONS OF THIS STYDY

- Our study highlights that managing critically ill patients in the emergency department should focus on more than just reducing ED waiting times.
- We used the latest data from the MIMIC-IV database, unaffected by COVID-19.
- We minimized baseline confounding and enhanced result reliability using three incrementally adjusted models and Propensity Score Matching.
- As a retrospective observational study, it cannot establish causality.
- The data are from the United States, and results may not generalize to other countries.
-

Introduction

The emergency department (ED) often serves as the primary entry point for critically ill patients requiring immediate care. Recent years have seen a marked increase in the capacity of ED clinicians to manage these patients over extended periods (1). Following initial stabilization in ED, patients are typically transported to appropriate departments for further diagnostic tests or therapeutic procedures. Most critically ill patients will be transported to specialized intensive care unit (ICU), where provides the highest level monitoring and care (2). During the last 2 decades, escalating acuity and volume of ED visits have resulted in greater demand for critical care services in the ED and ICUs (3). ED crowding seems to become an international problem and has long been recognised as posing significant patient safety threats (4). The impact of ED crowding on inpatient mortality remains contentious and appears to vary by region (5, 6).

ED length of stay (ED-LOS) is a critical ED performance indicator and is closely related to crowding, quality of care and patient outcome (7). However, the relationship between ED-LOS and in-hospital mortality, particularly for patients admitted directly to ICUs from the ED, remains a subject of debate. Groenland et al.

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(8) reported that a longer ED to ICUs time is associated with increased hospital mortality in the most severely ill patients. Conversely, Wessman et al. (9) found that patients with highest triage priority had a decreasing mortality risk with increasing quintile of ED-LOS for 7-day mortality, and the two highest quintile of ED-LOS had lower 30-day mortality. Saukkonen et al. (10) observed that the length of ED stay was not associated with the outcome of critically ill medical patients.

Given conflicting data and the need for a large sample size, our study utilizes the Medical Information Mart for Intensive Care IV (MIMIC-IV) database to explore the relationship between the ED to ICUs time and patient outcomes in a large cohort of critically ill individuals.

Methods

Study design

This retrospective observational study used data from the MIMIC-IV (version 2.2) database, which contains high-quality information on ICU admissions at Beth Israel Deaconess Medical Center from 2008 to 2019. Access to MIMIC-IV is limited to certified individuals; one author (JWQ) with access permission (Record ID 60311526) extracted the data. As all patient records in MIMIC-IV were fully de-identified, the institutional review board at Beth Israel Deaconess Medical Center waived the need for individual patient consent.

Cohort selection

Figure 1 outlined the inclusion and exclusion criteria for study participants. We initially considered 96,868 unique patients aged 18 and older admitted through the emergency room. We excluded who did not have an ICU stay or whose admission time to the ED was not recorded. Additionally, Patients with a gap of more than 6 hours between ED admission (*admittime*) and ICU admission (*intime*) recorded in the database were not considered to be transferred directly from the ED to the ICUs and were thus excluded. Additionally, we

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analyzed only the first ICU stay for patients who were admitted to the ICUs more than once. A total of 12703 patients were included in the final study cohort and divided into four groups based on the ED to ICUs time.

Data extraction and defnitions

All baseline characteristics were extracted within the first 24h of ICU admission from the MIMIC-IV database, including age, gender, marital status, ethnicity, admission type, first care unit, disease diagnoses, severity at admission measured by Sequential Organ Failure Assessment (SOFA) score, Logistic Organ Dysfunction System (LODS) score, Oxford Acute Severity of Illness Score (OASIS), Systemic infammatory response syndrome (SIRS) score, Charlson Comorbidity Index (CCI) score, Simplifed acute physiological score II (SAPSII), Acute physiology score III (APSIII). The ED to ICUs time was defined as the interval between *edregtime* (the time of registration in the ED) and *intime* (the time of transfer into the ICU) recorded in the MIMIC-IV database. *Admittime* provides the date and time the patient was admitted to the hospital. Data extraction was performed using pgAdmin4 PostgreSQL 16.1. All human studies adhered to the principles of the Declaration of Helsinki.

Primary outcomes and secondary outcomes

The primary outcome of this study was in hospital all-cause mortality. Of these, hospital death included ICU death and general ward inpatient death. The secondary outcomes included 28-day all-cause mortality, 1-year all-cause mortality and length of stay in ICU and hospital. Patient mortality information for discharged patients was accessed from the US Social Security Death Index.

Statistical analysis

Our study grouped patients into four categories based on their ED to ICU admission times. We assessed the

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distribution of continuous data using the Kolmogorov-Smirnov test and presented them as means with standard deviations or medians with interquartile ranges. Categorical data were shown in counts and percentages. The Mann-Whitney U test and the Pearson chi-square or Fisher's exact tests were used for comparisons. We adjusted for multiple comparisons using the false discovery rate, with a 5% threshold. To reduce bias, we matched patients in pairs based on their baseline characteristics using a method called propensity score matching (PSM), which was calibrated to a specific caliper width.

For statistical analyses, we created logistic regression models to calculate odds ratios (ORs) and 95% confidence intervals for the outcomes of hospital and ICU mortality. We analyzed these outcomes through three models: an unadjusted model, a model adjusted for disease severity only, and a fully adjusted model excluding variables prone to multicollinearity. We also evaluated the impact of ED to ICU admission time on secondary outcomes like 28-day and 1-year mortality using similar models and Kaplan Meier survival analysis, alongside tests to check for non-linear relationships and stratified analyses to verify consistency across different patient subgroups. Finally, we explored whether including ED to ICUs time improved the prediction accuracy of existing illness severity scores for in-hospital mortality using the area under the curve metric.

We performed statistical analyses with R software (version 4.3.3) and considered two-sided P values below 0.05 as significant. For multiple comparisons, we used a FDR correction to maintain this significance threshold.

Result

Baseline characteristics

Supplementary Table 1 showed the characteristics of 12,703 study participants categorized by ED to ICUs time quartiles. The median patient age was 64 years, and 56.51% were male. There were 4,509 all-cause deaths, 1,988 in-hospital deaths, and 4,013 deaths within one year. The median ED to ICUs time was 3.98 hours. The

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most common diagnoses included acute kidney failure and acute respiratory failure. Increasing the ED to ICUs time were associated with older age, a higher frequency of female patients, and certain diagnoses. Patients in later quartiles had generally lower severity scores.

With increasing ED to ICUs time, ICU length of stay (1.96 days vs. 1.80 days vs. 1.78 days vs. 1.70 days, $P < 0.001$), hospital mortality (17.60% vs. 16.52% vs. 16.24% vs. 12.21%, $P < 0.001$), ICU mortality (11.64% vs. 9.44% vs. 9.83% vs. 6.37%, $P < 0.001$) and 28-day mortality (20.64% vs. 20.43% vs. 20.47% vs. 16.85%, $P < 0.001$) decreased gradually. Conversely, 1-year mortality initially increased but then decreased in the longest duration quartile ($P = 0.005$).

Patients were divided into two groups based on the third quartile of ED to ICUs time (**Supplementary Table 2**). Before propensity-score matching, significant differences in age, gender, ethnicity, admission type, first care unit, and severity scores were observed between the groups. After matching, these differences were largely reduced, resulting in well-balanced baseline characteristics.

Incidence rate of all-cause mortality among different groups

Kaplan-Meier survival analysis assessed all-cause mortality across quartiles of ED to ICUs time, as shown in **Fig. 2**. In the overall cohort, the fourth quartile showed significantly lower 28-day mortality compared to the first quartile (20.64% vs. 16.85%, log-rank $P < 0.001$, Fig. 2A). However, there was no significant difference in 1-year mortality between these groups (29.39% vs. 31.90%, log-rank $P = 0.245$, Fig. 2B). Propensity score matching revealed a slight reduction in 28-day mortality for patients with ED to ICUs time ≥ 5.65 hours (17.81% vs. 18.23%, log-rank $P = 0.039$, Fig. 2C), but no significant difference in 1-year mortality in the matched cohort (18.54% vs. 16.80%, log-rank $P = 0.680$, Fig. 2D).

Association of the ED to ICUs time with all-cause mortality

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We evaluated whether ED to ICUs time were linked to in-hospital and ICU mortality. Findings indicate that longer times (5.65-34.55 hours) are associated with lower in-hospital mortality compared to shorter times (0.18-2.83 hours), with ORs indicating significant decreases: 0.651 initially, 0.710 after adjusting for OASIS and CCI scores, and 0.736 with further adjustments excluding variables with high variance inflation factors (VIF > 5). Similar trends were observed for ICU mortality, with the strongest reductions in the fourth quartile (OR 0.602) after adjustments (**Supplementary Fig.1**). **Table 1** and RCS regression analysis supported these findings, showing a nearly linear decrease in mortality risk with longer ED to ICUs time (P < 0.001) (**Fig.3**).

Additionally, stratified analyses revealed consistent inverse relationships between ED to ICUs time and mortality across various subgroups, including age, sex, ethnicity, and care unit type. Notably, significant reductions in in-hospital mortality were found among older adults (> 65 years), females, and both white and non-white races. Also, those not admitted through emergency wards, those transferred to units other than MICU/SICU, those with higher OASIS scores, and lower CCI scores showed lower mortality risks associated with longer ED to ICUs time (all P < 0.05) (**Fig. 4**).

Table 1 Odds Ratios (ORs) for All-Cause Mortality in-hospital and ICU mortality from Logistic Regression

Categories	Model 1		Model 2		Model 3	
	OR (95% CI)	*P*-value	OR (95% CI)	*P*-value	OR (95% CI)	*P*-value
Hospital death						
Continuous variable per 1 unit	0.935 (0.915, 0.954)	< 0.001	0.952 (0.930, 0.974)	< 0.001	0.966 (0.941, 0.990)	0.011
Quartilea[a]						
Q1 (N=3222)	Ref.		Ref.		Ref.	
Q2 (N=3147)	0.927 (0.813, 1.056)	0.255	1.007 (0.868, 1.169)	0.924	0.897 (0.762, 1.056)	0.236
Q3 (N=3165)	0.908 (0.796, 1.035)	0.148	0.961 (0.828, 1.116)	0.725	0.855 (0.726, 1.008)	0.089
Q4 (N=3169)	0.651 (0.566, 0.749)	< 0.001	0.710 (0.606, 0.832)	< 0.001	0.735 (0.618, 0.875)	< 0.001
ICU death						
Continuous variable per 1 unit	0.902 (0.876, 0.927)	< 0.001	0.930 (0.903, 0.958)	< 0.001	0.940 (0.910, 0.971)	< 0.001
Quartilea[a]						
Q1 (N=3222)	Ref.		Ref.		Ref.	
Q2 (N=3147)	0.791 (0.673, 0.929)	0.006	0.864 (0.723, 1.032)	0.129	0.765 (0.630, 0.927)	0.011
Q3 (N=3165)	0.827 (0.705, 0.970)	0.019	0.910 (0.793, 1.086)	0.296	0.815 (0.672, 0.988)	0.048

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Q4 (N=3169)	0.517 (0.432, 0.617)	< 0.001	0.602 (0.495, 0.732)	< 0.001	0.608 (0.491, 0.752)	< 0.001

Model 1: unadjusted; Model 2: adjusted for OASIS score, CCI score; Model 3: adjusted for age, gender, first care unit, OASIS score, CCI score, SOFA score, LODS score, SASPII score, APSIII score, acute kidney failure, acute respiratory failure, hypertension, pneumonia, urinary tract infection, acidosis and cerebral edema.

[a] ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr)

Incremental effect of ED to ICUs time on predictive value for primary outcomes

The inclusion of ED to ICUs time had an incremental effect on the AUC of existing severity of illness scores to predict all-cause mortality, including SOFA score (hospital: 0.645 vs. 0.657; ICU: 0.645 vs. 0.655), LODS score (hospital: 0.810 vs. 0.815; ICU: 0.810 vs. 0.811), OASIS score (hospital: 0.805 vs. 0.806; ICU: 0.805 vs. 0.805), SIRS score (hospital: 0.650 vs. 0.657; ICU: 0.650 vs. 0.656), CCI (hospital: 0.673 vs. 0.685; ICU: 0.673 vs. 0.680), SAPSII (hospital: 0.816 vs. 0.820; ICU: 0.816 vs. 0.819) and APSIII (hospital: 0.812 vs. 0.814; ICU: 0.812 vs. 0.813) and all P-values < 0.001 (**Supplementary table 3**).

Relationship between ED to ICUs time and length of stay

Linear regression analysis explored the impact of ED to ICUs time on hospital and ICU length of stay (LOS). Initially, a negative association was found between ED to ICUs time and in-hospital LOS among survivors (β = -0.099, P = 0.002). However, this association was not observed after adjustments in multiple linear models (β = 0.031, P = 0.285). In contrast, a consistent negative association remained in the fully adjusted model for ICU LOS among survivors, indicating shorter LOS with longer ED to ICUs time (β = -0.069, P < 0.001, **Supplementary Table 4**).

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Discussion

Our study aimed to explore the relationship between ED to ICU admission time and hospital mortality using the MIMIC-IV database. We developed three progressively adjusted models and employed PSM to reduce baseline confounding. Results showed that longer ED to ICUs time (5.65-34.55 hrs) were associated with lower in-hospital mortality, independent of other prognostic factors. Similar patterns were observed for ICU and 28-day mortality, but not for 1-year mortality.

Other investigators have studied that delayed time in ED is associated with hospital mortality. A large retrospective cohort study with data derived from MIMIC-III showed that every delayed hour in the ED increased the in-hospital mortality, and in particular patients with ≥6 h of transfer from ED to the ICUs had a significantly higher mortality rate than those with <6 h (11). However, this analysis defined delay as the time from initial ED admission to ICU admission, which inadvertently included patients initially moved to general wards before ICU transfer. A previous study showed that an increased hospital LOS before ICU admission was correlated with mortality (12, 13), because patients with longer hospital stays may have more comorbidities. Given the delayed recording in the MIMIC-IV database, we limited the time between recorded admission and ICU admission to 6 hours to accurately identify patients transferred directly from the ED to the ICUs.

The ED to ICUs time is multifactorially influenced, such as ED crowding (14), triage process(15), availability of ICU beds (16, 17) and so on. EDs are also increasingly implementing initiatives to improve ED timeliness and achieve faster throughput in several countries, such as bedside registration, self–check-in kiosks, immediate bedding, fast track, physician in triage, and other initiatives (18). Abdulai et al. (19) assessed trends in ED throughput in the United States from 2006 to 2016, and found that there were decreases in patient wait time, patients leaving without meeting the emergency physician, and boarding time after the implementation of a series of initiatives, especially compared with that of the previous decade. Increased throughput and

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reduced average wait times identify potentially critically ill patients as quickly as possible for targeted and effective treatment. An appropriate triage system is necessary to recognize, diagnose and treat the most severely ill patients, which could result in lower mortality. Several warning systems like “track and trigger”, “early warning scores (EWS)” have been developed to detect patient deterioration (20, 21). Fernandes ta al. (22) found that there was an improvement in decision-making of the health professionals' thereby leading to better clinical management and the outcomes of patients by using clinical decision support systems based on intelligent Systems in the ED. Another important factor which may influence ED to the ICUs time and worsen patient outcomes is availability of ICU beds (23). A previous study found that ICU admission decisions for critically ill ED patients are affected by Medical ICU bed availability (24). Various hospitals are adopting innovative approaches to enhance ICU capacity. Mitra et al. found that the creation of a high acuity units , which could provide patients with more acute care and closer monitoring than a general hospital ward but are not as resource intensive as an ICU, may be associated with reduced in-hospital mortality for ICU patients (25). Establishing a critical care resuscitation unit can handle more critically ill surgical patients and speed up the transfer of those with time-sensitive conditions to specialized centers. Furthermore, enhancing ICU capacity and improving its operational efficiency can reduce the time needed for patient admission to the ICU.

Intrahospital transport (IHT) of critically ill patients from the ED is inevitable and it could also result in life-threatening situations (26). Since IHT exposures the patients to a mobile environment, which can increase the risk of adverse events, ranging from equipment failure, device removal, poor monitoring to cardiovascular and respiratory instability (27). Serious adverse events can aggravate illness and even endanger patients’ lives, associated with significant mortality. Once critically ill patients being transported, they will be more vulnerable with an increasing probability of adverse events, as high as 79.8% in a study of 441 IHT patients admitted to ICU (28). Premature transport of critically ill patients in ED, despite shortening ED to ICUs time, resulted in a greatly reduced preparation time for IHT and increasing adverse events (29). Therefore, IHT of

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of critical patients from the ED requires careful consideration by emergency physicians. They must thoroughly assess the conditions of critical patients and balance the risks and benefits of transportation to determine the optimal solution.

Last, Trzeciak et al. (30) found that the ED staff may not have enough time to provide the required attention and medical care for some critical patients when ED overcrowding occurred. There is circumstantial evidence to suggest that mortality will increase among patients treated in locations that are unable to support the care needs of critically ill patients (31). The study presented by Kadar et al. showed that compared with patients in ED usual care group, those in ICU telemedicine support group had lower mortality and and lower ICU resource utilization (32), since ICU telemedicine team had observed a stable or improving course long enough to have confidence that the patient would be safe in a non-ICU setting. Efforts to improve the level of critical patient care in ED seems to be a significant approach, assuming that the level of ICUs care is achieved, then the time spent in ED may be less important.

Our study highlights that managing critically ill patients in the emergency department involves more than just minimizing ED wait times. Considering the patient's clinical condition, ED circumstances, transport risks, and ICU bed availability to determine the best timing for ICU admission is essential.

A key strength of our research is the use of the comprehensive MIMIC-IV database, which spans from 2008 to 2019. This dataset, the most recent before the onset of COVID-19, allows for an accurate analysis of the association between ED to ICU time and patient outcomes without the confounding effects of pandemic (33). The use of propensity score analysis further minimizes baseline confounding, enhancing the reliability of our findings. We also progressively adjusted our models to accurately estimate the impact of ED to ICU time on patient outcomes

This study has several limitations. First, it is a retrospective analysis from an observational study, which inherently cannot establish causality. Second, although we limited admission and ICU stays to no more than

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6 hours to exclude patients who were initially admitted to a general ward before ICU transfer, we cannot rule out that a small number of patients were transferred to the ICU within 6 hours from the ward. Third, the maximum wait time recorded in this study was 34.55 hours, limiting the applicability of our findings to patients with longer wait times. Lastly, the data were from the United States, so the results may not generalize to ED settings in other countries. Future studies should be well-designed, prospective, and multicenter to address these limitations.

Conclusions

This retrospective study found that patients transferred from the ED to the ICU within 5.65-34.55 hours had lower in-hospital, ICU, and 28-day mortality rates than those transferred in less than 5.65 hours. However, there was no significant impact on 1-year mortality. Additionally, longer ED to ICU times were associated with shorter ICU stays but did not significantly change overall hospital LOS.

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Abbreviations

APSIII Acute physiology score III

ASD Absolute standardized diference

AUC Area under the curve

CCI Charlson comorbidity index

CCU Coronary care unit

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CVICU Cardiac vascular intensive care unit

ED Emergency department

ICU Intensive care unit

IQR Interquartile range

LODS Logistic organ dysfunction system

LOS Length of stay

MICU Medical intensive care unit

MIMIC-IV Medical Information Mart for Intensive Care IV

Neuro SICU Neuro surgical intensive care unit

OASIS Oxford acute severity of illness score

OR Odds ratio

PSM Propensity score matching

SAPSII Simplifed acute physiological score II

SD Standard deviation

SICU Surgical intensive care unit

SIRS Systemic infammatory response syndrome

SOFA Sequential organ failure assessment

TSICU Trauma surgical intensive care unit

VIF Variance inflation factor

Author contributions

JWQ, YNY and ZMS designed the study, conducted the formal analysis, and drafted the manuscript. JWQ extracted the data from the MIMIC-IV database. YNY, ZMS and QXL reviewed and analyzed the data. KSZ,

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LYZ and WZZ guided the manuscript review and editing. XFJ and MQC instructed the conceptualization. All authors read and approved the final manuscript.

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Availability of data and materials

The datasets analyzed in this study are publicly available summary statistics. Data generated and analyzed can be available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was an analysis of the third-party anonymized databases with pre-existing IRB approval.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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Figure Legends

Fig. 1 Flow chart of patients selection for analytic

MIMIC-IV Medical Information Mart for Intensive Care-IV, ICU intensive care unit

Fig. 2 Kaplan-Meier survival analysis curves for all-cause mortality.

ED to ICUs time (hours): Q1 (≤2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

A. Cumulative probability of all-cause mortality at 28-day before PSM; B. At 1-year before PSM; C. At 28-day after PSM; D. At 1-year after PSM

Fig.3 Restricted cubic spline regression analysis of ED to ICUs time with in hospital all-cause mortality.

Red heavy central lines represent the estimated adjusted hazard ratios, with shaded ribbons denoting 95% confidence intervals. ED to ICUs time 5.65hr was selected as the reference level represented by the vertical dotted lines. The horizontal dotted lines represent the hazard ratio of 1.0.

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A. Restricted cubic spline for in-hospital death.

B. Restricted cubic spline for ICU death. HR hazard ratio, CI confidence interval

Fig. 4 Forest plots of odd ratios for the primary outcome in different subgroups.

A. ED to ICUs time and in-hospital death before PSM;

B. ED to ICUs time and ICU death before PSM;

C. ED to ICUs time and ICU death in the matched cohort after PSM;

D. ED to ICUs time and ICU death in the matched cohort after PSM.

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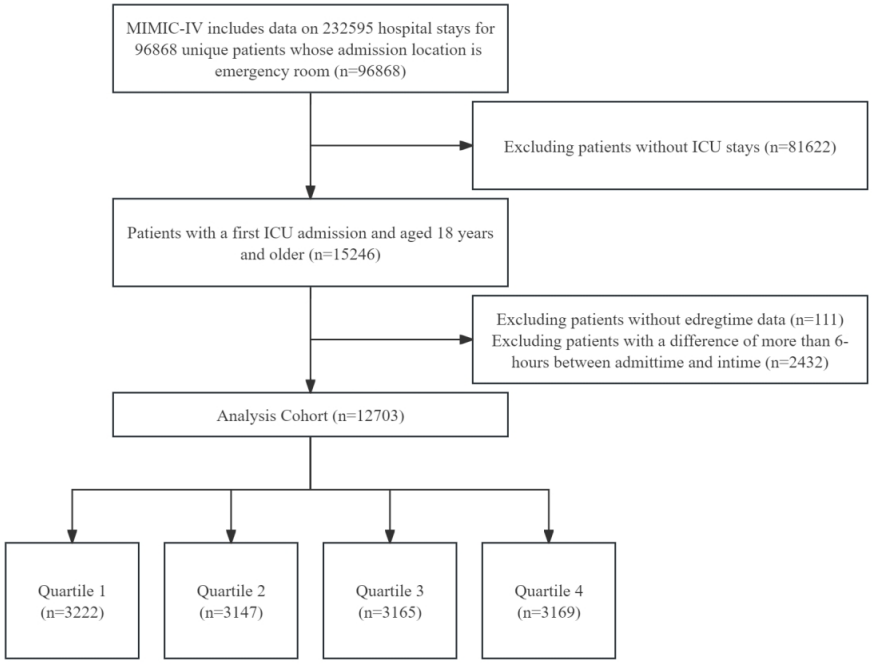


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MIMIC-IV Medical Information Mart for Intensive Care-IV, ICU intensive care unit

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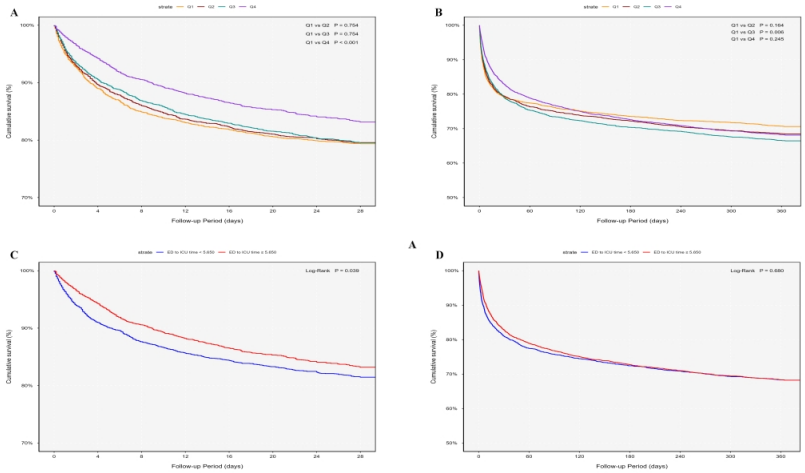


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2566x1443mm (76 x 76 DPI)

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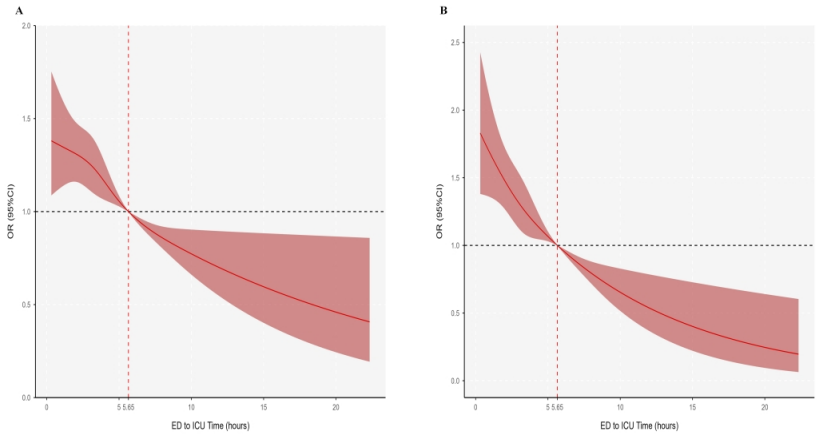


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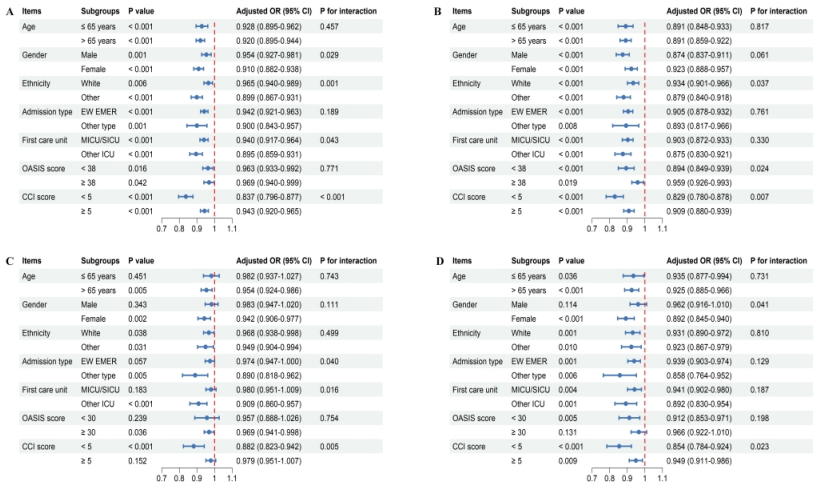


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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

Supplemental Digital Content: supplementary results

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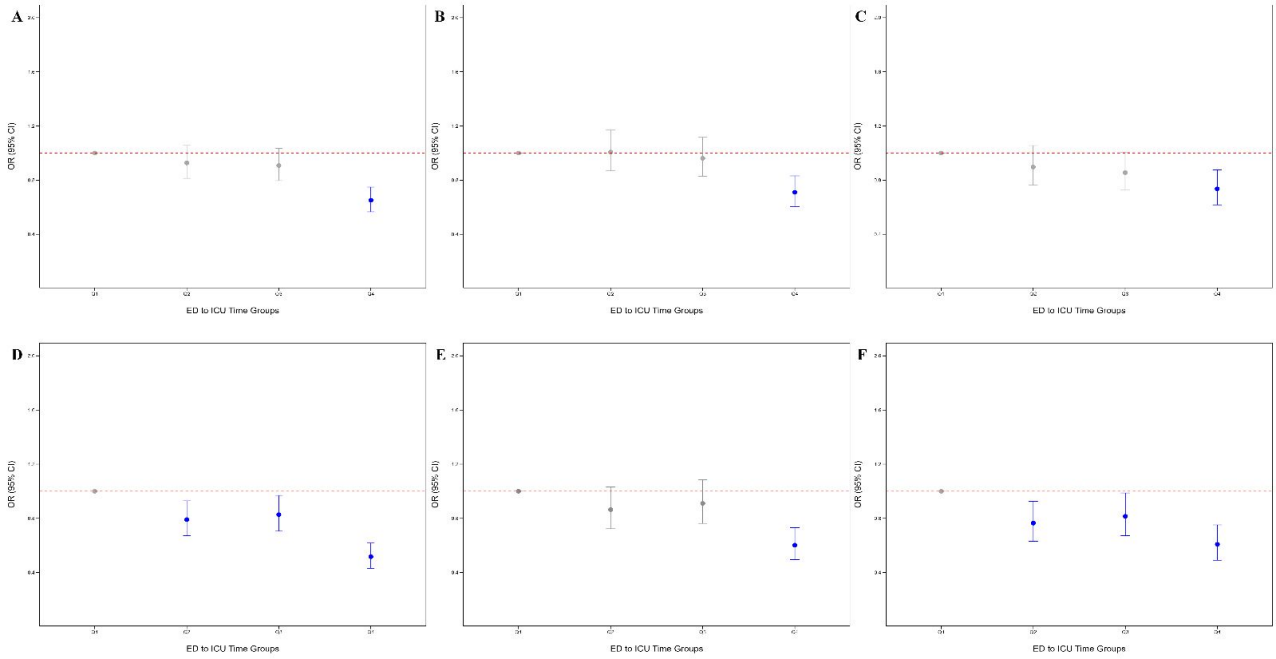
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Supplementary Figure 1



Supplementary Fig.1 Odds Ratios for all-cause in-hospital and ICU mortality among different groups.

ED to ICUs time (hours): Q1 (≤2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

A. ED to ICUs time and in-hospital mortality; unadjusted.

B. ED to ICUs time and in-hospital mortality; adjusted for OASIS Score and CCI score.

C. ED to ICUs time and in-hospital mortality; adjusted for age, gender, first care unit, SOFA score, LODS score, OASIS score, CCI score, SIRS score, SAPSII score, APSIII score, acute kidney failure, acute respiratory failure, hypertension, pneumonia, urinary tract infection, acidosis and cerebral edema.

D. ED to ICUs time and ICU mortality; unadjusted.

E. ED to ICUs time and ICU mortality; adjusted for OASIS Score and CCI score;

F. ED to ICUs time and ICU mortality; adjusted for age, gender, first care unit, SOFA score, LODS score, OASIS score, CCI score, SIRS score, SAPSII score, APSIII score, acute kidney failure, acute respiratory failure, hypertension, pneumonia, urinary tract infection, acidosis and cerebral edema.

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Supplementary Table 1

Baseline characteristics of ICU patients grouped according to ED to ICUs time quartiles

Categories	Overall (N=12703)	Q1 (N=3222)	Q2 (N=3147)	Q3 (N=3165)	Q4 (N=3169)	P-value
ED to ICUs time, hr, median (IQR)	3.98 (2.83, 5.65)	2.16 (1.66, 2.53)	3.41 (3.12, 3.68)	4.70 (4.3, 5.12)	7.22 (6.33, 8.72)	< 0.001
Demographic						
Age, years, median (IQR)	64 (49, 79)	61 (46, 77)	65 (49, 79)	66 (51, 80)	66 (51, 80)	< 0.001
Male, n (%)	7178 (56.51)	1993 (61.86)	1771 (56.28)	1784 (56.37)	1630 (51.43)	< 0.001
Marital status, n (%)						< 0.001
Married	4822 (37.96)	1149 (35.66)	1215 (38.61)	1205 (38.07)	1253 (39.54)	
Single	3777 (29.73)	892 (27.68)	905 (28.76)	948 (29.95)	1032 (32.57)	
Widowed	1541 (12.13)	295 (9.16)	378 (12.01)	427 (13.49)	441 (13.92)	
Divorced	767 (6.04)	172 (5.34)	157 (4.99)	201 (6.35)	237 (7.48)	
Unkown	1796 (14.14)	714 (22.16)	492 (15.63)	384 (12.13)	206 (6.50)	
Ethnicity, n (%)						< 0.001
White	8149 (64.15)	1799 (55.83)	1986 (63.11)	2122 (67.05)	2242 (70.75)	
Black	1103 (8.68)	249 (7.91)	249 (7.91)	277 (8.75)	365 (11.52)	
Asian	375 (2.95)	87 (2.76)	87 (2.76)	105 (3.32)	117 (3.69)	
Hispanic/Latino	463 (3.64)	112 (3.48)	115 (3.65)	117 (3.70)	119 (3.76)	
Other	636 (5.01)	195 (6.05)	146 (4.64)	149 (4.71)	146 (4.61)	
Unknown	1977 (15.56)	838 (26.01)	564 (17.92)	395 (12.48)	180 (5.68)	
Weight, Kg, median (IQR)	76.3 (64.0, 90.7)	78.0 (66.0, 93.0)	76.9 (64.0, 91.0)	76.2 (64.0, 90.0)	74.7 (62.5, 88.9)	< 0.001
Admission type, n (%)						< 0.001
EW EMER	11357 (89.40)	2912 (90.38)	2883 (91.61)	2834 (89.54)	2728 (86.08)	
OBSERVATION ADMIT	1194 (9.40)	269 (8.35)	236 (7.50)	300 (9.48)	389 (12.28)	
EU OBSERVATION	147 (1.16)	40 (1.24)	27 (0.86)	30 (0.95)	50 (1.58)	
DIRECT OBSERVATION	4 (0.03)	1 (0.03)	1 (0.03)	0 (0.00)	2 (0.06)	
URGENT	1 (0.01)	0 (0.00)	0 (0.00)	1 (0.03)	0 (0.00)	

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First care unit, n (%)						< 0.001
MICU	3301 (25.99)	683 (21.20)	798 (25.36)	863 (27.47)	957 (30.20)	
SICU	2569 (20.22)	553 (17.16)	732 (23.26)	712 (22.50)	572 (18.05)	
TSICU	2709 (21.33)	941 (29.21)	707 (22.47)	571 (18.04)	490 (15.46)	
MICU/SICU	2123 (16.71)	365 (11.33)	447 (14.20)	602 (19.02)	709 (22.37)	
CCU	1053 (8.29)	417 (12.94)	260 (8.26)	200 (6.32)	176 (5.55)	
CVICU	445 (3.50)	164 (5.09)	93 (2.96)	92 (2.91)	96 (3.03)	
Neuro SICU	503 (3.96)	99 (3.07)	110 (3.50)	125 (3.95)	169 (5.33)	
Most common diagnoses, n (%)						
Acute kidney failure, unspecified	1928 (20.80)	412 (18.38)	448 (19.01)	474 (19.01)	594 (25.92)	< 0.001
Acute respiratory failure	1678 (18.10)	473 (21.11)	453 (19.22)	431 (18.10)	321 (14.01)	< 0.001
Unspecified essential hypertension	931 (10.04)	237 (10.58)	254 (10.78)	253 (10.63)	187 (8.16)	0.003
Acidosis	907 (9.78)	198 (8.84)	212 (8.99)	207 (8.69)	290 (12.65)	< 0.001
Cerebral edema	897 (9.68)	224 (10.00)	267 (11.33)	256 (10.75)	150 (6.54)	< 0.001
Urinary tract infection, site not specified	879 (9.48)	180 (8.03)	205 (8.70)	236 (9.91)	258 (11.26)	< 0.001
Pneumonia, organism unspecified	856 (9.23)	200 (8.92)	198 (8.40)	237 (9.95)	221 (9.64)	0.135
Severity scores						
SOFA score, mean (SD)	1.15 (1.74)	1.19 (1.80)	1.06 (1.63)	1.16 (1.79)	1.19 (1.76)	0.042
LODS score, mean (SD)	4.33 (3.26)	4.56 (3.48)	4.32 (3.27)	4.41 (3.40)	4.02 (3.05)	< 0.001
OASIS score, mean (SD)	31.81 (9.28)	32.84 (9.72)	31.81 (9.32)	31.70 (9.20)	30.86 (8.74)	< 0.001
SIRS score, mean (SD)	2.51 (0.99)	2.63 (0.99)	2.53 (0.99)	2.47 (1.00)	2.41 (0.98)	< 0.001
CCI, mean (SD)	4.80 (3.07)	4.35 (3.01)	4.67 (3.03)	5.02 (3.12)	5.16 (3.12)	< 0.001
SAPSII, mean (SD)	33.26 (14.69)	32.97 (15.37)	32.98 (14.88)	33.69 (14.48)	33.41 (13.99)	0.002
APSIII, mean (SD)	46.37 (24.00)	47.61 (25.71)	46.04 (23.82)	47.24 (23.95)	44.57 (22.26)	< 0.001
Outcome						
Follow-up death, n (%)	4509 (35.50)	1031 (32.00)	1105 (35.11)	1196 (37.79)	1177 (37.14)	< 0.001
Hospital death, n (%)	1988 (15.65)	567 (17.60)	520 (16.52)	514 (16.24)	387 (12.21)	< 0.001
ICU death, n (%)	1185 (9.33)	375 (11.64)	297 (9.44)	311 (9.82)	202 (6.37)	< 0.001
28-days death, n (%)	2490 (19.60)	665 (20.64)	643 (20.43)	648 (20.47)	534 (16.85)	< 0.001
1-year death, n (%)	4013 (31.59)	947 (29.39)	993 (31.55)	1062 (33.55)	1011 (31.90)	0.005

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Hosptial-LOS, days, median (IQR)	5.0 (2.8, 9.4)	5.1 (2.7, 9.7)	5.0 (2.8, 9.1)	5.0 (2.8, 9.3)	5.3 (3.0, 9.3)	0.054
ICU-LOS, median (IQR)	1.80 (0.96, 3.59)	1.96 (1.11, 4.15)	1.80 (0.97, 3.64)	1.78 (0.95, 3.53)	1.70 (0.91, 2.98)	< 0.001

ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr)

ED emergency department, ICU intensive care unit, IQR interquartile range, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit, CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, SOFA sequential organ failure assessment, LODS logistic organ dysfunction system, OASIS Oxford acute severity of illness score, SIRS systemic infammatory response syndrome, CCI Charlson Comorbidity Index, SAPSII, simplifed acute physiological score II, APSIII acute physiology score III, LOS length of stay

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Supplementary Table 2

Baseline characteristics between two groups before and after PSM

Categories	Original Cohort		*P*-value	Matched Cohort		*P*-value
	ED to ICUs Time <5.65 hr (n=9517)	ED to ICUs Time >= 5.65 hr (n=3186)		ED to ICUs Time <5.65 hr (n=3160)	ED to ICUs Time >= 5.65 hr (n=3160)	
Demographic						
Age, years, median (IQR)	64 (49, 79)	66 (51, 80)	< 0.001	67 (52, 81)	66 (51, 80)	0.257
Male, n (%)	5541 (58.22)	1637 (51.38)	< 0.001	1590 (50.32)	1631 (51.61)	0.707
Ethnicity, n (%)			< 0.001			0.879
White	5896 (61.95)	2253 (70.72)		2246 (71.08)	2242 (70.95)	
Black	736 (7.73)	367 (11.52)		353 (11.17)	358 (11.33)	
Asian	256 (2.69)	119 (3.74)		122 (3.86)	114 (3.61)	
Hispanic/Latino	344 (3.61)	119 (3.74)		102 (3.23)	119 (3.77)	
Other	489 (5.14)	147 (4.61)		149 (4.72)	146 (4.62)	
Unknown	1796 (18.87)	181 (5.68)		188 (5.95)	181 (5.73)	
Admission type, n (%)			< 0.001			0.970
EW EMER	8613 (90.50)	2744 (86.13)		2736 (86.58)	2743 (86.80)	
OBSERVATION ADMIT	804 (8.45)	390 (12.24)		376 (11.90)	370 (11.71)	
EU OBSERVATION	97 (1.02)	50 (1.57)		47 (1.49)	47 (1.49)	
DIRECT OBSERVATION	2 (0.02)	2 (0.06)		1 (0.03)	0 (0.00)	
URGENT	1 (0.01)	0 (0.00)		0 (0.00)	0 (0.00)	
First care unit, n (%)			< 0.001			0.922
MICU	2341 (24.60)	960 (30.13)		913 (28.89)	954 (30.19)	
SICU	1988 (20.89)	581 (18.24)		593 (18.77)	581 (18.39)	
TSICU	2218 (23.31)	491 (15.41)		483 (15.28)	491 (15.54)	
MICU/SICU	1411 (14.83)	712 (22.35)		722 (22.85)	698 (22.09)	
CCU	877 (9.22)	176 (5.52)		174 (5.51)	176 (5.57)	
CVICU	349 (3.67)	96 (3.01)		104 (3.29)	96 (3.04)	

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Neuro SICU	333 (3.50)	170 (5.34)		171 (5.41)	164 (5.19)	
Severity scores						
SOFA score, mean (SD)	1.14 (1.73)	1.19 (1.76)	0.054	1.17 (1.79)	1.18 (1.75)	0.443
LODS score, mean (SD)	4.43 (3.32)	4.02 (3.05)	< 0.001	4.07 (3.05)	4.03 (3.05)	0.600
OASIS score, mean (SD)	32.12 (9.43)	30.87 (8.73)	< 0.001	31.22 (9.11)	30.91 (8.73)	0.438
SIRS score, mean (SD)	2.55 (0.99)	2.41 (0.98)	< 0.001	2.41 (0.98)	2.41 (0.98)	0.985
CCI, mean (SD)	4.68 (3.05)	5.16 (3.12)	< 0.001	5.21 (3.11)	5.14 (3.11)	0.269
SAPSII, mean (SD)	33.20 (14.92)	33.43 (14.00)	0.047	33.59 (14.47)	33.41 (14.02)	0.909
APSIII, mean (SD)	46.97 (24.53)	44.59 (22.24)	< 0.001	44.79 (22.19)	44.65 (22.26)	0.732
Outcome						
Follow-up death, n (%)	3326 (34.95)	1183 (37.13)	0.031	1140 (36.08)	1169 (36.99)	0.465
Hospital death, n (%)	1598 (16.79)	390 (12.24)	< 0.001	449 (14.21)	386 (12.22)	0.021
ICU death, n (%)	981 (10.31)	204 (6.40)	< 0.001	272 (8.61)	202 (6.39)	< 0.001
28-days death, n (%)	1952 (20.51)	538 (16.89)	< 0.001	586 (18.54)	531 (16.80)	0.075
1-year death, n (%)	2996 (31.48)	1017 (31.92)	0.659	1004 (31.77)	1004 (31.77)	1.000
Hospital-LOS, days, median (IQR)	5.0 (2.8, 9.5)	5.3 (3.0, 9.3)	0.011	4.9 (2.8, 8.7)	5.3 (3.0, 9.3)	0.016
ICU-LOS, median (IQR)	1.8 (1.0, 3.8)	1.7 (0.9, 3.0)	< 0.001	1.8 (0.9, 3.3)	1.7 (0.9, 3.0)	< 0.001

ED emergency department, ICU intensive care unit, IQR interquartile range, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit, CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, SOFA sequential organ failure assessment, LODS logistic organ dysfunction system, OASIS Oxford acute severity of illness score, SIRS systemic infammatory response syndrome, CCI Charlson Comorbidity Index, SAPSII, simplifed acute physiological score II, APSIII acute physiology score III, LOS length of stay

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Supplementary Table 3

Discrimination of each predictive model for outcomes using AUC

Models	AUC (95% CI)	P-value	Models	AUC (95% CI)	P-value
Hospital death					
SOFA Score	0.645 (0.632-0.659)	< 0.001	+ ED to ICUs time	0.657 (0.643-0.671)	< 0.001
LODS Score	0.810 (0.799-0.820)	< 0.001	+ ED to ICUs time	0.815 (0.801-0.822)	< 0.001
OASIS Score	0.805 (0.795-0.815)	< 0.001	+ ED to ICUs time	0.806 (0.795-0.816)	< 0.001
SIRS Score	0.650 (0.638-0.662)	< 0.001	+ ED to ICUs time	0.657 (0.644-0.670)	< 0.001
CCI	0.673 (0.661-0.685)	< 0.001	+ ED to ICUs time	0.685 (0.673-0.697)	< 0.001
SAPSII	0.816 (0.807-0.826)	< 0.001	+ ED to ICUs time	0.820 (0.810-0.829)	< 0.001
APSIII	0.812 (0.801-0.823)	< 0.001	+ ED to ICUs time	0.814 (0.803-0.824)	< 0.001
ICU death					
SOFA Score	0.645 (0.632-0.659)	< 0.001	+ ED to ICUs time	0.655 (0.642-0.669)	< 0.001
LODS Score	0.810 (0.799-0.820)	< 0.001	+ ED to ICUs time	0.811 (0.800-0.822)	< 0.001
OASIS Score	0.805 (0.795-0.815)	< 0.001	+ ED to ICUs time	0.805 (0.795-0.816)	< 0.001
SIRS Score	0.650 (0.638-0.662)	< 0.001	+ ED to ICUs time	0.656 (0.644-0.669)	< 0.001
CCI	0.673 (0.661-0.685)	< 0.001	+ ED to ICUs time	0.680 (0.668-0.692)	< 0.001
SAPSII	0.816 (0.807-0.826)	< 0.001	+ ED to ICUs time	0.819 (0.810-0.829)	< 0.001
APSIII	0.812 (0.801-0.823)	< 0.001	+ ED to ICUs time	0.813 (0.802-0.824)	< 0.001

ED emergency department, ICU intensive care unit, SOFA sequential organ failure assessment, LODS logistic organ dysfunction system, OASIS Oxford acute severity of illness score, SIRS systemic infammatory response syndrome, CCI Charlson Comorbidity Index, SAPSII, simplifed acute physiological score II, APSIII acute physiology score III

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Supplementary Table 4

Relationship between ED to ICUs time and length of stay

	Coef.	S.E	t-value	*P*-value
Hospital-LOS[a]				
Model 1	-0.099	0.032	-3.153	0.002
Model 2	-0.032	0.030	-1.070	0.285
Model 3	0.031	0.029	1.069	0.285
ICU-LOS[b]				
Model 1	-0.157	0.017	-9.439	< 0.001
Model 2	-0.097	0.016	-6.287	< 0.001
Model 3	-0.069	0.015	-4.503	< 0.001

Model 1: unadjusted; Model 2: adjusted for OASIS score, Charlson Comorbidity Index; Model 3: adjusted for age, gender, admission type, first care unit, OASIS score, Charlson Comorbidity Index, acute kidney failure, acute respiratory failure, Hypertension, pneumonia, urinary tract infection, acidosis, cerebral edema.

[a] The relationship between ED to ICUs time and length of hospital stay was analyzed in a cohort that survived the hospital stay (n=10701)

[b] The relationship between ED to ICUs time index and length of ICU stay was analyzed in a cohort that survived the ICU stay (n=11518)

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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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Objectives: The association between the duration from the emergency department (ED) to the intensive care units (ICUs) and in-hospital mortality among patients admitted directly to the ICUs from the ED remains controversial. This study aimed to utilize data from the Medical Information Mart for Intensive Care IV database to explore the relationship between the ED to ICUs time and patient outcomes.

Design: Retrospective observational study.

Setting: Admissions to the Beth Israel Deaconess Medical Center intensive care from 2008 to 2019.

Participants: Adult patients in the MIMIC-IV database admitted directly from the ED to the ICUs, excluding those with missing data or a hospital-to-ICU admission interval over 6 hours.

Primary and secondary outcome measures: The primary outcome was in-hospital all-cause mortality. Secondary outcomes included 28-day all-cause mortality and length of stay in ICU and hospital.

Results: Out of 96,868 initially identified patients, 12,703 were included after exclusions. The median ED to ICUs time was 3.98 hours. Longer ED to ICUs times were associated with lower in-hospital mortality, decreasing from 17.6% in the shortest to 12.2% in the longest interval group, and shorter ICU stays. This association remained significant in adjusted models (OR: 0.61, 95% CI: 0.62-0.84, $p < 0.01$). Kaplan-Meier curves indicated consistently better survival in the longest interval group. Propensity score-matched analysis supported these findings, with the longest ED to ICUs time showing the highest 28-day survival rates ($p < 0.01$). Sensitivity analysis, reintroducing patients with hospital to ICU times over 6 hours, confirmed the robustness of these results.

Conclusions: Longer ED to ICUs time was linked to lower mortality and shorter ICU length of stay, suggesting potential benefits of extended ED stabilization time for critically ill patients.

Keywords: Emergency Department; Intensive Care Units; Mortality; Length of Stay; Patient Care

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STRENGTHS AND LIMITATIONS OF THIS STYDY

- The study utilizes a comprehensive and high-quality dataset from the MIMIC-IV database, offering detailed clinical and demographic data on ICU admissions.
- The use of robust statistical methods, including propensity score matching, sensitivity analysis, and subgroup analysis, enhances the validity and reliability of the results.
- Restricted cubic spline analysis was used to explore non-linear associations, offering a more nuanced understanding of ED to ICU transfer time effects.
- Lack of data on specific ED interventions may introduce information bias, as we could not assess their potential impact on the outcomes.
- Findings are based on data from a single healthcare system in the United States, which may limit the generalizability of results to other international settings.

Introduction

The emergency department (ED) often serves as the primary entry point for critically ill patients requiring immediate care. In recent years, there has been a marked increase in the capacity of ED clinicians to manage these patients over extended periods [1]. Following initial stabilization in ED, patients are typically transported to appropriate departments for further diagnostic tests or therapeutic procedures. Most critically ill patients will be transported to specialized intensive care units (ICUs), where provide the highest level monitoring and care [2]. During the last 2 decades, escalating acuity and volume of ED visits have resulted in greater demand for critical care services in the ED and ICUs [3]. ED crowding seems to become an international problem and has long been recognised as posing significant patient safety threats [4]. The impact of ED crowding and ED length of stay (ED-LOS) on inpatient mortality remains contentious and appears to vary by region [5, 6].

ED-LOS is a critical ED performance indicator and is closely related to crowding, quality of care and

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patient outcome [7]. However, the relationship between ED-LOS and in-hospital mortality, particularly for patients admitted directly to ICUs from the ED, remains a subject of debate. Groenland et al. [8] reported that a longer ED to ICUs time is associated with increased hospital mortality in the most severely ill patients. Conversely, Wessman et al. [9] found that patients with highest triage priority had a decreasing mortality risk with increasing quintile of ED-LOS for 7-day mortality, and the two highest quintile of ED-LOS had lower 30-day mortality. Saukkonen et al. [10] observed that the length of ED stay was not associated with the outcome of critically ill medical patients.

In light of these conflicting findings and the requirement for a large cohort to discern subtle outcome differences, our study leverages the extensive Medical Information Mart for Intensive Care IV (MIMIC-IV) database to examine the association between ED to ICUs time and mortality among critically ill patients. This study offers a unique perspective on the potential benefits of extended ED stabilization time, addressing gaps in the current understanding of optimal ED management duration prior to ICU transfer.

Methods

Study design

This retrospective observational study used data from the MIMIC-IV (version 2.2) database, which contains high-quality information on ICU admissions at Beth Israel Deaconess Medical Center from 2008 to 2019. Access to MIMIC-IV is limited to certified individuals; one author (JWQ) with access permission (Record ID 60311526) extracted the data. As all patient records in MIMIC-IV were fully de-identified, the institutional review board at Beth Israel Deaconess Medical Center waived the need for individual patient consent.

Cohort selection

We initially identified all patients aged 18 years and older who were admitted through the emergency room.

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Patients were excluded if they did not have a recorded ICU stay or if their admission time to the ED was missing. Additionally, patients with a gap of exceeding 6 hours between hospital admission (*admittime*) and ICU admission (*intime*) recorded in the database were not considered to be transferred directly from the ED to the ICUs and were thus excluded. For patients with multiple ICU admissions, only the first ICU stay was analyzed. After applying these criteria, the final study cohort was divided into four quartiles based on the ED to ICUs time.

Data extraction and defnitions

All baseline characteristics were extracted within the first 24h of ICU admission from the MIMIC-IV database, including age, gender, ethnicity, admission type, first care unit, chronic conditions, severity scores (Charlson Comorbidity Index (CCI) score, Acute physiology score III (APSIII)), and intervention during ICU. The ED to ICUs time was defined as the interval between *edregtime* (the time of registration in the ED) and *intime* (the time of transfer into the ICU) recorded in the MIMIC-IV database. *Admittime* provides the date and time the patient was admitted to the hospital. Data extraction was performed using pgAdmin4 PostgreSQL 16.1.

Primary outcomes and secondary outcomes

The primary outcome of this study was in hospital all-cause mortality. Of these, hospital death included ICU death and general ward inpatient death. The secondary outcomes included 28-day all-cause mortality, LOS in ICU and LOS in hospital. Patient mortality information for discharged patients was accessed from the US Social Security Death Index.

Statistical analysis

Our study divided patients into four quartiles based on their ED to ICUs admission times. We assessed the

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distribution of continuous data using the Kolmogorov-Smirnov test and presented them as means with standard deviations or medians with interquartile ranges. Categorical data were shown in counts and percentages. The Mann-Whitney U test and the Pearson chi-square or Fisher's exact tests were used for comparisons. We adjusted for multiple comparisons using the false discovery rate, with a 5% threshold.

We used logistic regression models to estimate odds ratios (ORs) and 95% confidence intervals (CIs) for hospital and ICU mortality. Analyses were conducted using three models: an unadjusted model, a model adjusted for illness severity alone, and a fully adjusted model addressing potential multicollinearity. The fully adjusted model included a comprehensive set of covariates to assess the relationship between ED to ICUs time and in-hospital mortality, while controlling for key confounders. Age and sex were included to account for demographic differences, while the APSIII and CCI scores provided a detailed assessment of acute illness severity and chronic comorbidity burden, respectively. Additional covariates included diabetes, congestive heart failure, renal disease, and malignancy, given their known impact on critically ill patients' outcomes. Adjustments were also made for invasive mechanical ventilation, dialysis, and vasopressor use, as these reflect the severity of respiratory failure, renal impairment, and hemodynamic instability. This selection of covariates aimed to minimize residual confounding and ensure precise estimates of the effect of ED to ICUs time on in-hospital mortality. Secondary outcomes, including 28-day mortality, were analyzed using similar models and Kaplan-Meier survival analysis, with additional tests for non-linear relationships and stratified analyses to ensure consistency across patient subgroups.

Restricted cubic spline (RCS) models were applied to further explore the non-linear association between ED to ICUs time and both in-hospital and ICU mortality. Pairwise propensity score matching (PSM) was then conducted to compare patients in the highest quartile of ED to ICUs time (Q4) with those in the lower quartiles (Q1, Q2, Q3). Propensity scores were derived using logistic regression, incorporating baseline characteristics such as age, gender, race, first care unit, diabetes, congestive heart failure, renal disease, malignancy, CCI

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score, APSIII score, invasive mechanical ventilation, dialysis and vasopressors. Matching employed a nearest-neighbor approach without replacement, with a caliper of 0.05 to ensure comparability. Balance before and after matching was assessed using standardized mean differences (SMD), complemented by balance plots for visual assessment.

A sensitivity analysis was also performed by reintroducing patients with a time difference greater than six hours between hospital admission and ICU admission. This analysis aimed to assess the robustness of the study findings by evaluating whether the association between ED to ICUs time and in-hospital mortality persisted when including patients potentially admitted to non-ICU wards before ICU transfer.

We performed statistical analyses with R software (version 4.3.3) and considered two-sided P values below 0.05 as significant. For multiple comparisons, we used a FDR correction to maintain this significance threshold.

Result

Baseline characteristics

From 96,868 patients initially identified, we excluded those without ICU stays, missing ED admission times, or whose hospital admission time and ICU admission time differed by more than 6 hours, leaving a final cohort of 12,703 patients. Details are shown in **Figure 1**. **Supplementary Table 1** showed the characteristics of 12,703 study participants categorized by ED to ICUs time quartiles. The median patient age was 64 years, and 56.51% were male. There were 1,185 ICU deaths, 1,988 in-hospital deaths, and 2,490 deaths within 28 days. The median ED to ICUs time was 3.98 hours. Among the chronic conditions in the cohort, diabetes was present in 2,938 patients (23.13%), congestive heart failure in 2,401 (18.90%), renal disease in 1,753 (13.80%), and malignant cancer in 1,090 (8.58%). The median age increased from 61 years in Q1 to 66 years in Q3 and Q4, with an overall median of 64 years ($p < 0.01$). Male patients comprised 56.51% of the study population, with

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their proportion decreasing from 61.86% in Q1 to 51.43% in Q4 ($p < 0.01$). The mean CCI score showed a significant increase across quartiles, ranging from 4.35 in Q1 to 5.16 in Q4 ($p < 0.01$). The mean APSIII score demonstrated a slight variation, decreasing from 46.37 in Q1 to 44.57 in Q4, though this difference remained statistically significant ($p < 0.01$).

In terms of outcomes, hospital mortality occurred in 1,988 patients (15.65%), with rates ranging from 17.60% in Q1 to 12.21% in Q4 ($p < 0.01$). ICU mortality was observed in 1,185 patients (9.33%), with the highest rate in Q1 (11.64%) and the lowest in Q4 (6.37%) ($p < 0.01$). The 28-day mortality rate was 19.60% overall, varying from 20.64% in Q1 to 16.85% in Q4 ($p < 0.01$). The median hospital LOS was 5.0 days (IQR: 2.8-9.4) across all patients, with a slight difference among quartiles, from 5.0 days in Q1 to 5.3 days in Q4 ($p = 0.05$). The median ICU LOS was 1.80 days, ranging from 1.96 days in Q1 to 1.70 days in Q4 ($p < 0.01$), indicating significant differences between groups.

Association of the ED to ICUs time with all-cause mortality

In the unadjusted model (**Fig. 2A**), patients in the longest ED to ICUs time group (Q4) had a significantly lower in-hospital mortality risk compared to those in the shortest time group (Q1), with an OR of 0.65 (95% CI: 0.57-0.75, $p < 0.01$) (**Table 1**). This association remained robust after adjusting for APSIII and CCI scores (Model 2: OR 0.64, 95% CI: 0.54-0.75, $p < 0.01$) and persisted after further adjustment for age, gender, first care unit, comorbidities, and ICU interventions (Model 3: OR 0.61, 95% CI: 0.62-0.84, $p < 0.01$) (**Fig. 2B-C**). A similar trend was observed for ICU mortality. In Model 1, Q4 was associated with a significantly lower ICU mortality risk compared to Q1, with an OR of 0.52 (95% CI: 0.43-0.62, $p < 0.01$) (**Fig. 2D**). This relationship remained significant in Model 2 (OR 0.53, 95% CI: 0.44-0.65, $p < 0.01$) and Model 3 (OR 0.63, 95% CI: 0.51-0.77, $p < 0.01$) (**Fig. 2E-F**). Additionally, the VIF values for the fully adjusted model remained below 5, indicating minimal multicollinearity among the covariates (**Supplementary Fig. 1-2**). RCS

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regression analysis indicated a decreasing trend in both in-hospital and ICU mortality risk with longer ED to ICUs time ($p < 0.01$) (**Supplementary Fig. 3**), though some non-linearity may be present.

Among the secondary outcomes, **Fig. 3A** presents the Kaplan-Meier survival curve for 28-day cumulative survival across the quartiles of ED to ICUs time. Throughout the follow-up period, patients in Q4, representing the longest ED to ICUs time, consistently showed higher survival rates compared to the other groups. The survival advantage of Q4 over Q1 was statistically significant, as indicated by a log-rank p-value of < 0.001.

Table 1 Odds Ratios (ORs) for All-Cause Mortality in-hospital and ICU mortality from Logistic Regression

Categories	Model 1		Model 2		Model 3	
	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value
Hospital death						
Continuous variable per 1 unit	0.94 (0.92, 0.95)	< 0.01	0.94 (0.92, 0.95)	< 0.001	0.95 (0.93, 0.97)	< 0.01
Quartilea[a]						
Q1 (N=3222)	Ref.		Ref.		Ref.	
Q2 (N=3147)	0.93 (0.81, 1.06)	0.26	0.97 (0.83, 1.13)	0.70	0.94 (0.81, 1.10)	0.58
Q3 (N=3165)	0.91 (0.80, 1.04)	0.15	0.85 (0.72, 0.99)	0.04	0.87 (0.76, 1.03)	0.21
Q4 (N=3169)	0.65 (0.57, 0.75)	< 0.01	0.64 (0.54, 0.75)	< 0.01	0.61 (0.62, 0.84)	< 0.01
ICU death						
Continuous variable per 1 unit	0.90 (0.88, 0.93)	< 0.01	0.92 (0.89, 0.94)	< 0.01	0.93 (0.90, 0.96)	< 0.01
Quartilea[a]						
Q1 (N=3222)	Ref.		Ref.		Ref.	
Q2 (N=3147)	0.79 (0.67, 0.93)	0.01	0.83 (0.69, 0.99)	0.04	0.82 (0.69, 0.99)	0.04
Q3 (N=3165)	0.83 (0.71, 0.97)	0.02	0.80 (0.67, 0.96)	0.02	0.86 (0.71, 1.02)	0.09
Q4 (N=3169)	0.52 (0.43, 0.62)	< 0.01	0.53 (0.44, 0.65)	< 0.01	0.63 (0.51, 0.77)	< 0.01

Model 1: unadjusted; Model 2: adjusted for APSIII score, CCI score; Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, invasive mechanical ventilation, analysis and vasopressors.

[a] ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr)

Propensity Score Matched Analysis

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After performing PSM with a 1:1 ratio and a caliper of 0.05, factors such as age, gender, race, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, CCI score, APSIII score, invasive mechanical ventilation, dialysis, and vasopressor use were balanced between groups. The effectiveness of the matching was confirmed through standardized mean difference plots (**Supplementary Fig. 4-6**) and further supported by the balanced baseline characteristics shown in **Supplementary Table 2-4**. Post-matching survival analysis revealed that patients in Q4 consistently exhibited higher 28-day survival rates compared to those in Q1, Q2, and Q3 (**Fig. 3B-D**). Kaplan-Meier survival curves demonstrated that Q4 maintained the highest survival rates throughout the follow-up period across all matched pairs, with significant differences observed in log-rank tests ($p < 0.05$ for all comparisons), underscoring the association between longer ED to ICUs times and improved short-term survival outcomes.

Relationship between ED to ICUs time and length of stay

Supplementary Table 5 shows the relationship between ED to ICUs time and both hospital and ICU LOS. For hospital LOS, the unadjusted model indicated a significant negative association with ED to ICUs time (Coef. = -0.099, P = 0.002); however, this association was not significant after adjustment in Models 2 and 3 ($p > 0.05$). In contrast, for ICU LOS, a longer ED to ICUs time was consistently associated with a shorter duration of stay across all models, with coefficients of -0.157 ($p < 0.01$) in Model 1, -0.097 ($p < 0.001$) in Model 2, and -0.069 ($p < 0.01$) in Model 3. These results indicate a sustained inverse relationship between ED to ICUs time and ICU LOS, even after adjustments.

Subgroup Analysis

The association between longer ED to ICUs time and reduced in-hospital mortality remains significant across most subgroups, with adjusted ORs below 1 (**Fig. 4A**). Significant interactions were observed with gender (p

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= 0.03), CCI score ($p < 0.01$), and malignant cancer ($p = 0.01$), indicating that the impact of extended ED to ICUs time may vary depending on these factors. Similar trends were noted for ICU mortality, with adjusted ORs generally showing a reduction in risk with prolonged ED stays across most subgroups (**Fig. 4B**). However, none of the interactions reached statistical significance for gender ($p = 0.06$). Significant interaction effects were observed for CCI score ($p < 0.01$) and congestive heart failure ($p = 0.03$), suggesting variability in the association depending on the severity of comorbidities. These findings underscore the robustness of the protective association between longer ED to ICUs time and improved survival, while highlighting certain patient characteristics that may influence this relationship.

Sensitivity Analysis

In the sensitivity analysis, patients initially excluded due to a time gap of more than 6 hours between hospital admission (*admittime*) and ICU admission (*intime*) were reintroduced to evaluate the impact of this criterion on the study results. Re-inclusion of these patients did not alter the overall trends observed in the primary analysis. **Supplementary Table 6** presents the baseline characteristics of this expanded cohort. The adjusted odds ratios from the logistic regression models (**Supplementary Fig. 7**) continued to indicate lower odds of in-hospital and ICU mortality for patients in Q4 compared to those in the other groups. Similarly, the Kaplan-Meier survival curves (**Supplementary Fig. 8**) confirmed that patients in Q4 (ED to ICUs time >7.54 hours) consistently maintained the highest 28-day survival rates compared to Q1, Q2, and Q3, with significant differences in the log-rank tests (Q1 vs. Q4, $P < 0.001$; Q1 vs. Q3, $P = 0.047$). These findings suggest that the exclusion of patients with longer pre-ICU stays did not materially affect the relationship between extended ED stays and better survival outcomes, thereby reinforcing the robustness of the results.

Discussion

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Our study aimed to explore the relationship between ED to ICUs admission time and hospital mortality using the MIMIC-IV database. Through three progressively adjusted models and propensity score matching to minimize baseline confounding, we unexpectedly found that longer ED to ICUs time (5.65-34.55 hours) were associated with lower in-hospital mortality, independent of other prognostic factors. Similar patterns were observed for ICU and 28-day mortality. Subgroup analysis revealed that this association held across most patient groups, although certain characteristics, such as malignant cancer, appeared to modify the effect. This may be because patients with malignancy often present with complex conditions, and acute issues are typically stabilized in the ED. In advanced cases, treatment focuses on symptom relief over intensive rescue, as further ICU intervention may not improve prognosis [11, 12].

Previous studies have shown an association between prolonged ED stay and increased hospital mortality. A large retrospective cohort study with data derived from MIMIC-III showed that every delayed hour in the ED increased the in-hospital mortality, and in particular patients with ≥ 6h of transfer from ED to the ICUs had a significantly higher mortality rate than those with < 6h (13). However, this analysis may have included patients initially transferred from the ED to general wards before ICU admission, which may not fully reflect outcomes for those transferred directly from the ED to the ICU. Other studies have similarly shown that a longer hospital stay before ICU admission was correlated with mortality, because patients with longer hospital stays may have more comorbidities [14, 15]. We limited the time between recorded hospital admission and ICU admission to 6 hours to identify patients transferred directly from the ED to the ICUs. However, this 6-hour threshold may introduce selection bias, as some patients initially transferred from the ED to non-ICU wards may have stayed there for less than six hours before ICU transfer. To address this, we conducted a sensitivity analysis reintroducing patients with hospital to ICUs times over six hours, confirming the robustness of our findings.

One possible explanation for the observed association between longer ED to ICUs time and lower

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mortality is that sufficient waiting time in ED allows for the initiation of critical care interventions on-site, providing stabilizing treatment. Previous studies have demonstrated that early application of critical care interventions, such as goal-directed therapy in the initial stages of severe sepsis and septic shock, can improve patient outcomes by stabilizing clinical status and reducing deterioration risks prior to ICU transfer [16]. Similarly, the study presented by Kadar et al. showed that compared with patients in ED usual care group, those in ICU telemedicine support group had lower mortality and and lower ICU resource utilization [17], since ICU telemedicine team had observed a stable or improving course long enough to have confidence that the patient would be safe in a non-ICU setting. Collectively, these findings underscore the importance of timely critical care interventions in the ED for reducing mortality and improving resource efficiency. Trzeciak et al. [18] further noted that during periods of ED overcrowding, staff may struggle to provide the necessary care for critically ill patients, a limitation associated with increased mortality when patients receive care in settings unable to fully support their critical needs [19]. Thus, extended ED to ICUs time may not simply reflect delays but rather opportunities for essential stabilization, supporting the notion that early critical care in the ED can be pivotal in reducing mortality.

An alternative explanation may be that premature intrahospital transport (IHT) of critically ill patients carries significant risks. While IHT from the ED is often unavoidable, it exposes patients to a mobile environment, increasing the likelihood of adverse events, including equipment failure, device dislodgement, inadequate monitoring, and cardiovascular or respiratory instability [20, 21]. Such serious adverse events can worsen patient conditions, heightening the risk of mortality. Once critically ill patients being transported, they will be more vulnerable with an increasing probability of adverse events, as high as 79.8% in a study of 441 IHT patients admitted to ICU [22]. Premature transport of critically ill patients from the ED, while potentially reducing ED to ICUs time, often limits essential preparation time, increasing the risk of complications [23]. Therefore, IHT requires careful consideration by emergency physicians, who must thoroughly evaluate the

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clinical status and balance the risks of premature transfer against the potential benefits of reducing ED to ICUs time to determine the safest timing and approach.

However, the ED to ICUs time is multifactorially influenced, such as ED crowding [24], triage process [25], availability of ICU beds [26, 27] and so on. To address these challenges, EDs worldwide are increasingly implementing initiatives to improve timeliness and throughput, such as bedside registration, self–check-in kiosks, immediate bedding, fast-track systems, and physician-in-triage protocols [28]. Abdulai et al. [29] examined trends in ED throughput in the United States from 2006 to 2016, and found that there were decreases in patient wait time, patients leaving without meeting the emergency physician, and boarding time after the implementation of a series of initiatives, especially compared with that of the previous decade. Enhanced throughput and reduced wait times enable more timely identification and treatment of critically ill patients, potentially improving outcomes. An effective triage system is necessary to recognize, diagnose and treat the most severely ill patients, which could result in lower mortality. To this end, various warning systems, such as ‘track and trigger’ and early warning scores, have been developed to detect early signs of deterioration [30, 31]. Fernandes ta al. [32] found that there was an improvement in decision-making of the health professionals' thereby leading to better clinical management and the outcomes of patients by using clinical decision support systems based on intelligent Systems in the ED. Another important factor which may influence ED to the ICUs time and worsen patient outcomes is availability of ICU beds [33]. A previous study found that ICU admission decisions for critically ill ED patients are affected by Medical ICU bed availability [34]. To address these limitations, hospitals are adopting innovative strategies to expand ICU capacity. For instance, Mitra et al. [35] found that establishing high-acuity units, which offer close monitoring and acute care comparable to the ICU but with fewer resources, was associated with reduced in-hospital mortality for patients requiring intensive care. Establishing a critical care resuscitation unit can handle more critically ill surgical patients and speed up the transfer of those with time-sensitive conditions to specialized centers. Furthermore, enhancing

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ICU capacity and improving its operational efficiency can reduce the time needed for patient admission to the ICU.

Our study highlights that managing critically ill patients in the emergency department involves more than just minimizing ED wait times. Considering the patient's clinical condition, ED circumstances, transport risks, and ICU bed availability to determine the best timing for ICU admission is essential. A key strength of our research is the use of the comprehensive MIMIC-IV database, which spans from 2008 to 2019. This dataset, the most recent before the onset of COVID-19, allows for an accurate analysis of the association between ED to ICU time and patient outcomes without the confounding effects of pandemic [36]. The use of propensity score analysis further minimizes baseline confounding, enhancing the reliability of our findings. We also progressively adjusted our models to accurately estimate the impact of ED to ICU time on patient outcomes.

This study has several limitations. First, it is a retrospective analysis from an observational study, which inherently cannot establish causality. Second, although we limited the hospital-to-ICU admission interval to 6 hours to exclude patients initially admitted to general wards before ICU transfer, a small subset of patients may still have been transferred from a ward within this timeframe, introducing potential selection bias. Third, the MIMIC database does not capture information on critical care interventions administered in the ED, which limits our analysis. This lack of data prevents us from evaluating how early interventions may have influenced the observed association between longer ED stays and lower in-hospital mortality, potentially introducing information bias. Lastly, the data were from the United States, so the results may not generalize to ED settings in other countries. Future studies should be well-designed, prospective, and multicenter to address these limitations.

Conclusions

Patients transferred from the ED to the ICUs within 5.65-34.55 hours experienced lower in-hospital, ICU, and

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28-day mortality rates compared to those transferred sooner. These findings suggest that extended ED stays may allow for critical stabilization that improves outcomes and underscore ED to ICUs time as an important factor in patient survival, warranting further research to optimize transfer protocols.

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Author contributions

JWQ, YNY and ZMS designed the study, conducted the formal analysis, and drafted the manuscript. JWQ extracted the data from the MIMIC-IV database. YNY, ZMS and QXL reviewed and analyzed the data. KSZ, LYZ and WZZ guided the manuscript review and editing. XFJ and MQC instructed the conceptualization. MQC is the guarantor of this study. All authors read and approved the final manuscript.

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Data availability statement

Data generated and analyzed can be available from the corresponding author on reasonable request.

Declarations

Ethics approval

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Given that this study analyzed an anonymous public database with prior Institutional Review Board approval, no ethical review was necessary.

Patient consent for publication

Not required.

Patient and public involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Competing interests

The authors declare that they have no competing interests.

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Figure Legends

Fig. 1 Flow chart of patients selection for analytic

MIMIC-IV Medical Information Mart for Intensive Care-IV, ICU intensive care unit

Fig. 2 Odds Ratios for all-cause in-hospital and ICU mortality among different groups

ED to ICUs time (hours): Q1 (≤2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

A. Association between ED-to-ICUs time and in-hospital mortality, unadjusted.

B. Association between ED-to-ICUs time and in-hospital mortality, adjusted for APSIII score and CCI score.

C. Association between ED-to-ICUs time and in-hospital mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors.

D. Association between ED-to-ICUs time and ICU mortality, unadjusted.

E. Association between ED-to-ICUs time and ICU mortality, adjusted for APSIII score and CCI score.

F. Association between ED-to-ICUs time and ICU mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors.

Fig. 3 Kaplan-Meier survival analysis curves for all-cause mortality.

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ED to ICUs time (hours): Q1 (≤2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

A. Cumulative probability of all-cause mortality at 28-day before PSM; B. Cumulative probability of 28-day survival for Q4 vs. Q1 after PSM; C. Cumulative probability of 28-day survival for Q4 vs. Q2 after PSM; D. Cumulative probability of 28-day survival for Q4 vs. Q3 after PSM.

Fig. 4 Forest plots of odd ratios for the primary outcome in different subgroups.

A. Association between ED to ICUs time and in-hospital mortality;

B. Association between ED to ICUs time and ICU mortality.

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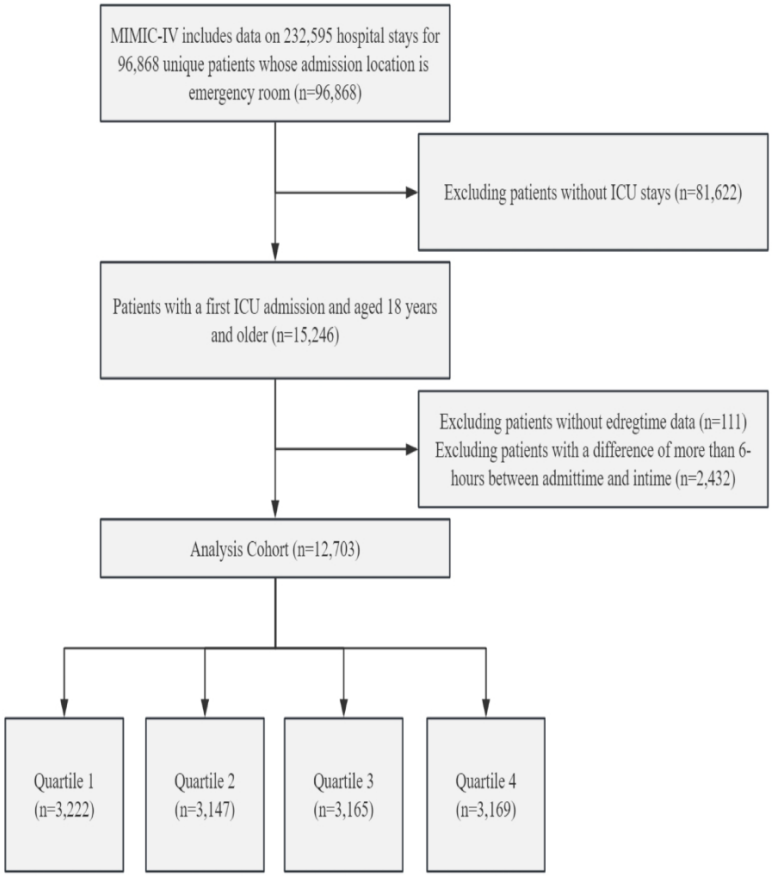


Fig. 1 Flow chart of patients selection for analytic
MIMIC-IV Medical Information Mart for Intensive Care-IV, ICU intensive care unit

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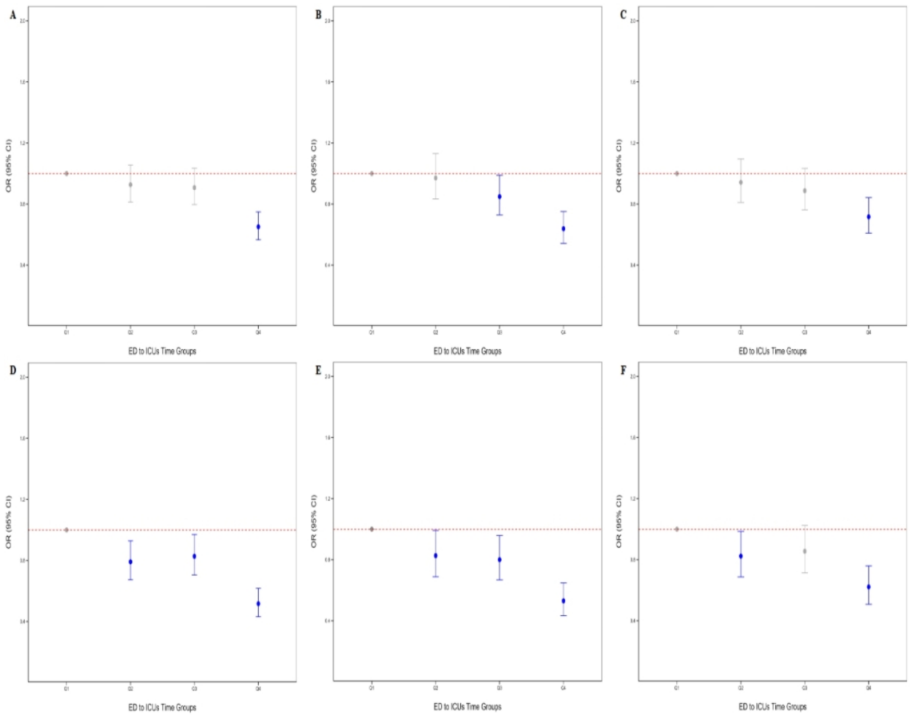


Fig. 2 Odds Ratios for all-cause in-hospital and ICU mortality among different groups
ED to ICUs time (hours): Q1 (≤2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).
A. Association between ED-to-ICUs time and in-hospital mortality, unadjusted.
B. Association between ED-to-ICUs time and in-hospital mortality, adjusted for APSIII score and CCI score.
C. Association between ED-to-ICUs time and in-hospital mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors.
D. Association between ED-to-ICUs time and ICU mortality, unadjusted.
E. Association between ED-to-ICUs time and ICU mortality, adjusted for APSIII score and CCI score.
F. Association between ED-to-ICUs time and ICU mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors.

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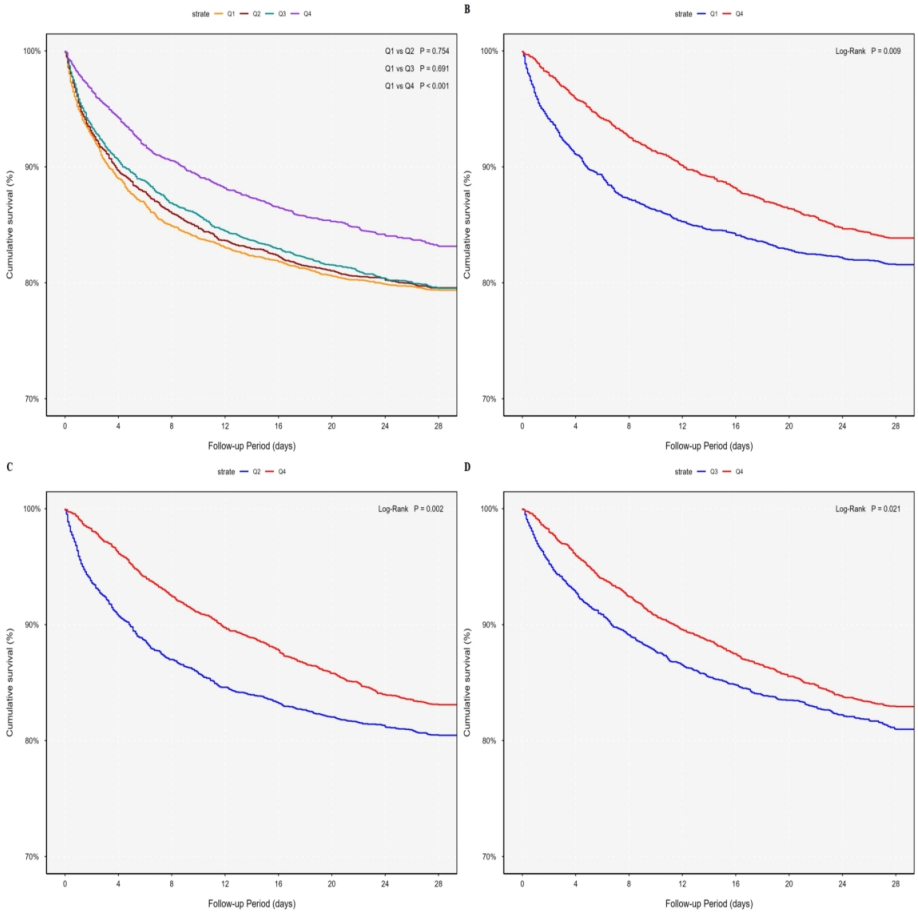


Fig. 3 Kaplan-Meier survival analysis curves for all-cause mortality.
ED to ICUs time (hours): Q1 (≤2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).
A. Cumulative probability of all-cause mortality at 28-day before PSM; B. Cumulative probability of 28-day survival for Q4 vs. Q1 after PSM; C. Cumulative probability of 28-day survival for Q4 vs. Q2 after PSM; D. Cumulative probability of 28-day survival for Q4 vs. Q3 after PSM.

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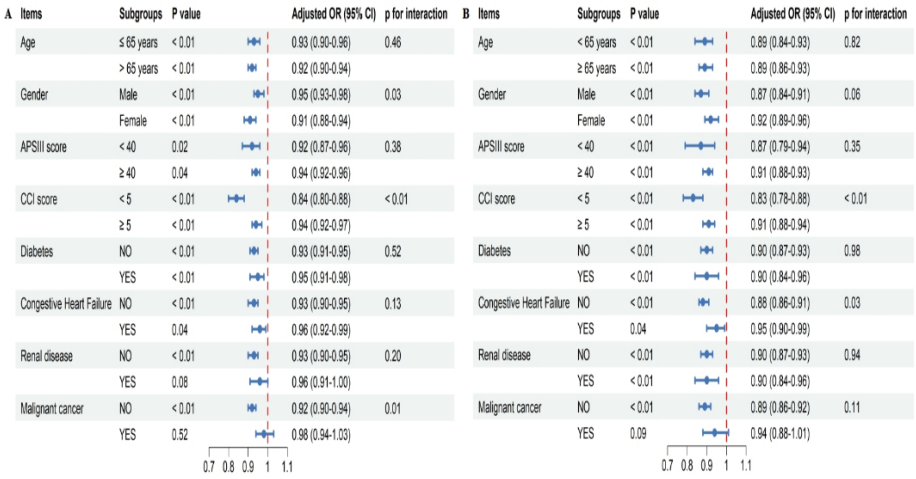


Fig. 4 Forest plots of odd ratios for the primary outcome in different subgroups.
A. Association between ED to ICUs time and in-hospital mortality;
B. Association between ED to ICUs time and ICU mortality.

90x50mm (300 x 300 DPI)

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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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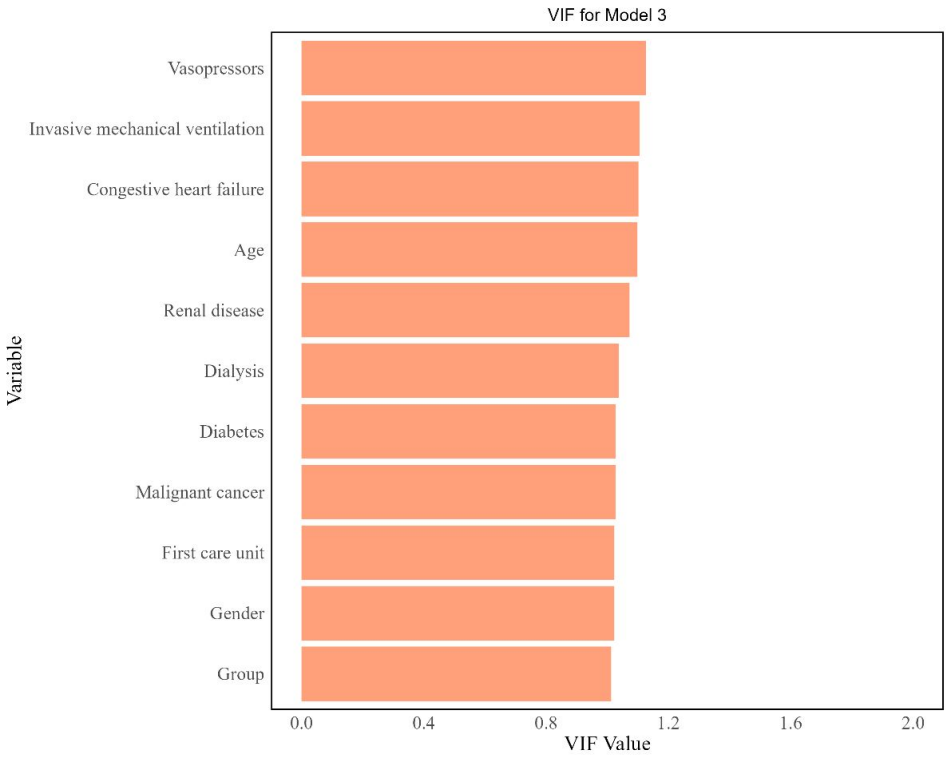
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Supplementary Figure



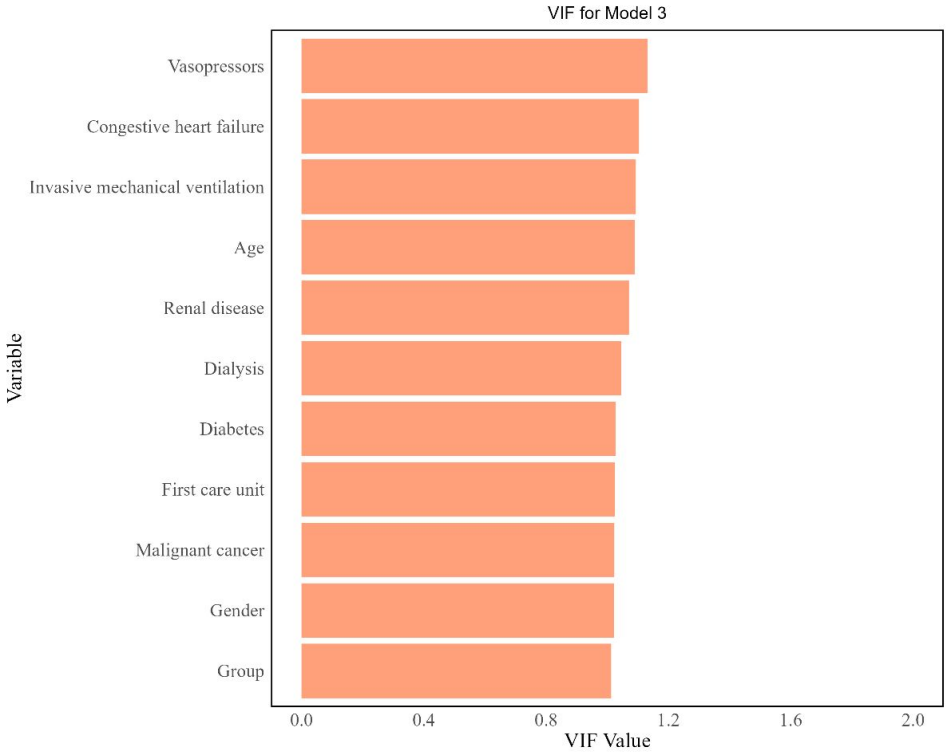
Supplementary Fig. 1 Variance Inflation Factor for Variables in Model 3 for In-Hospital Mortality

Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors.

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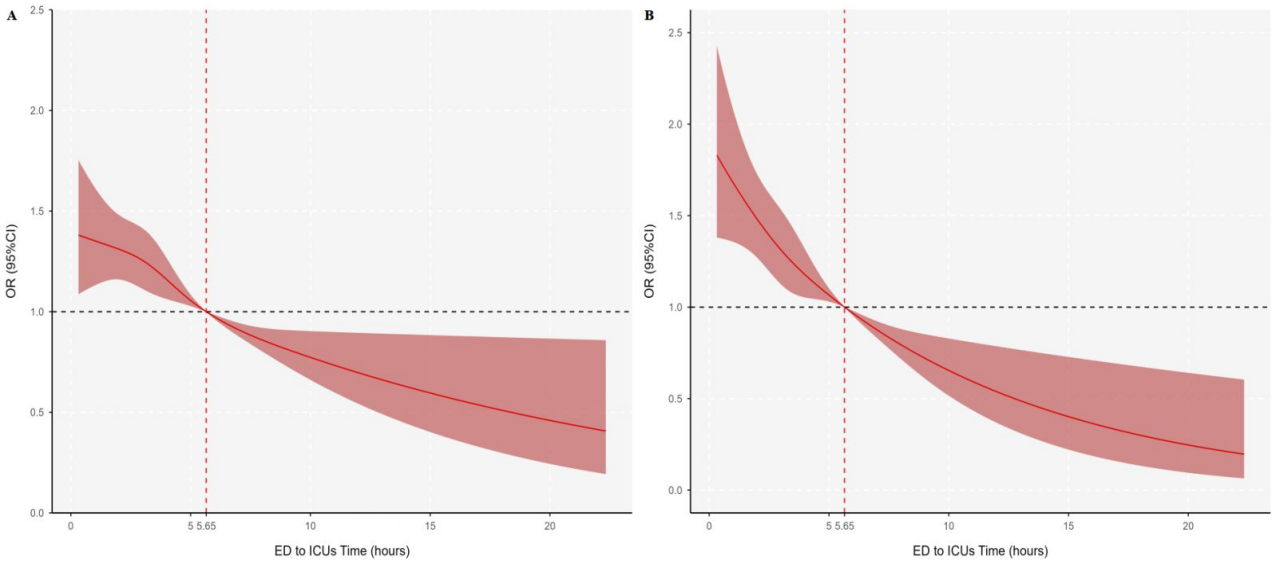
Supplementary Fig. 2 Variance Inflation Factor for Variables in Model 3 for ICU Mortality

Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis, acidosis and vasopressors.

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Supplementary Fig. 3 Restricted cubic spline regression analysis of ED to ICUs time with in hospital all-cause mortality

Red heavy central lines represent the estimated adjusted hazard ratios, with shaded ribbons denoting 95% confidence intervals. ED to ICUs time 5.65hr was selected as the reference level represented by the vertical dotted lines. The horizontal dotted lines represent the hazard ratio of 1.0.

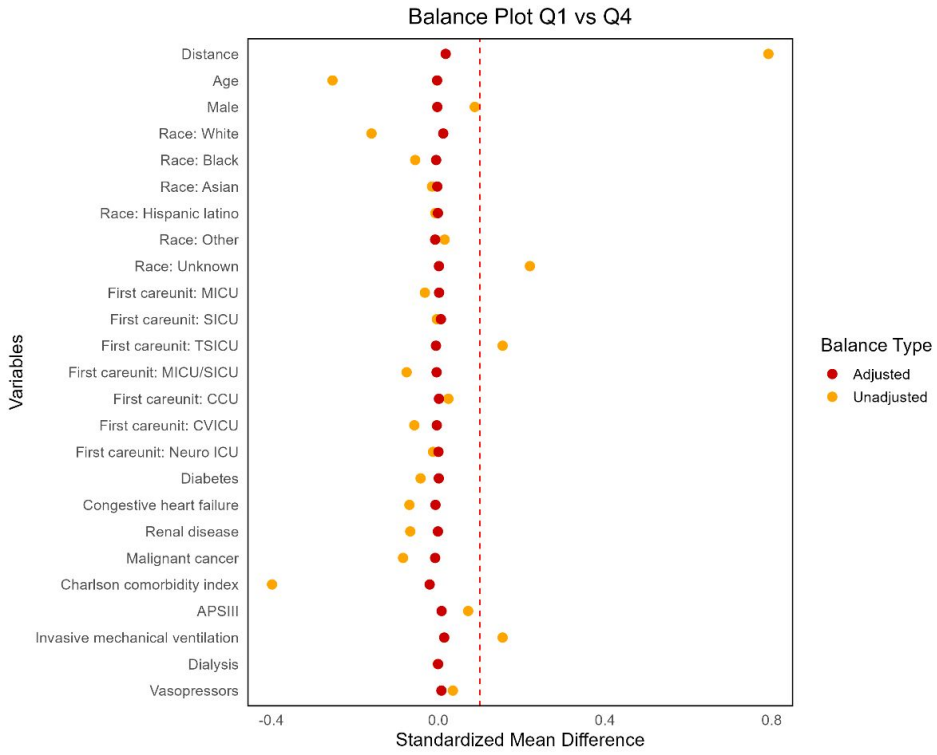
A. Restricted cubic spline for in-hospital death.

B. Restricted cubic spline for ICU death. HR hazard ratio, CI confidence interval

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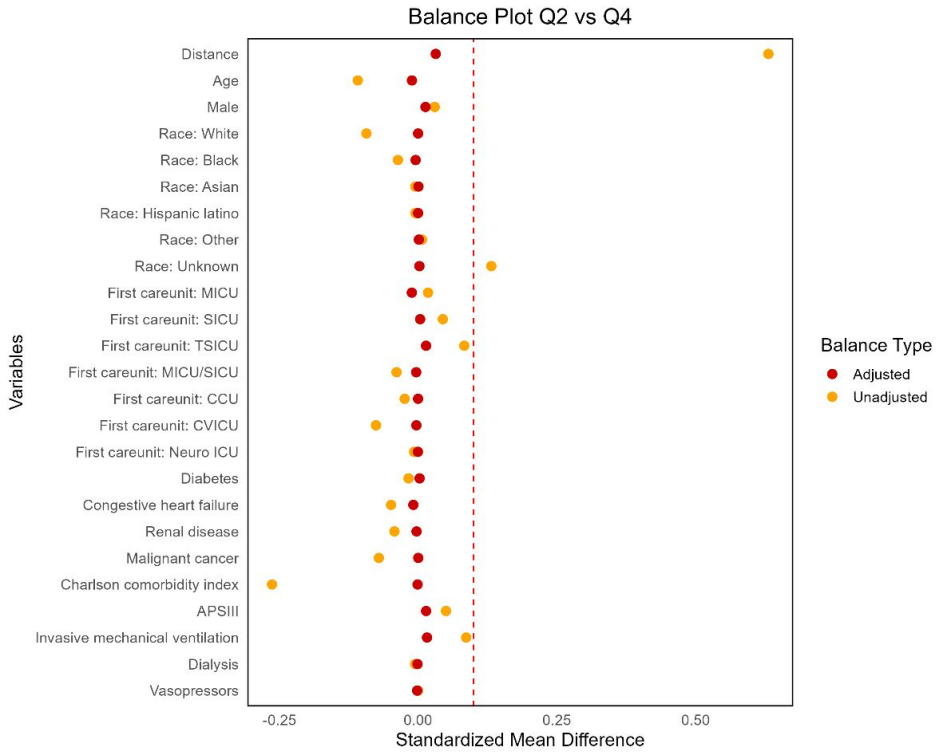


Supplementary Fig. 4 Standardized Mean Differences for Covariates Before and After Propensity Score Matching Between Q1 and Q4

Adjusted: Standardized mean difference after matching.
Unadjusted: Standardized mean difference before matching.
The dashed red line represents the threshold of acceptable imbalance (0.1).

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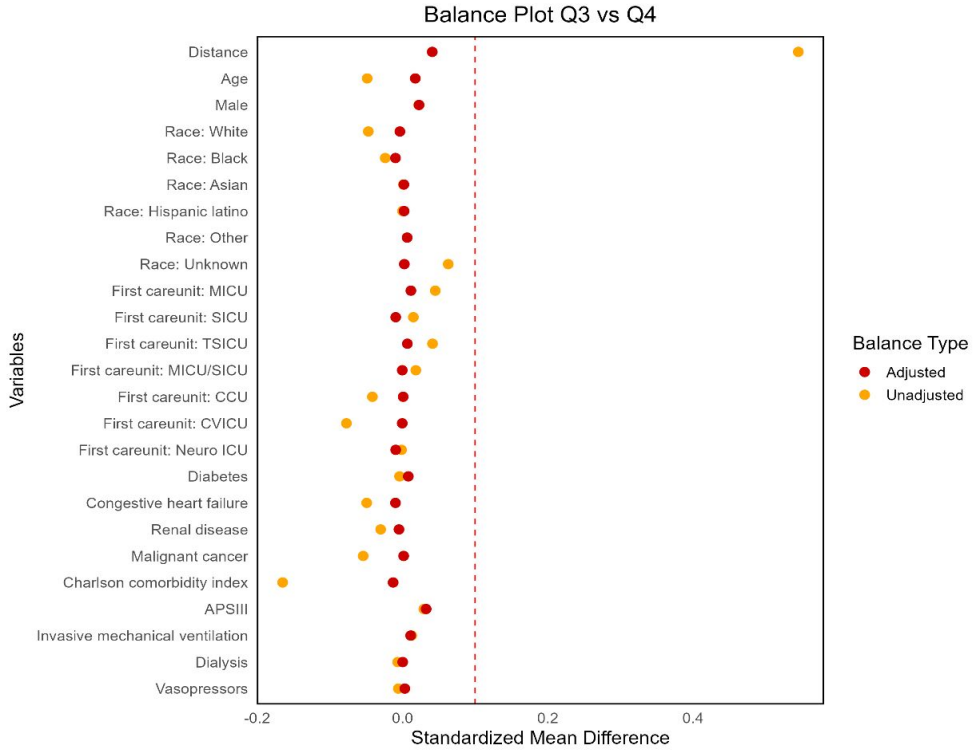


Supplementary Fig. 5 Standardized Mean Differences for Covariates Before and After Propensity Score Matching Between Q2 and Q4

Adjusted: Standardized mean difference after matching.
Unadjusted: Standardized mean difference before matching.
The dashed red line represents the threshold of acceptable imbalance (0.1).

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Supplementary Fig. 6 Standardized Mean Differences for Covariates Before and After Propensity Score Matching Between Q3 and Q4

Adjusted: Standardized mean difference after matching.
Unadjusted: Standardized mean difference before matching.
The dashed red line represents the threshold of acceptable imbalance (0.1).

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Supplementary Table

Supplementary Table 1 Baseline characteristics of ICU patients grouped according to ED to ICUs time quartiles

Categories	Overall (N=12703)	Q1 (N=3222)	Q2 (N=3147)	Q3 (N=3165)	Q4 (N=3169)	P-value
ED to ICUs time, hr, median (IQR)	3.98 (2.83, 5.65)	2.16 (1.66, 2.53)	3.41 (3.12, 3.68)	4.70 (4.3[illegible] 5.12)	7.22 (6.33, 8.72)	< 0.01
Demographic						
Age, years, median (IQR)	64 (49, 79)	61 (46, 77)	65 (49, 79)	66 (51, [illegible])	66 (51, 80)	< 0.01
Male, n (%)	7178 (56.51)	1993 (61.86)	1771 (56.28)	1784 (5[illegible]7)	1630 (51.43)	< 0.01
Ethnicity, n (%)						< 0.01
White	8149 (64.15)	1799 (55.83)	1986 (63.11)	2122 (6[illegible]5)	2242 (70.75)	
Black	1103 (8.68)	249 (7.91)	249 (7.91)	277 (8.[illegible])	365 (11.52)	
Asian	375 (2.95)	87 (2.76)	87 (2.76)	105 (3.[illegible])	117 (3.69)	
Hispanic/Latino	463 (3.64)	112 (3.48)	115 (3.65)	117 (3.[illegible])	119 (3.76)	
Other	636 (5.01)	195 (6.05)	146 (4.64)	149 (4.[illegible])	146 (4.61)	
Unknown	1977 (15.56)	838 (26.01)	564 (17.92)	395 (12.[illegible])	180 (5.68)	
Admission type, n (%)						< 0.01
EW EMER	11357 (89.40)	2912 (90.38)	2883 (91.61)	2834 (8[illegible]4)	2728 (86.08)	
OBSERVATION ADMIT	1194 (9.40)	269 (8.35)	236 (7.50)	300 (9.4[illegible])	389 (12.28)	
EU OBSERVATION	147 (1.16)	40 (1.24)	27 (0.86)	30 (0.9[illegible])	50 (1.58)	
DIRECT OBSERVATION	4 (0.03)	1 (0.03)	1 (0.03)	0 (0.00)	2 (0.06)	
URGENT	1 (0.01)	0 (0.00)	0 (0.00)	1 (0.03)	0 (0.00)	
First care unit, n (%)						< 0.01
MICU	3301 (25.99)	683 (21.20)	798 (25.36)	863 (27.27)	957 (30.20)	
SICU	2569 (20.22)	553 (17.16)	732 (23.26)	712 (22.5[illegible])	572 (18.05)	
TSICU	2709 (21.33)	941 (29.21)	707 (22.47)	571 (18.0[illegible])	490 (15.46)	
MICU/SICU	2123 (16.71)	365 (11.33)	447 (14.20)	602 (19.0[illegible])	709 (22.37)	

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CCU	1053 (8.29)	417 (12.94)	260 (8.26)	200 (6.3[illegible])	176 (5.55)	
CVICU	445 (3.50)	164 (5.09)	93 (2.96)	92 (2.9[illegible])	96 (3.03)	
Neuro SICU	503 (3.96)	99 (3.07)	110 (3.50)	125 (3.9[illegible])	169 (5.33)	
Chronic conditions, n (%)						
Diabetes	2938 (23.13%)	674 (20.92%)	723 (22.97%)	767 (24.[illegible]%)	774 (24.42%)	< 0.01
Congestive heart failure	2401 (18.90%)	567 (17.60%)	605 (19.22%)	596 (18.[illegible]%)	633 (19.97%)	0.10
Renal disease	1753 (13.80%)	375 (11.64%)	388 (12.33%)	485 (15.[illegible]%)	505 (15.94%)	< 0.01
Malignant cancer	1090 (8.58%)	208 (6.46%)	240 (7.63%)	297 (9.[illegible]%)	345 (10.89%)	< 0.01
Severity scores						
CCI, mean (SD)	4.80 (3.07)	4.35 (3.01)	4.67 (3.03)	5.02 (3.[illegible])	5.16 (3.12)	< 0.01
APSIII, mean (SD)	46.37 (24.00)	47.61 (25.71)	46.04 (23.82)	47.24 (2[illegible].95)	44.57 (22.26)	< 0.01
Intervention during ICU, n (%)						
Invasive mechanical ventilation	4196 (33.03%)	1366 (42.4%)	1122 (35.65%)	988 (31.[illegible]%)	720 (22.72%)	< 0.01
Dialysis	235 (1.85%)	73 (2.27%)	48 (1.53%)	60 (1.9[illegible])	54 (1.70%)	0.15
Vasopressors	2425 (19.09%)	734 (22.78%)	562 (17.86%)	600 (18.[illegible]%)	529 (16.69%)	< 0.01
Outcome						
Hospital death, n (%)	1988 (15.65)	567 (17.60)	520 (16.52)	514 (16.[illegible])	387 (12.21)	< 0.01
ICU death, n (%)	1185 (9.33)	375 (11.64)	297 (9.44)	311 (9.8[illegible])	202 (6.37)	< 0.01
28-days death, n (%)	2490 (19.60)	665 (20.64)	643 (20.43)	648 (20.[illegible])	534 (16.85)	< 0.01
Hosptial-LOS, days, median (IQR)	5.0 (2.8, 9.4)	5.1 (2.7, 9.7)	5.0 (2.8, 9.1)	5.0 (2.8[illegible].3)	5.3 (3.0, 9.3)	0.05
ICU-LOS, median (IQR)	1.80 (0.96, 3.59)	1.96 (1.11, 4.15)	1.80 (0.97, 3.64)	1.78 (0.[illegible] 3.53)	1.70 (0.91, 2.98)	< 0.01

ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr)

ED emergency department, ICU intensive care unit, IQR interquartile range, *EW EMER* emergency ward emergency, EU OBSERVATION: emergency unit observation, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit, CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, CCI Charlson Comorbidity Index, APSIII acute physiology score III, LOS length of stay

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Supplementary Table 2 Baseline Characteristics of Q4 and Q1 Groups After Propensity Score Matching

Categories	Matched Cohort (N=5056)	Q4 (N=2528)	Q1 (N=2528)	*p*-value
ED to ICUs time, hr, median (IQR)	7.54 (5.6, 18.91)	18.91 (9.6, 57.48)	5.6 (4.98, 6.4)	< 0.01
Demographic				
Age, years, median (IQR)	66 (53, 80)	66 (52, 80)	66 (53, 80)	0.38
Male, n (%)	2873 (56.82%)	1439 (56.92%)	1434 (56.72%)	0.91
Ethnicity, n (%)				0.83
White	3741 (73.99%)	1855 (73.38%)	1886 (74.6%)	
Black	472 (9.34%)	242 (9.57%)	230 (9.1%)	
Asian	147 (2.91%)	76 (3.01%)	71 (2.81%)	
Hispanic/Latino	199 (3.94%)	100 (3.96%)	99 (3.92%)	
Other	254 (5.02%)	136 (5.38%)	118 (4.67%)	
Unknown	243 (4.81%)	119 (4.71%)	124 (4.91%)	
First care unit, n (%)				0.99
MICU	1212 (23.97%)	603 (23.85%)	609 (24.09%)	
SICU	1037 (20.51%)	510 (20.17%)	527 (20.85%)	
TSICU	913 (18.06%)	463 (18.31%)	450 (17.8%)	
MICU/SICU	771 (15.25%)	390 (15.43%)	381 (15.07%)	
CCU	631 (12.48%)	313 (12.38%)	318 (12.58%)	
CVICU	322 (6.37%)	165 (6.53%)	157 (6.21%)	
Neuro SICU	170 (3.36%)	84 (3.32%)	86 (3.4%)	
Chronic conditions, n (%)				
Diabetes	1186 (23.46%)	591 (23.38%)	595 (23.54%)	0.92
Congestive heart failure	1078 (21.32%)	547 (21.64%)	531 (21%)	0.59
Renal disease	693 (13.71%)	347 (13.73%)	346 (13.69%)	1.00
Malignant cancer	448 (8.86%)	233 (9.22%)	215 (8.5%)	0.38
Severity scores				
CCI, mean (SD)	5 (3, 7)	5 (3, 7)	5 (3, 7)	0.67
APSIII, mean (SD)	41 (30, 56)	40 (30, 56)	41 (31, 56)	0.12
Intervention during ICU, n (%)				
Invasive mechanical ventilation	1533 (30.32%)	748 (29.59%)	785 (31.05%)	0.27
Dialysis	110 (2.18%)	55 (2.18%)	55 (2.18%)	1.00
Vasopressors	936 (18.51%)	458 (18.12%)	478 (18.91%)	0.52
Outcome				
Hospital death, n (%)	725 (14.34%)	340 (13.45%)	385 (15.23%)	0.08
ICU death, n (%)	407 (8.05%)	177 (7%)	230 (9.1%)	0.01
28-days death, n (%)	874 (17.29%)	408 (16.14%)	466 (18.43%)	0.03
ICU-LOS, median (IQR)	1.74 (0.96, 3.15)	1.74 (0.99, 3.15)	1.73 (0.92, 3.15)	0.32
Hospital-LOS, days, median (IQR)	6.1 (3.6, 10.8)	7.4 (4.38, 12.4)	5.1 (2.9, 9.2)	< 0.01

ED emergency department, ICU intensive care unit, IQR interquartile range, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit,

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CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, CCI Charlson Comorbidity Index, APSIII acute physiology score III, LOS length of stay

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Supplementary Table 3 Baseline Characteristics of Q4 and Q2 Groups After Propensity Score Matching

Categories	Matched Cohort (N=5844)	Q4 (N=2922)	Q2 (N=2922)	*p*-value
ED to ICUs time, hr, median (IQR)	6.02 (3.73, 19.35)	19.35 (9.72, 57.3)	3.73 (3.4, 4.1)	< 0.01
Demographic				
Age, years, median (IQR)	66 (52, 80)	66 (52, 80)	66 (51, 80)	0.98
Male, n (%)	3129 (53.54%)	1545 (52.87%)	1584 (54.21%)	0.32
Ethnicity, n (%)				0.99
White	4274 (73.13%)	2137 (73.13%)	2137 (73.13%)	
Black	605 (10.35%)	309 (10.57%)	296 (10.13%)	
Asian	206 (3.52%)	102 (3.49%)	104 (3.56%)	
Hispanic/Latino	234 (4%)	117 (4%)	117 (4%)	
Other	282 (4.83%)	139 (4.76%)	143 (4.89%)	
Unknown	243 (4.16%)	118 (4.04%)	125 (4.28%)	
First care unit, n (%)				0.82
MICU	1615 (27.64%)	824 (28.2%)	791 (27.07%)	
SICU	1301 (22.26%)	645 (22.07%)	656 (22.45%)	
TSICU	984 (16.84%)	471 (16.12%)	513 (17.56%)	
MICU/SICU	1028 (17.59%)	519 (17.76%)	509 (17.42%)	
CCU	484 (8.28%)	242 (8.28%)	242 (8.28%)	
CVICU	225 (3.85%)	117 (4%)	108 (3.7%)	
Neuro SICU	207 (3.54%)	104 (3.56%)	103 (3.52%)	
Chronic conditions, n (%)				
Diabetes	1424 (24.37%)	708 (24.23%)	716 (24.5%)	0.84
Congestive heart failure	1273 (21.78%)	649 (22.21%)	624 (21.36%)	0.46
Renal disease	922 (15.78%)	465 (15.91%)	457 (15.64%)	0.81
Malignant cancer	569 (9.74%)	284 (9.72%)	285 (9.75%)	1.00
Severity scores				
CCI, mean (SD)	5 (3, 7)	5 (3, 7)	5 (3, 7)	0.95
APSIII, mean (SD)	40 (30, 56)	40 (30, 56)	40 (30, 56)	0.85
Intervention during ICU, n (%)				
Invasive mechanical ventilation	1583 (27.09%)	768 (26.28%)	815 (27.89%)	0.17
Dialysis	107 (1.83%)	55 (1.88%)	52 (1.78%)	0.84
Vasopressors	1019 (17.44%)	512 (17.52%)	507 (17.35%)	0.89
Outcome				
Hospital death, n (%)	852 (14.58%)	397 (13.59%)	455 (15.57%)	0.04
ICU death, n (%)	454 (7.77%)	204 (6.98%)	250 (8.56%)	0.03
28-days death, n (%)	1065 (18.22%)	494 (16.91%)	571 (19.54%)	0.01
ICU-LOS, median (IQR)	1.73 (0.96, 3.25)	1.76 (1, 3.21)	1.72 (0.93, 3.32)	0.53
Hospital-LOS, days, median (IQR)	6.1 (3.5, 10.8)	7.5 (4.4, 12.5)	4.9 (2.8, 8.9)	< 0.01

ED emergency department, ICU intensive care unit, IQR interquartile range, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit,

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CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, CCI Charlson Comorbidity Index, APSIII acute physiology score III, LOS length of stay

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Supplementary Table 4 Baseline Characteristics of Q4 and Q3 Groups After Propensity Score Matching

Categories	Matched Cohort (N=6224)	Q4 (N=3112)	Q3 (N=3112)	*p*-value
ED to ICUs time, hr, median (IQR)	7.54 (5.6, 18.91)	18.91 (9.6, 57.48)	5.6 (4.98, 6.4)	< 0.01
Demographic				
Age, years, median (IQR)	66 (53, 80)	66 (52, 80)	66 (53, 80)	0.38
Male, n (%)	3271 (52.55%)	1600 (51.41%)	1671 (53.7%)	0.08
Ethnicity, n (%)				0.64
White	4521 (72.64%)	2266 (72.81%)	2255 (72.46%)	
Black	692 (11.12%)	361 (11.6%)	331 (10.64%)	
Asian	230 (3.7%)	112 (3.6%)	118 (3.79%)	
Hispanic/Latino	267 (4.29%)	130 (4.18%)	137 (4.4%)	
Other	284 (4.56%)	132 (4.24%)	152 (4.88%)	
Unknown	230 (3.7%)	111 (3.57%)	119 (3.82%)	
First care unit, n (%)				0.54
MICU	1778 (28.57%)	871 (27.99%)	907 (29.15%)	
SICU	1263 (20.29%)	646 (20.76%)	617 (19.83%)	
TSICU	965 (15.50%)	472 (15.17%)	493 (15.84%)	
MICU/SICU	1303 (20.94%)	652 (20.95%)	651 (20.92%)	
CCU	427 (6.86%)	212 (6.81%)	215 (6.91%)	
CVICU	213 (3.42%)	107 (3.44%)	106 (3.41%)	
Neuro SICU	275 (4.42%)	152 (4.88%)	123 (3.95%)	
Chronic conditions, n (%)				
Diabetes	1543 (24.79%)	759 (24.39%)	784 (25.19%)	0.48
Congestive heart failure	1306 (20.98%)	668 (21.47%)	638 (20.50%)	0.37
Renal disease	1013 (16.28%)	514 (16.52%)	499 (16.03%)	0.63
Malignant cancer	707 (11.36%)	351 (11.28%)	356 (11.44%)	0.87
Severity scores				
CCI, mean (SD)	5 (3, 7)	5 (3, 7)	5 (3, 7)	0.67
APSIII, mean (SD)	41 (30, 56)	40 (30, 56)	41 (31, 56)	0.12
Intervention during ICU, n (%)				
Invasive mechanical ventilation	1545 (24.82%)	755 (24.26%)	790 (25.39%)	0.33
Dialysis	113 (1.82%)	56 (1.80%)	57 (1.83%)	1.00
Vasopressors	1084 (17.42%)	537 (17.26%)	547 (17.58%)	0.78
Outcome				
Hospital death, n (%)	860 (13.82%)	426 (13.69%)	434 (13.95%)	0.81
ICU death, n (%)	463 (7.44%)	211 (6.78%)	252 (8.10%)	0.06
28-days death, n (%)	1123 (18.04%)	531 (17.06%)	592 (19.02%)	0.05
ICU-LOS, median (IQR)	1.74 (0.96, 3.15)	1.74 (0.99, 3.15)	1.73 (0.92, 3.15)	0.32
Hospital-LOS, days, median (IQR)	6.1 (3.6, 10.8)	7.4 (4.38, 12.4)	5.1 (2.9, 9.2)	< 0.01

ED emergency department, ICU intensive care unit, IQR interquartile range, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit,

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CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, CCI Charlson Comorbidity Index, APSIII acute physiology score III, LOS length of stay

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Supplementary Table 5 Relationship between ED to ICUs time and length of stay

	Coef.	S.E	t-value	*p*-value
Hospital-LOS[a]				
Model 1	-0.099	0.032	-3.153	0.002
Model 2	-0.032	0.030	-1.070	0.285
Model 3	0.031	0.029	1.069	0.285
ICU-LOS[b]				
Model 1	-0.157	0.017	-9.439	< 0.001
Model 2	-0.097	0.016	-6.287	< 0.001
Model 3	-0.069	0.015	-4.503	< 0.001

Model 1: unadjusted; Model 2: adjusted for APSIII score, Charlson Comorbidity Index; Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, invasive mechanical ventilation, dialysis and vasopressors.

[a] The relationship between ED to ICUs time and length of hospital stay was analyzed in a cohort that survived the hospital stay (n=10701)

[b] The relationship between ED to ICUs time index and length of ICU stay was analyzed in a cohort that survived the ICU stay (n=11518)

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Supplementary Sensitivity Analysis

Supplementary Table 6 Baseline characteristics of ICU patients grouped according to ED to ICUs time quartiles in Sensitivity Analysis

Categories	Overall (N=15135)	Q1 (N=3797)	Q2 (N=3775)	Q3 (N=3780)	Q4 (N=3783)	*p*-value
ED to ICUs time, hr, median (IQR)	4.48 (3.03, 7.54)	2.3 (1.77, 2.7)	3.72 (3.38, 4.08)	5.58 (4.8, 6.4)	22.46 (9.95, 72.78)	< 0.01
Demographic						
Age, years, median (IQR)	65 (50, 79)	62 (46, 77)	65 (50, 80)	66 (52, 80)	67 (53, 80)	< 0.01
Male, n (%)	8489 (56.09%)	2330 (61.36%)	2098 (55.58%)	2072 (54.81%)	1989 (52.58%)	< 0.01
Ethnicity, n (%)						< 0.01
White	9963 (65.83%)	2178 (57.36%)	2417 (64.03%)	2594 (68.62%)	2774 (73.33%)	
Black	1369 (9.05%)	244 (6.43%)	313 (8.29%)	361 (9.55%)	451 (11.92%)	
Asian	455 (3.01%)	76 (2.00%)	112 (2.97%)	136 (3.60%)	131 (3.46%)	
Hispanic/Latino	566 (3.74%)	128 (3.37%)	134 (3.55%)	152 (4.02%)	152 (4.02%)	
Other	734 (4.85%)	216 (5.69%)	182 (4.82%)	180 (4.76%)	156 (4.12%)	
Unknown	2048 (13.53%)	955 (25.15%)	617 (16.34%)	357 (9.44%)	119 (3.15%)	
Admission type, n (%)						< 0.01
EW EMER	13343 (88.16%)	3439 (90.57%)	3445 (91.26%)	3355 (88.76%)	3104 (82.05%)	
OBSERVATION ADMIT	1630 (10.77%)	311 (8.19%)	291 (7.71%)	387 (10.24%)	641 (16.94%)	
EU OBSERVATION	157 (1.04%)	45 (1.19%)	38 (1.01%)	38 (1.01%)	36 (0.95%)	
DIRECT OBSERVATION	4 (0.03%)	2 (0.05%)	0 (0%)	0 (0%)	2 (0.05%)	
URGENT	1 (0.01%)	0 (0%)	1 (0.03%)	0 (0%)	0 (0%)	
First care unit, n (%)						< 0.01
MICU	3815 (25.21%)	807 (21.25%)	990 (26.23%)	1094 (28.94%)	924 (24.43%)	
SICU	3038 (20.07%)	697 (18.36%)	873 (23.13%)	762 (20.16%)	706 (18.66%)	
TSICU	3014 (19.91%)	1078 (28.39%)	802 (21.25%)	645 (17.06%)	489 (12.93%)	
MICU/SICU	2541 (16.79%)	442 (11.64%)	578 (15.31%)	795 (21.03%)	726 (19.19%)	

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CCU	1354 (8.95%)	472 (12.43%)	285 (7.55%)	220 (5.83%)	377 (9.97%)	
CVICU	798 (5.27%)	183 (4.82%)	111 (2.94%)	106 (2.81%)	398 (10.52%)	
Neuro SICU	575 (3.80%)	118 (3.11%)	136 (3.60%)	158 (4.19%)	163 (4.31%)	
Chronic conditions, n (%)						
Diabetes	3577 (23.63%)	798 (21.02%)	887 (23.50%)	938 (24.88%)	954 (25.22%)	< 0.01
Congestive heart failure	3053 (20.17%)	663 (17.46%)	735 (19.47%)	734 (19.47%)	921 (24.35%)	< 0.01
Renal disease	2217 (14.65%)	435 (11.46%)	524 (13.88%)	572 (15.17%)	686 (18.13%)	< 0.01
Malignant cancer	1494 (9.87%)	254 (6.69%)	303 (8.03%)	366 (9.71%)	571 (15.09%)	< 0.01
Severity scores						
CCI, mean (SD)	5 (3, 7)	4 (2, 6)	5 (3, 7)	5 (3, 7)	5 (3, 8)	< 0.01
APSIII, mean (SD)	40 (30, 57)	40 (30, 60)	41 (30, 57)	41 (30, 57)	40 (30, 56)	0.26
Intervention during ICU, n (%)						
Invasive mechanical ventilation	4927 (32.55%)	1582 (41.66%)	1316 (34.86%)	1038 (27.56%)	991 (26.20%)	< 0.01
Dialysis	300 (1.98%)	84 (2.21%)	67 (1.77%)	62 (1.64%)	87 (2.30%)	0.11
Vasopressors	2905 (19.19%)	835 (21.99%)	696 (18.44%)	676 (17.93%)	698 (18.45%)	< 0.01
Outcome						
Hospital death, n (%)	2367 (15.64%)	661 (17.41%)	641 (16.98%)	535 (14.20%)	530 (14.01%)	< 0.01
ICU death, n (%)	1374 (9.08%)	427 (11.25%)	379 (10.04%)	306 (8.12%)	262 (6.93%)	< 0.01
28-days death, n (%)	2947 (19.47%)	777 (20.46%)	784 (20.77%)	719 (19.08%)	667 (17.63%)	< 0.01
Hosptial-LOS, days, median (IQR)	5.7 (3.1, 10.4)	5 (2.8, 9.7)	4.9 (2.8, 9.1)	5.1 (2.9, 9.4)	7.7 (4.6, 12.8)	< 0.01
ICU-LOS, median (IQR)	1.81 (0.98, 3.56)	1.94 (1.1, 4.12)	1.76 (0.95, 3.64)	1.76 (0.93, 3.28)	1.78 (1.03, 3.19)	< 0.01

ED to ICUs time (hours): Q1 (≤ 3.03 hr), Q2 (3.03-4.48 hr), Q3 (4.48-7.54 hr), Q4 (> 7.54 hr)

ED emergency department, ICU intensive care unit, IQR interquartile range, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit, CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, CCI Charlson Comorbidity Index, APSIII acute physiology score III, LOS length of stay

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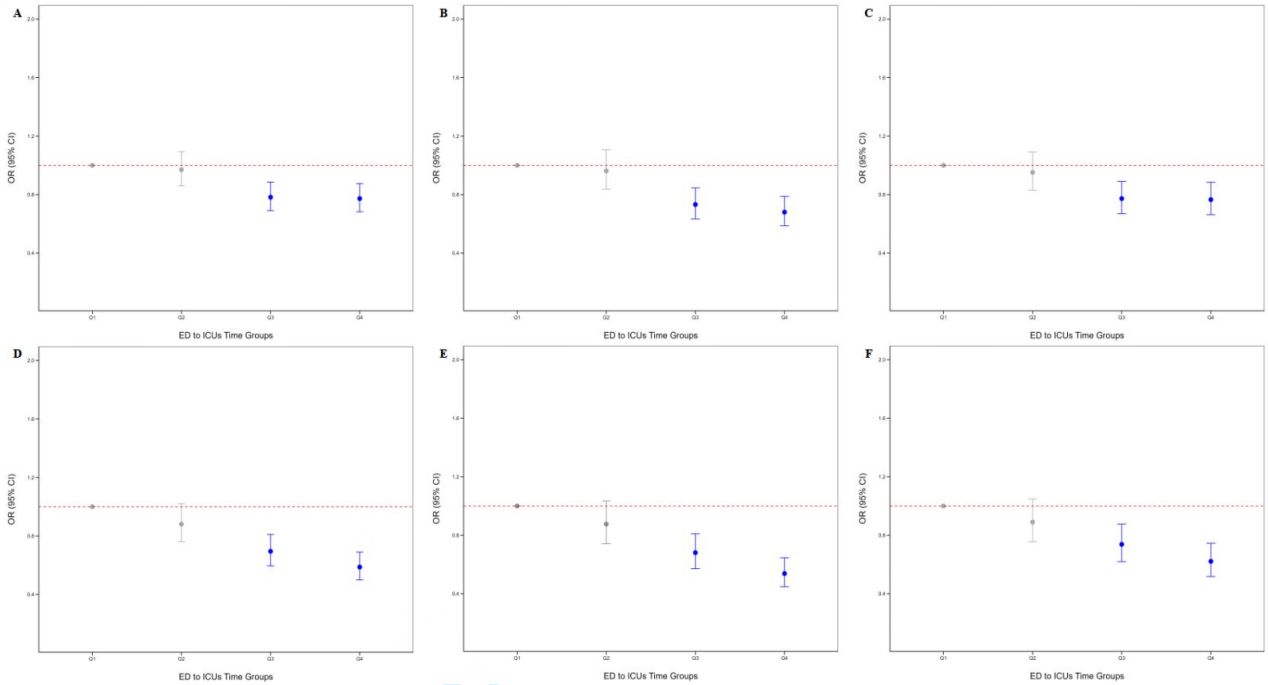
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Supplementary Fig.7 Odds Ratios for All-Cause In-Hospital and ICU Mortality Across Different ED-to-ICUs Time Groups in Sensitivity Analysis

ED to ICUs time (hours): Q1 (≤ 3.03 hr), Q2 (3.03-4.48 hr), Q3 (4.48-7.54 hr), Q4 (> 7.54 hr)

A. Association between ED-to-ICUs time and in-hospital mortality, unadjusted.

B. Association between ED-to-ICUs time and in-hospital mortality, adjusted for APSIII score and CCI score.

C. Association between ED-to-ICUs time and in-hospital mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors.

D. Association between ED-to-ICUs time and ICU mortality, unadjusted.

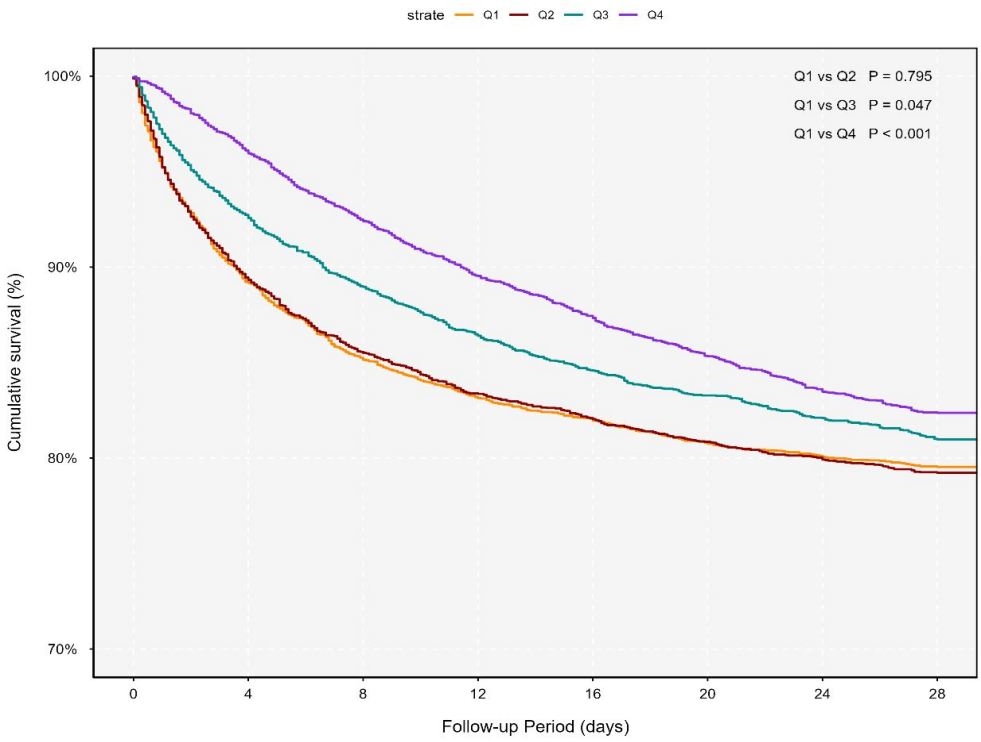
E. Association between ED-to-ICUs time and ICU mortality, adjusted for OASIS score and CCI score.

F. Association between ED-to-ICUs time and ICU mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors.

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Supplementary Fig.8 Kaplan-Meier Survival Curves for All-Cause Mortality in Sensitivity Analysis

ED to ICUs time (hours): Q1 (≤ 3.03 hr), Q2 (3.03-4.48 hr), Q3 (4.48-7.54 hr), Q4 (> 7.54 hr)

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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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Objectives: The association between the duration from the emergency department (ED) to the intensive care units (ICUs) and in-hospital mortality among patients admitted directly to the ICUs from the ED remains controversial. This study aimed to utilize data from the Medical Information Mart for Intensive Care IV database to explore the relationship between the ED to ICUs time and patient outcomes.

Design: Retrospective observational study.

Setting: Admissions to the Beth Israel Deaconess Medical Center intensive care from 2008 to 2019.

Participants: Adult patients in the MIMIC-IV database admitted directly from the ED to the ICUs, excluding those with missing data or a hospital-to-ICU admission interval over 6 hours.

Primary and secondary outcome measures: The primary outcome was in-hospital all-cause mortality. Secondary outcomes included 28-day all-cause mortality and length of stay in ICU and hospital.

Results: Out of 96,868 initially identified patients, 12,703 were included after exclusions. The median ED to ICUs time was 3.98 hours. Longer ED to ICUs times were associated with lower in-hospital mortality, decreasing from 17.6% in the shortest to 12.2% in the longest interval group, and shorter ICU stays. After propensity score weighting, adjusted logistic regression models confirmed the inverse association between longer ED to ICUs time and in-hospital mortality (OR: 0.75, 95% CI: 0.69-0.82, $p < 0.01$). Restricted cubic spline analysis showed a nonlinear decline in mortality risk with increasing ED to ICUs time, with a sharper reduction after 5.65 hours. Kaplan-Meier curves indicated consistently better survival in the longest interval group ($p < 0.01$). Sensitivity analysis, reintroducing patients with hospital to ICU times over 6 hours, confirmed the robustness of these results.

Conclusions: Longer ED to ICUs time was linked to lower mortality and shorter ICU length of stay, suggesting that appropriately extending ED stays may benefit critically ill patients.

Keywords: Emergency Department; Intensive Care Units; Mortality; Generalized Boosted Model; Patient Care

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STRENGTHS AND LIMITATIONS OF THIS STYDY

- The study utilizes a comprehensive and high-quality dataset from the MIMIC-IV database, offering detailed clinical and demographic data on ICU admissions.
- The use of robust statistical methods, including propensity score weighting based on generalized boosted models, sensitivity analysis, and subgroup analysis, ensures comprehensive covariate balance and enhances the validity and reliability of the results.
- Restricted cubic spline analysis was used to explore non-linear associations, offering a more nuanced understanding of ED to ICUs time effects.
- Lack of data on specific ED interventions may introduce information bias, as we could not assess their potential impact on the outcomes.
- Findings are based on data from a single healthcare system in the United States, which may limit the generalizability of results to other international settings.

Introduction

The emergency department (ED) often serves as the primary entry point for critically ill patients requiring immediate care. In recent years, there has been a marked increase in the capacity of ED clinicians to manage these patients over extended periods [1]. Following initial stabilization in the ED, patients are typically transported to appropriate departments for further diagnostic tests or therapeutic procedures. Most critically ill patients will be transported to specialized intensive care units (ICUs), which provide the highest level of monitoring and care [2]. During the last two decades, escalating acuity and volume of ED visits have resulted in greater demand for critical care services in the ED and ICUs [3]. ED crowding has become an international issue and has long been recognized as posing significant patient safety threats [4]. The impact of ED crowding and ED length of stay (ED-LOS) on inpatient mortality remains contentious and appears to vary by region [5, 6].

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ED-LOS is a critical ED performance indicator and is closely related to crowding, quality of care and patient outcomes [7]. However, the relationship between ED-LOS and in-hospital mortality, particularly for patients admitted directly to ICUs from the ED, remains a subject of debate. Groenland et al. [8] reported that a longer ED to ICUs time is associated with increased hospital mortality in the most severely ill patients. Conversely, Wessman et al. [9] found that patients with the highest triage priority had a decreasing mortality risk with increasing quintiles of ED-LOS for 7-day mortality, and the two highest quintile of ED-LOS had lower 30-day mortality. Saukkonen et al. [10] observed that the length of ED stay was not associated with the outcome of critically ill medical patients.

In light of these conflicting findings and the need for a large cohort to discern subtle outcome differences, our study leverages the extensive Medical Information Mart for Intensive Care IV (MIMIC-IV) database to examine the association between ED to ICUs time and mortality among critically ill patients. This study offers a unique perspective on the potential benefits of extended ED time, addressing gaps in the current understanding of optimal ED management duration before ICU transfer.

Methods

Study design

This retrospective observational study used data from the MIMIC-IV (version 2.2) database, which contains high-quality information on ICU admissions at Beth Israel Deaconess Medical Center from 2008 to 2019. Access to MIMIC-IV is limited to certified individuals; one author (JWQ) with access permission (Record ID 60311526) extracted the data. As all patient records in MIMIC-IV were fully de-identified, the institutional review board at Beth Israel Deaconess Medical Center waived the need for individual patient consent.

Cohort selection

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We initially identified all patients aged 18 years and older who were admitted through the emergency room. Patients were excluded if they did not have a recorded ICU stay or if their admission time to the ED was missing. Additionally, patients with a gap of exceeding 6 hours between hospital admission time (*admittime*) and ICU admission time (*intime*) recorded in the database were excluded as they were not directly transferred from the ED to the ICUs and were thus excluded. For patients with multiple ICU admissions, only the first ICU stay was analyzed. After applying these criteria, the final study cohort was divided into four quartiles based on the ED to ICUs time.

Data extraction and definitions

All baseline characteristics were extracted within the first 24h of ICU admission from the MIMIC-IV database, including age, gender, ethnicity, admission type, first care unit, chronic conditions, severity scores (Charlson Comorbidity Index (CCI) score, Acute physiology score III (APSIII)), and intervention during ICU. The ED to ICUs time was defined as the interval between *edregtime* (the time of registration in the ED) and *intime* (the time of transfer into the ICU) recorded in the MIMIC-IV database. *Admittime* provides the date and time the patient was admitted to the hospital. Data extraction was performed using pgAdmin4 PostgreSQL 16.1.

Primary outcomes and secondary outcomes

The primary outcome of this study was in hospital all-cause mortality. Of these, hospital death included ICU death and general ward inpatient death. The secondary outcomes included 28-day all-cause mortality, LOS in ICU and LOS in hospital. Patient mortality information for discharged patients was accessed from the US Social Security Death Index.

Statistical analysis

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Our study divided patients into four quartiles based on their ED to ICUs admission times. We assessed the distribution of continuous data using the Kolmogorov-Smirnov test and presented them as means with standard deviations or medians with interquartile ranges. Categorical data were shown in counts and percentages. The Mann-Whitney U test and the Pearson chi-square or Fisher's exact tests were used for comparisons.

We used logistic regression models to estimate odds ratios (ORs) and 95% confidence intervals (CIs) for hospital and ICU mortality. Analyses were conducted using three models: an unadjusted model, a model adjusted for illness severity alone, and a fully adjusted model addressing potential multicollinearity. After fitting the logistic models, pairwise comparisons of the four quartiles were performed, and we applied Tukey's method to adjust for multiple comparisons. The fully adjusted model included a comprehensive set of covariates to assess the relationship between ED to ICUs time and in-hospital mortality, while controlling for key confounders. Age and sex were included to account for demographic differences, while the APSIII and CCI scores provided a detailed assessment of acute illness severity and chronic comorbidity burden, respectively. Additional covariates included diabetes, congestive heart failure, renal disease, and malignancy, given their known impact on critically ill patients' outcomes. Adjustments were also made for invasive mechanical ventilation, dialysis, and vasopressor use, as these reflect the severity of respiratory failure, renal impairment, and hemodynamic instability. This selection of covariates aimed to minimize residual confounding and precise effect estimation of ED to ICUs time on in-hospital mortality. Secondary outcomes, including 28-day mortality, were analyzed using similar models and Kaplan-Meier survival analysis, with additional tests for non-linear relationships and stratified analyses to ensure consistency across patient subgroups. Restricted cubic spline (RCS) models were applied to further explore the non-linear association between ED to ICUs time and both in-hospital and ICU mortality.

Generalized Boosted Models (GBM) were utilized to estimate generalized propensity scores (GPS) to

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achieve covariate balance among the four quartiles of ED to ICU admission times (Q1, Q2, Q3, Q4). The GBM model was implemented using the *twang* R package with the following hyperparameters: 10,000 boosting iterations, an interaction depth of 3, and a shrinkage parameter of 0.01. The GBM model estimated the average treatment effect (ATE) for all four quartiles, ensuring comparability between groups. Covariates included age, gender, race, first care unit, CCI scores, APSIII scores, diabetes, congestive heart failure, renal disease, malignancy, invasive mechanical ventilation, dialysis, and vasopressor use. The balancing process was guided by minimizing the maximum standardized effect size across all covariates. Balance diagnostics were conducted using standardized mean differences (SMD) and Kolmogorov-Smirnov statistics to evaluate the effectiveness of the model in achieving covariate balance before and after adjustment. Sample weights from the GBM model were used to estimate effective sample sizes (ESS) for quartiles.

A sensitivity analysis was also performed by reintroducing patients with a time difference greater than six hours between hospital admission and ICU admission. This analysis aimed to assess the robustness of the study findings by evaluating whether the association between ED to ICUs time and in-hospital mortality persisted when including patients potentially admitted to non-ICU wards before ICU transfer.

We performed statistical analyses with R software (version 4.3.3) and considered two-sided P values below 0.05 as significant.

Result

Baseline characteristics

From 96,868 patients initially identified, we excluded those without ICU stays, missing ED admission times, or whose hospital admission time and ICU admission time differed by more than 6 hours, leaving a final cohort of 12,703 patients. Details are shown in **Fig. 1**. **Supplementary Table 1** summarizes characteristics of 12,703 study participants categorized by ED to ICUs time quartiles. The median patient age was 64 years, and 56.51%

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were male. There were 1,185 ICU deaths, 1,988 in-hospital deaths, and 2,490 deaths within 28 days. The median ED to ICUs time was 3.98 hours. Among the chronic conditions in the cohort, diabetes was present in 2,938 patients (23.13%), congestive heart failure in 2,401 (18.90%), renal disease in 1,753 (13.80%), and malignant cancer in 1,090 (8.58%). The median age increased from 61 years in Q1 to 66 years in Q3 and Q4, with an overall median of 64 years ($p < 0.01$). Male patients comprised 56.51% of the study population, with their proportion decreasing from 61.86% in Q1 to 51.43% in Q4 ($p < 0.01$). The mean CCI score showed a significant increase across quartiles, ranging from 4.35 in Q1 to 5.16 in Q4 ($p < 0.01$). The mean APSIII score demonstrated a slight variation, decreasing from 46.37 in Q1 to 44.57 in Q4, though this difference remained statistically significant ($p < 0.01$).

In terms of outcomes, hospital mortality occurred in 1,988 patients (15.65%), with rates ranging from 17.60% in Q1 to 12.21% in Q4 ($p < 0.01$). ICU mortality was observed in 1,185 patients (9.33%), with the highest rate in Q1 (11.64%) and the lowest in Q4 (6.37%) ($p < 0.01$). The 28-day mortality rate was 19.60% overall, varying from 20.64% in Q1 to 16.85% in Q4 ($p < 0.01$). The median hospital LOS was 5.0 days (IQR: 2.8-9.4) across all patients, with a slight difference among quartiles, from 5.0 days in Q1 to 5.3 days in Q4 ($p = 0.05$). The median ICU LOS was 1.80 days, ranging from 1.96 days in Q1 to 1.70 days in Q4 ($p < 0.01$), indicating significant differences between groups.

Association of the ED to ICUs time with all-cause mortality

Following the application of a GBM and propensity score weighting, we achieved significant enhancement in baseline comparability across the ED to ICUs time quartiles. As demonstrated in **Supplementary Fig. 1** and **Supplementary Table 2**, the maximum standardized mean difference was reduced from 0.56 before weighting to 0.04 after weighting, and the maximum Kolmogorov-Smirnov statistic improved from 0.20 to 0.02, with the minimum Kolmogorov-Smirnov p-value increasing from < 0.01 to 0.22. These improvements

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in covariate balance were achieved through iterative optimization, as illustrated in **Supplementary Fig. 2**. Effective sample sizes remained robust post-weighting: for instance, Q1 retained an effective sample size of 2651.1, and Q4 achieved 2475.5, ensuring sufficient analytical power for subsequent modeling (**Supplementary Table 3**).

Under these more balanced conditions, three weighted logistic regression models were constructed to assess the association between ED to ICUs time and all-cause mortality (**Table 1**). In Model 1, which incorporated weighting without additional adjustments, patients in the longest ED to ICUs time quartile (Q4) already exhibited significantly lower odds of in-hospital mortality compared to those in Q1 (OR 0.80, 95% CI: 0.74-0.86, $p < 0.01$) (**Fig. 2**). Further adjustment for illness severity (Model 2) and comprehensive adjustment for severity scores, comorbidities, demographics, first care unit type, and ICU-level interventions (Model 3) reinforced this inverse relationship. In the fully adjusted Model 3, the odds of in-hospital mortality in Q4 relative to Q1 were 0.75 (95% CI: 0.69-0.82, $p < 0.01$), while ICU mortality odds were 0.63 (95% CI: 0.61-0.74, $p < 0.01$). Variance inflation factor analyses demonstrated no significant collinearity issues within Model 3, supporting the stability and interpretability of the final model (**Supplementary Fig. 3**).

Building on these findings, restricted cubic spline analyses revealed a gradual, nonlinear decline in both in-hospital and ICU mortality risk as ED to ICUs time increased, with a more pronounced reduction observed after 5.65 hours (**Supplementary Fig. 4**).

The 28-day survival was further evaluated as a secondary outcome using Kaplan-Meier estimates (**Fig. 3**). Throughout the follow-up period, Q4 consistently exhibited a higher cumulative survival rate compared to Q1, as well as Q2 and Q3. While the survival differences between Q1 and the intermediate quartiles were not statistically significant ($P = 0.60$ and $P = 0.95$, respectively), Q4 demonstrated a significantly improved 28-day cumulative survival compared to Q1 ($P = 0.03$).

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Table 1 Odds Ratios (ORs) for All-Cause Mortality in-hospital and ICU mortality from Logistic Regression after Generalized Boosted Models

Categories	Model 1		Model 2		Model 3	
	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value
Hospital death						
Quartilea[a]						
Q1	Ref.		Ref.		Ref.	
Q2	1.01 (0.93, 1.08)	0.86	1.01 (0.93, 1.09)	0.78	1.01 (0.94, 1.09)	0.75
Q3	0.98 (0.92, 1.05)	0.66	0.96 (0.89, 1.04)	0.33	0.95 (0.88, 1.03)	0.23
Q4	0.80 (0.74, 0.86)	< 0.01	0.76 (0.70, 0.82)	< 0.01	0.75 (0.69, 0.82)	< 0.01
ICU death						
Quartilea[a]						
Q1	Ref.		Ref.		Ref.	
Q2	0.89 (0.81, 0.97)	< 0.01	0.87 (0.79, 0.95)	< 0.01	0.82 (0.79, 0.96)	< 0.01
Q3	0.95 (0.87, 1.03)	0.02	0.92 (0.83, 1.01)	0.08	0.86 (0.83, 0.99)	0.04
Q4	0.71 (0.65, 0.78)	< 0.01	0.67 (0.60, 0.74)	< 0.01	0.63 (0.61, 0.74)	< 0.01

Model 1: unadjusted; Model 2: adjusted for APSIII score, CCI score; Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, invasive mechanical ventilation, analysis and vasopressors.

[a] ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

Relationship between ED to ICUs time and length of stay

Supplementary Table 4 shows the relationship between ED to ICUs time and both hospital and ICU LOS. For hospital LOS, the unadjusted model indicated a significant negative association with ED to ICUs time (Coef. = -0.099, P = 0.002); however, this association lost significance after adjustment in Models 2 and 3 ($p > 0.05$). In contrast, for ICU LOS, a longer ED to ICUs time was consistently associated with a shorter duration of stay across all models, with coefficients of -0.157 ($p < 0.01$) in Model 1, -0.097 ($p < 0.01$) in Model 2, and -0.069 ($p < 0.01$) in Model 3. These results indicate a sustained inverse relationship between ED to ICUs time and ICU LOS, even after adjustments.

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Subgroup Analysis

The protective association between longer ED to ICUs time and reduced in-hospital mortality risk was broadly consistent across most examined subgroups in the subgroup analyses hospital mortality, with ORs remaining below 1 (**Fig. 4A**). However, the magnitude of this protective effect differed by specific patient characteristics. Significant interactions were observed with gender ($p = 0.03$), CCI score ($p < 0.01$), and malignant cancer ($p = 0.01$), indicating that the strength of the inverse association increased among patients with higher comorbidity burdens or malignancies, and varied by gender. Similar trends were noted for ICU mortality, with ORs generally showing a reduction in risk with prolonged ED stays across most subgroups (**Fig. 4B**). Although the interaction with gender did not reach significance ($p = 0.06$), CCI score ($p < 0.01$) and congestive heart failure ($p = 0.03$) emerged as significant effect modifiers, suggesting potential variability in the association depending on the severity of comorbidities. These findings underscore the robustness of the protective association between longer ED to ICUs time and improved survival, while highlighting certain patient characteristics that may influence this relationship.

Sensitivity Analysis

In the sensitivity analysis, patients initially excluded due to a time gap of more than 6 hours between hospital admission (*admittime*) and ICU admission (*intime*) were reintroduced to evaluate the impact of this criterion on the study results. **Supplementary Table 5** presents the baseline characteristics of this expanded cohort. After applying GBM and propensity score weighting, baseline balance remained markedly improved in this expanded cohort. As shown in **Supplementary Table 6**, the maximum standardized mean difference decreased from 0.64 before weighting to 0.06 after weighting, and the maximum Kolmogorov-Smirnov statistic declined from 0.22 to 0.03, with the minimum Kolmogorov-Smirnov p-value increasing from < 0.01 to 0.07. Correspondingly, the effective sample sizes remained robust after adjustment, as detailed in

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Supplementary Table 7. **Supplementary Fig. 5** further illustrates the substantial reduction in standardized mean differences following weighting. The odds ratios from the logistic regression models (**Supplementary Table 8** and **Supplementary Fig. 6**) continued to indicate lower odds of in-hospital and ICU mortality for patients in Q4 compared to those in the other groups. Moreover, Kaplan-Meier survival curves (**Supplementary Fig. 7**) showed the highest 28-day survival rates for Q4, with Q1 vs. Q4 comparisons remaining significant ($p < 0.01$). These findings suggest that the exclusion of patients with longer pre-ICU stays did not materially affect the relationship between extended ED stays and better survival outcomes, thereby reinforcing the robustness of the results.

Discussion

Our study, employing GBM-based propensity score weighting to achieve enhanced baseline comparability, provides a more nuanced and reliable insight into the relationship between ED to ICUs time and mortality outcomes. By leveraging GBM's capability to optimize balance across groups through iterative tree-based modeling, as detailed by McCaffrey et al. [11], we ensured robust minimization of baseline confounding. After establishing near-equivalence of patient characteristics across quartiles, we unexpectedly found that longer ED to ICUs time (5.65-34.55 hours) were consistently associated with lower in-hospital, ICU, and 28-day mortality. Notably, this inverse relationship persisted through sequential adjustments for severity of illness, comorbidity burden, demographic factors, and critical interventions. Subgroup analysis further demonstrated that the protective association remained significant across most patient groups, but did not achieve statistical significance among those with malignant cancer. This lack of significance may be attributed to the fact that patients with malignancy often present with complex conditions, where acute issues are typically addressed and stabilized in the ED. In more advanced cases, treatment focuses on symptom relief over intensive rescue, as further ICU intervention may not improve prognosis [12, 13]. Altogether, by employing GBM-based

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weighting to minimize confounding and achieve comprehensive covariate balance, our findings challenge the conventional assumption that prolonged ED stays necessarily worsen outcomes [8, 14, 15], instead suggesting that, under certain conditions, extended ED management may confer survival benefits.

A large retrospective cohort study with data derived from MIMIC-III showed that every delayed hour in the ED increased the in-hospital mortality, and in particular patients with ≥ 6h of transfer from ED to the ICUs had a significantly higher mortality rate than those with < 6h [14]. However, this analysis may have included patients initially transferred from the ED to general wards before ICU admission, which may not fully reflect outcomes for those transferred directly from the ED to the ICU. Other studies have similarly shown that a longer hospital stay before ICU admission was correlated with mortality, because patients with longer hospital stays may have more comorbidities [16, 17]. We limited the time between recorded hospital admission and ICU admission to 6 hours to identify patients transferred directly from the ED to the ICUs. However, this 6-hour threshold may introduce selection bias, as some patients initially transferred from the ED to non-ICU wards may have stayed there for less than six hours before ICU transfer. To address this, we conducted a sensitivity analysis reintroducing patients with hospital to ICUs times over six hours, confirming the robustness of our findings.

The ED to ICUs time is multifactorially influenced, including ED crowding [18], triage process [19], availability of ICU beds [20, 21], which may shape clinical outcome. Trzeciak et al. [22] noted that ED overcrowding, staff may struggle to provide the necessary care for critically ill patients, a limitation associated with increased mortality when patients receive care in settings unable to fully support their critical needs [23]. To address these challenges, EDs worldwide are adopting measures to enhance timeliness and throughput, including bedside registration, self-check-in kiosks, immediate bedding, fast-track systems, and physician-in-triage protocols [24]. Abdulai et al. [25] examined trends in ED throughput in the United States from 2006 to 2016 and found after implementing a series of measures, improved ED throughput allowed healthcare

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providers more time to assess and manage critically ill patients who are stranded in the emergency department, facilitating appropriate treatment decisions and potentially improving outcomes. Additionally, clinical decision support systems based on intelligent technologies, as noted by Fernandes et al. [26] , have been shown to enhance healthcare professionals' decision-making. These systems help ED staff identify, diagnose, and treat critically ill patients, ensuring appropriate ICU transfers while stable patients remain in the ED. This underscores that extended ED stays are not delays but a rational use of resources to improve overall outcomes and reduce mortality. Availability of ICU beds is a crucial factor influencing ED to ICUs time and patient outcomes [27]. A previous study found that ICU admission decisions for critically ill ED patients are affected by medical ICU bed availability [28]. To address these limitations, hospitals are adopting innovative strategies to expand ICU capacity. For instance, Mitra et al. [29] found that establishing high-acuity units, which offer close monitoring and acute care comparable to the ICU but with fewer resources, was associated with reduced in-hospital mortality for patients requiring intensive care. Establishing a critical care resuscitation unit can handle more critically ill surgical patients and speed up the transfer of those with time-sensitive conditions to specialized centers. Furthermore, enhancing ICU capacity and improving its operational efficiency can reduce the time needed for patient admission to the ICU.

This association between longer ED to ICUs time and lower hospital mortality can be explained by several possible mechanisms. One potential explanation is that extended waiting time in the ED provides a crucial period for assessing patient conditions and implementing necessary stabilization measures before ICU transfer. Extended waiting time in the ED may serve as a critical period to stabilize patients by initiating critical care interventions, such as goal-directed therapy, which stabilize patients' clinical status and reduce the risk of deterioration before ICU transfer [30]. Alternatively, extended ED to ICU transfer times may help critically ill patients avoid the significant risks associated with premature intrahospital transport (IHT). Although often unavoidable, IHT exposes patients to a mobile environment, increasing the likelihood of

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adverse events, including equipment failure, device dislodgement, inadequate monitoring, and cardiovascular or respiratory instability [31, 32]. These adverse events, reported in up to 79.8% of ICU-bound IHT cases in a study of 441 patients [33], can exacerbate clinical deterioration and elevate mortality risk. Premature transport of critically ill patients from the ED, while potentially reducing ED to ICUs time, often limits essential preparation time, increasing the risk of complications [34]. By allowing patients to avoid the risks of premature transfer and receive appropriate critical care interventions in the ED, extended ED to ICUs time may provide a vital window for stabilization and improved outcomes. Thus, extended ED to ICUs time may represent opportunities for essential stabilization rather than mere delays, supporting the notion that early critical care in the ED can be pivotal in reducing mortality. Another potential explanation is the significant advancements in the ability of emergency department clinicians to manage critically ill patients over extended periods, combined with the development of advanced emergency medical technologies [35]. For example, the study by Kadar et al. [36] demonstrated that patients receiving ICU telemedicine support in the ED had lower mortality rates and reduced ICU resource utilization compared to those receiving standard ED care. This benefit was attributed to telemedicine teams monitoring patient stability over time, enabling safe non-ICU management and optimized resource use.

Our study highlights that managing critically ill patients in the emergency department involves more than just minimizing ED wait times. Considering the patient's clinical condition, ED circumstances, ED capabilities, transport risks, and ICU bed availability to determine the best timing for ICU admission is essential. A key strength of our research is the use of the comprehensive MIMIC-IV database, which spans from 2008 to 2019. This dataset, the most recent before the onset of COVID-19, allows for an accurate analysis of the association between ED to ICUs time and patient outcomes without the confounding effects of pandemic [37]. The use of GBM-based propensity score weighting minimized baseline confounding and ensured covariate balance, enabling a precise analysis of the unexpected finding that longer ED to ICUs time were

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associated with lower mortality rates.

Limitations

This study has several limitations. First, as a retrospective analysis drawn from the MIMIC database, these findings cannot establish causality and may not be generalizable to other countries. Second, we excluded patients who died in the ED to focus on those successfully transferred to the ICUs. Although necessary, this may introduce selection bias by underrepresenting the most critically ill and potentially underestimating mortality associated with shorter ED to ICUs time. Third, the MIMIC database does not record critical interventions performed in the ED and records surgeries with daily resolution rather than hourly granularity. This limited temporal detail prevents determining whether interventions preceded ICU admission, complicating the interpretation of ED to ICUs time while possibly leading to an underestimation of pre-admission mortality. Lastly, the retrospective design prevented a detailed examination of the components influencing ED to ICUs time, including orientation decisions, clinical assessments, diagnostic workups, boarding durations. Future research should employ prospective, multicenter designs with more comprehensive and temporally precise data collection to better elucidate the impact of ED to ICUs time on patient outcomes.

Conclusions

Patients transferred from the ED to the ICUs within 5.65-34.55 hours experienced lower in-hospital, ICU, and 28-day mortality rates compared to those transferred earlier. These findings suggest that appropriately extended ED stays may facilitate critical stabilization, improving outcomes and emphasizing the need for further research to optimize transfer protocols and ED management strategies.

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Author contributions

JWQ, YNY and ZMS designed the study, conducted the formal analysis, and drafted the manuscript. JWQ extracted the data from the MIMIC-IV database. YNY, ZMS and QXL reviewed and analyzed the data. KSZ, LYZ and WZZ guided the manuscript review and editing. XFJ and MQC instructed the conceptualization. MQC is the guarantor of this study. All authors read and approved the final manuscript.

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Data availability statement

Data generated and analyzed can be available from the corresponding author on reasonable request.

Declarations

Ethics approval

Given that this study analyzed an anonymous public database with prior Institutional Review Board approval, no ethical review was necessary.

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Patient consent for publication

Not required.

Patient and public involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Competing interests

The authors declare that they have no competing interests.

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Figure Legends

Fig. 1 Flow chart of patients selection for analytic

MIMIC-IV Medical Information Mart for Intensive Care-IV, ICU intensive care unit

Fig. 2 Odds Ratios for all-cause in-hospital and ICU mortality after GBM-based propensity score weighting

ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

A. Association between ED to ICUs time and in-hospital mortality, unadjusted (weighted).

B. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score and CCI score (weighted).

C. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

D. Association between ED to ICUs time and ICU mortality, unadjusted (weighted).

E. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score and CCI score (weighted).

F. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

Fig. 3 Kaplan-Meier survival analysis curves for 28-day all-cause mortality

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Kaplan-Meier survival curves for patients stratified into four quartiles based on ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

Fig. 4 Forest plots of odd ratios for the primary outcome in different subgroups

A. Association between ED to ICUs time and in-hospital mortality; B. Association between ED to ICUs time and ICU mortality. ORs with 95% CIs are displayed for each subgroup.

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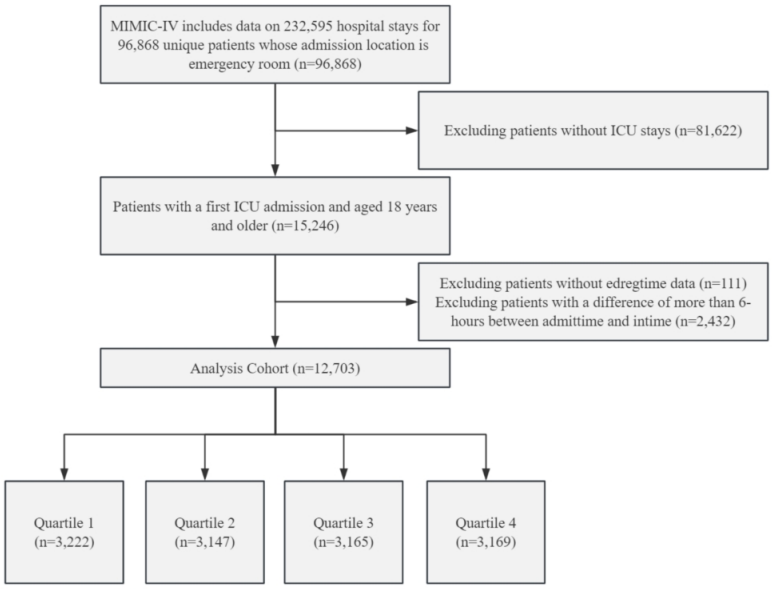


Fig. 1 Flow chart of patients selection for analytic
MIMIC-IV Medical Information Mart for Intensive Care-IV, ICU intensive care unit

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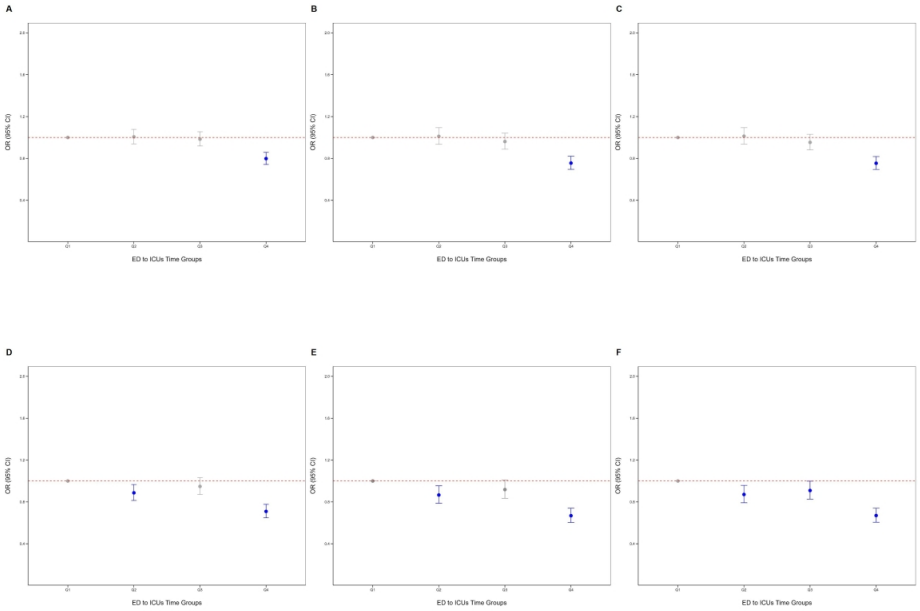


Fig. 2 Odds Ratios for all-cause in-hospital and ICU mortality after GBM-based propensity score weighting
ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).
A. Association between ED to ICUs time and in-hospital mortality, unadjusted (weighted).
B. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score and CCI score (weighted).
C. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).
D. Association between ED to ICUs time and ICU mortality, unadjusted (weighted).
E. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score and CCI score (weighted).
F. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

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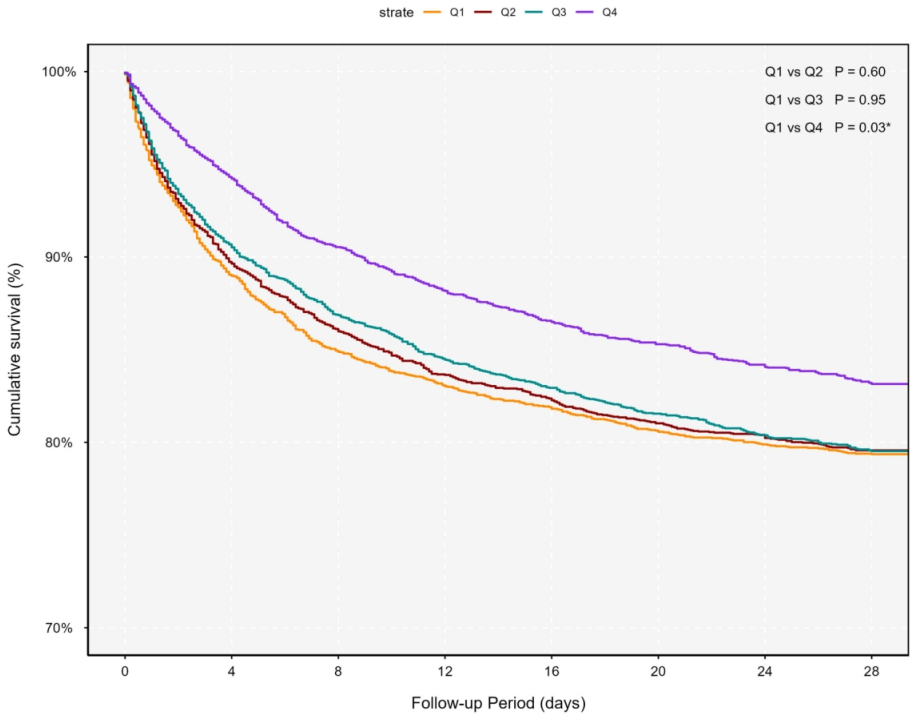


Fig. 3 Kaplan-Meier survival analysis curves for 28-day all-cause mortality
Kaplan-Meier survival curves for patients stratified into four quartiles based on ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

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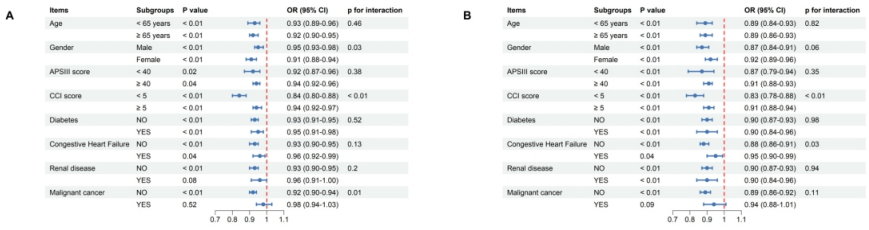


Fig. 4 Forest plots of odd ratios for the primary outcome in different subgroups
A. Association between ED to ICUs time and in-hospital mortality; B. Association between ED to ICUs time and ICU mortality. ORs with 95% CIs are displayed for each subgroup.

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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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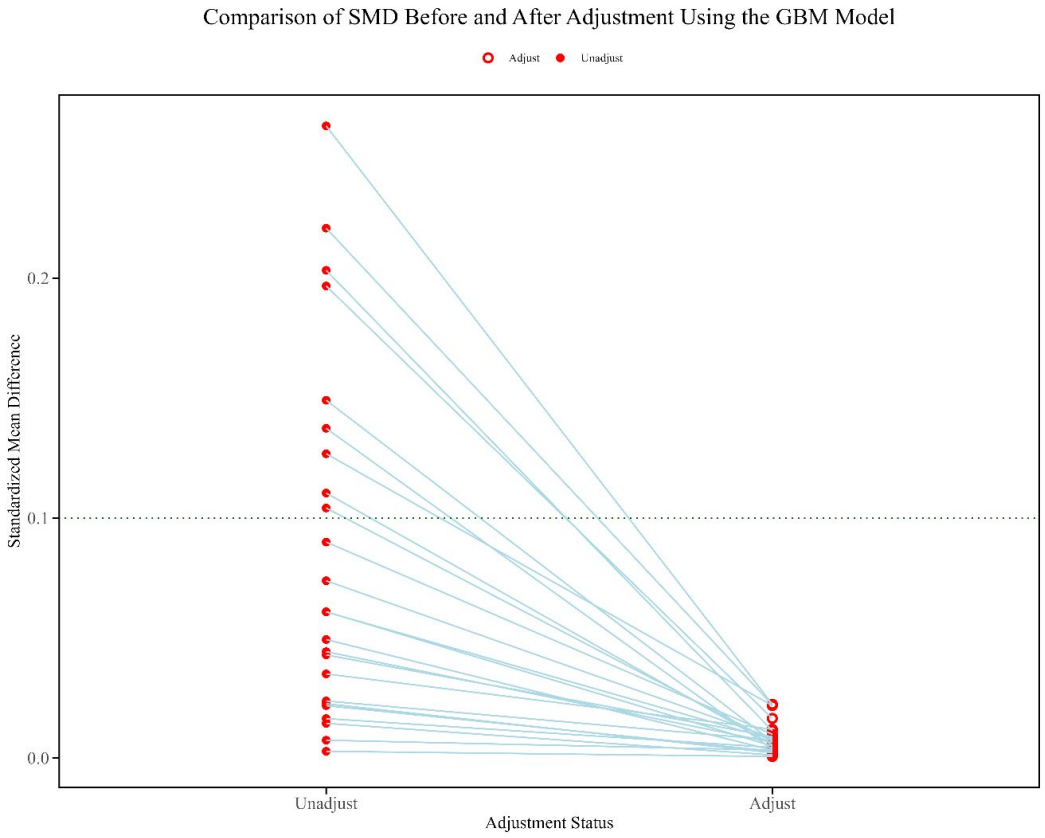
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Supplementary Figure

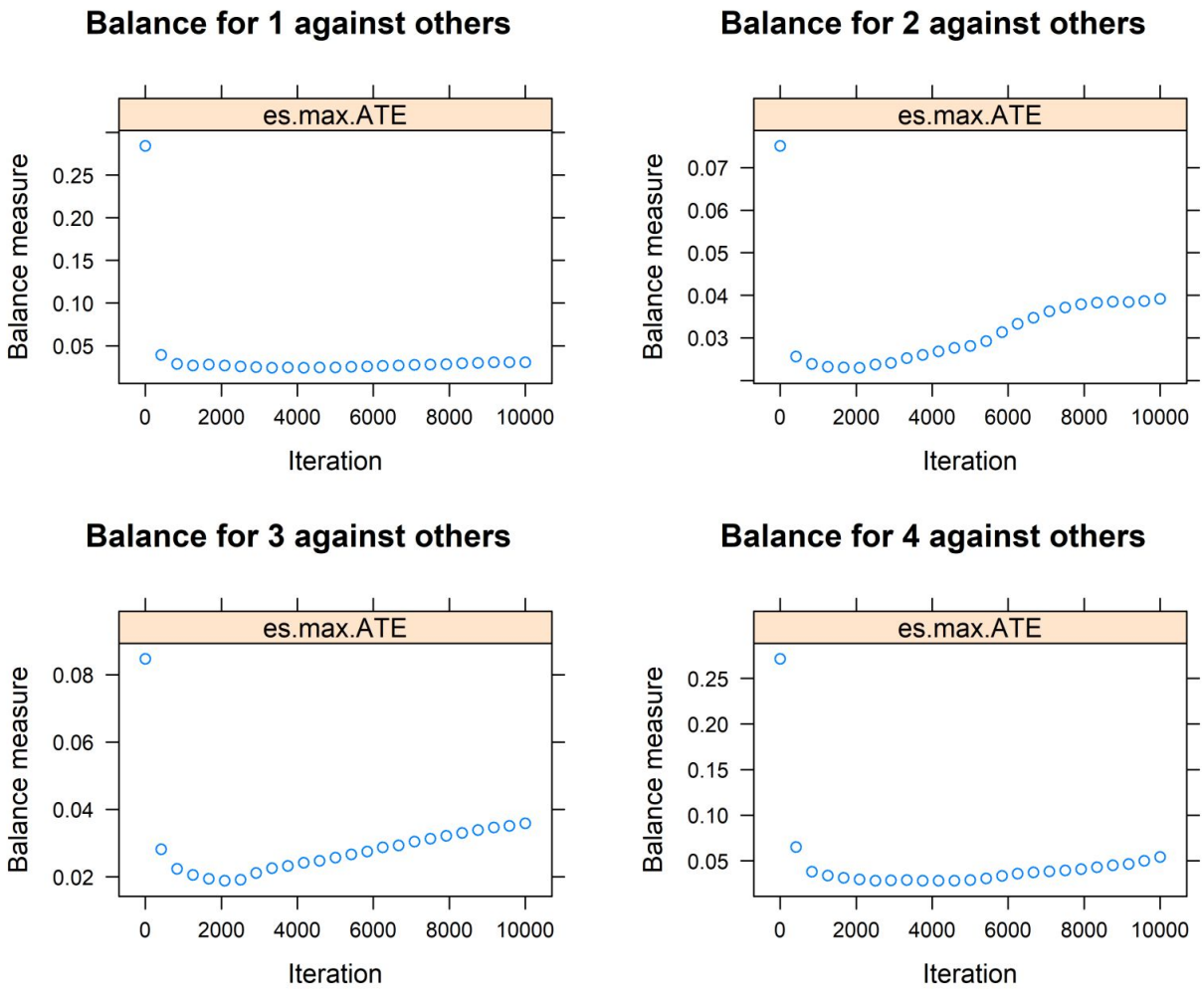


Supplementary Fig. 1 Comparison of Standardized Mean Differences Before and After GBM-Based Propensity Score Weighting

GBM: Generalized Boosted Model, SMD: Standardized Mean Differences. Red circles represent the unadjusted SMD values, and open circles represent the adjusted SMD values after weighting. The horizontal dashed line at SMD = 0.1 indicates the threshold for acceptable covariate balance.
Covariates include age, gender, race, first care unit, CCI scores, APSIII scores, diabetes, congestive heart failure, renal disease, malignancy, invasive mechanical ventilation, dialysis, and vasopressor use.

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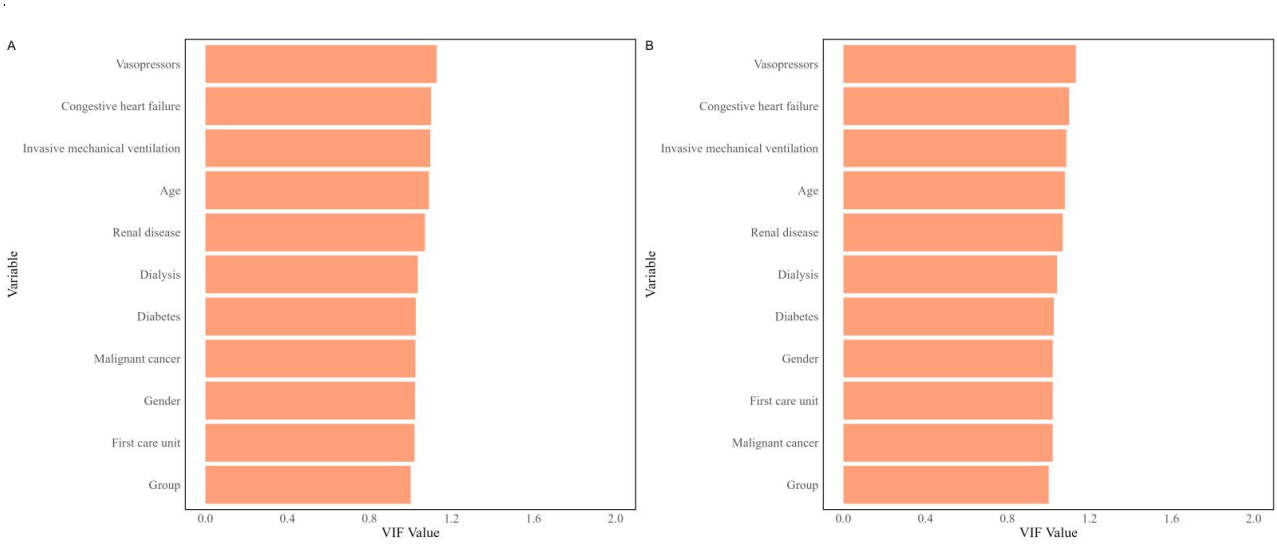


Supplementary Fig. 2 Balance Measures Across Iterations for Propensity Score Weighting Using GBM

ATE: Average Treatment Effect. The figure shows the balance measure (es.max.ATE) across iterations during propensity score weighting using Generalized Boosted Model for each comparison group: Q1 vs others, Q2 vs others, Q3 vs others, and Q4 vs others. Each panel represents the balance assessment for a specific group, with the x-axis indicating the number of iterations and the y-axis showing the balance measure.

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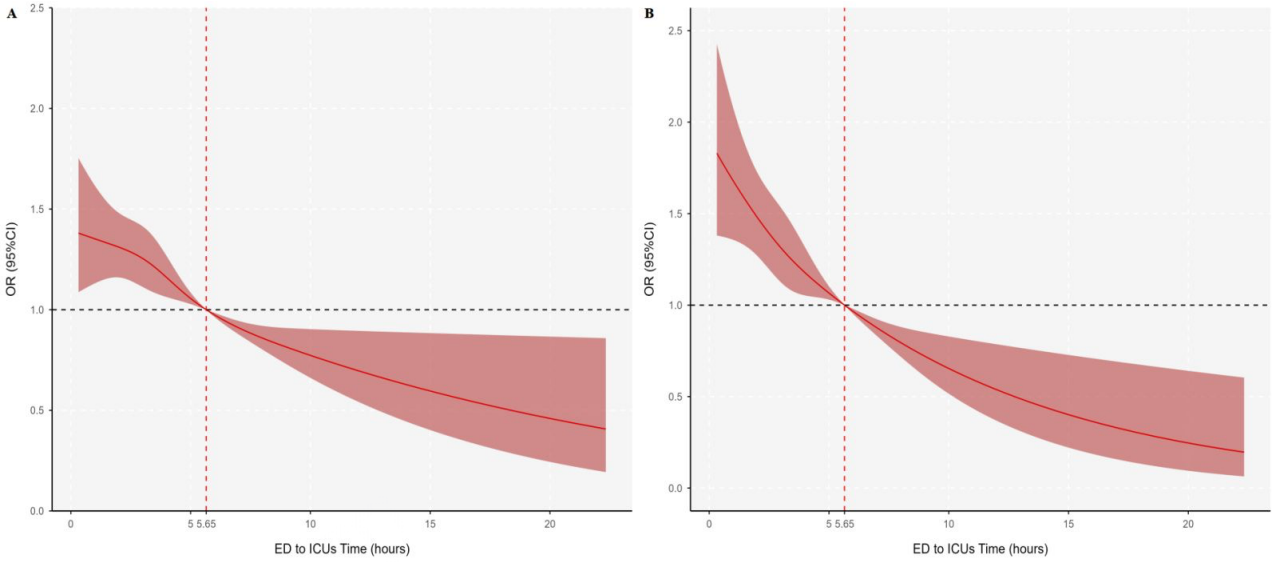


Supplementary Fig. 3 Variance Inflation Factor Analysis for GBM-Weighted Logistic Models: Hospital and ICU Mortality

A. Model analyzing predictors of hospital mortality. B. Model analyzing predictors of ICU mortality. Variables include vasopressors, congestive heart failure, invasive mechanical ventilation, age, renal disease, dialysis, diabetes, malignant cancer, gender, first care unit, and group. All variables have VIF values less than 5, indicating no significant multicollinearity.

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Supplementary Fig. 4 Restricted cubic spline regression analysis of ED to ICUs time with in hospital all-cause mortality

Red heavy central lines represent the estimated adjusted hazard ratios, with shaded ribbons denoting 95% confidence intervals. ED to ICUs time 5.65hr was selected as the reference level represented by the vertical dotted lines. The horizontal dotted lines represent the hazard ratio of 1.0. HR hazard ratio, CI confidence interval

A. Restricted cubic spline for in-hospital death. B. Restricted cubic spline for ICU death.

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Supplementary Table

Supplementary Table 1 Baseline characteristics of ICU patients grouped according to ED to ICUs time quartiles

Categories	Overall (N=12703)	Q1 (N=3222)	Q2 (N=3147)	Q3 (N=3165)	Q4 (N=3169)	P-value
ED to ICUs time, hr, median (IQR)	3.98 (2.83, 5.65)	2.16 (1.66, 2.53)	3.41 (3.12, 3.68)	4.70 (4.[illegible] 5.12)	7.22 (6.33, 8.72)	< 0.01
Demographic						
Age, years, median (IQR)	64 (49, 79)	61 (46, 77)	65 (49, 79)	66 (51, [illegible])	66 (51, 80)	< 0.01
Male, n (%)	7178 (56.51)	1993 (61.86)	1771 (56.28)	1784 (5[illegible]7)	1630 (51.43)	< 0.01
Ethnicity, n (%)						< 0.01
White	8149 (64.15)	1799 (55.83)	1986 (63.11)	2122 (67.05)	2242 (70.75)	
Black	1103 (8.68)	249 (7.91)	249 (7.91)	277 (8.75)	365 (11.52)	
Asian	375 (2.95)	87 (2.76)	87 (2.76)	105 (3.32)	117 (3.69)	
Hispanic/Latino	463 (3.64)	112 (3.48)	115 (3.65)	117 (3.70)	119 (3.76)	
Other	636 (5.01)	195 (6.05)	146 (4.64)	149 (4.71)	146 (4.61)	
Unknown	1977 (15.56)	838 (26.01)	564 (17.92)	395 (12.48)	180 (5.68)	
Admission type, n (%)						< 0.01
EW EMER	11357 (89.40)	2912 (90.38)	2883 (91.61)	2834 (89.54)	2728 (86.08)	
OBSERVATION ADMIT	1194 (9.40)	269 (8.35)	236 (7.50)	300 (9.48)	389 (12.28)	
EU OBSERVATION	147 (1.16)	40 (1.24)	27 (0.86)	30 (0.95)	50 (1.58)	
DIRECT OBSERVATION	4 (0.03)	1 (0.03)	1 (0.03)	0 (0.00)	2 (0.06)	
URGENT	1 (0.01)	0 (0.00)	0 (0.00)	1 (0.03)	0 (0.00)	
First care unit, n (%)						< 0.01
MICU	3301 (25.99)	683 (21.20)	798 (25.36)	863 (27.26)	957 (30.20)	
SICU	2569 (20.22)	553 (17.16)	732 (23.26)	712 (22.50)	572 (18.05)	
TSICU	2709 (21.33)	941 (29.21)	707 (22.47)	571 (18.04)	490 (15.46)	

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MICU/SICU	2123 (16.71)	365 (11.33)	447 (14.20)	602 (19.[illegible])	709 (22.37)	
CCU	1053 (8.29)	417 (12.94)	260 (8.26)	200 (6.3[illegible])	176 (5.55)	
CVICU	445 (3.50)	164 (5.09)	93 (2.96)	92 (2.9[illegible])	96 (3.03)	
Neuro SICU	503 (3.96)	99 (3.07)	110 (3.50)	125 (3.9[illegible])	169 (5.33)	
Chronic conditions, n (%)						
Diabetes	2938 (23.13%)	674 (20.92%)	723 (22.97%)	767 (24.[illegible]%)	774 (24.42%)	< 0.01
Congestive heart failure	2401 (18.90%)	567 (17.60%)	605 (19.22%)	596 (18.[illegible]%)	633 (19.97%)	0.10
Renal disease	1753 (13.80%)	375 (11.64%)	388 (12.33%)	485 (15.[illegible]%)	505 (15.94%)	< 0.01
Malignant cancer	1090 (8.58%)	208 (6.46%)	240 (7.63%)	297 (9.[illegible]%)	345 (10.89%)	< 0.01
Severity scores						
CCI, mean (SD)	4.80 (3.07)	4.35 (3.01)	4.67 (3.03)	5.02 (3.[illegible])	5.16 (3.12)	< 0.01
APSIII, mean (SD)	46.37 (24.00)	47.61 (25.71)	46.04 (23.82)	47.24 (23.[illegible]95)	44.57 (22.26)	< 0.01
Intervention during ICU, n (%)						
Invasive mechanical ventilation	4196 (33.03%)	1366 (42.4%)	1122 (35.65%)	988 (31.[illegible]%)	720 (22.72%)	< 0.01
Dialysis	235 (1.85%)	73 (2.27%)	48 (1.53%)	60 (1.9[illegible]%)	54 (1.70%)	0.15
Vasopressors	2425 (19.09%)	734 (22.78%)	562 (17.86%)	600 (18.[illegible]%)	529 (16.69%)	< 0.01
Outcome						
Hospital death, n (%)	1988 (15.65)	567 (17.60)	520 (16.52)	514 (16.[illegible])	387 (12.21)	< 0.01
ICU death, n (%)	1185 (9.33)	375 (11.64)	297 (9.44)	311 (9.8[illegible])	202 (6.37)	< 0.01
28-day death, n (%)	2490 (19.60)	665 (20.64)	643 (20.43)	648 (20.[illegible])	534 (16.85)	< 0.01
Hosptial-LOS, days, median (IQR)	5.0 (2.8, 9.4)	5.1 (2.7, 9.7)	5.0 (2.8, 9.1)	5.0 (2.8[illegible]3)	5.3 (3.0, 9.3)	0.05
ICU-LOS, median (IQR)	1.80 (0.96, 3.59)	1.96 (1.11, 4.15)	1.80 (0.97, 3.64)	1.78 (0.9[illegible]3.53)	1.70 (0.91, 2.98)	< 0.01

ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr)

ED emergency department, ICU intensive care unit, IQR interquartile range, *EW EMER* emergency ward emergency, EU OBSERVATION: emergency unit observation, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit, CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, CCI Charlson Comorbidity Index, APSIII acute

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physiology score III, LOS length of stay

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Supplementary Table 2 Covariate Balance Diagnostics Before and After Generalized Boosted Model Adjustment

Metrics	Maximum SMD	Maximum KS	Minimum KS *p-value*
Before GBM	0.56	0.20	< 0.01
After GBM	0.04	0.02	0.22

KS: Kolmogorov-Smirnov statistic.

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Supplementary Table 3 Sample Sizes and Effective Sample Sizes Before and After GBM Adjustment

Quartile	Original Sample Size (N)	Effective Sample Size (N)
Q1	3222	2651.1
Q2	3147	3058.7
Q3	3165	3031.7
Q4	3169	2475.5

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Supplementary Table 4 Relationship between ED to ICUs time and length of stay

	Coef.	S.E	t-value	*p*-value
Hospital-LOS[a]				
Model 1	-0.099	0.032	-3.153	0.002
Model 2	-0.032	0.030	-1.070	0.285
Model 3	0.031	0.029	1.069	0.285
ICU-LOS[b]				
Model 1	-0.157	0.017	-9.439	< 0.001
Model 2	-0.097	0.016	-6.287	< 0.001
Model 3	-0.069	0.015	-4.503	< 0.001

Model 1: unadjusted; Model 2: adjusted for APSIII score, Charlson Comorbidity Index; Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, invasive mechanical ventilation, dialysis and vasopressors.

[a] The relationship between ED to ICUs time and length of hospital stay was analyzed in a cohort that survived the hospital stay (n=10701)

[b] The relationship between ED to ICUs time index and length of ICU stay was analyzed in a cohort that survived the ICU stay (n=11518)

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Supplementary Sensitivity Analysis

Supplementary Table 5 Baseline characteristics of ICU patients grouped according to ED to ICUs time quartiles in Sensitivity Analysis

Categories	Overall (N=15135)	Q1 (N=3797)	Q2 (N=3775)	Q3 (N=3780)	Q4 (N=3783)	*p*-value
ED to ICUs time, hr, median (IQR)	4.48 (3.03, 7.54)	2.3 (1.77, 2.7)	3.72 (3.38, 4.08)	5.58 (4.7, 6.4)	22.46 (9.95, 72.78)	< 0.01
Demographic						
Age, years, median (IQR)	65 (50, 79)	62 (46, 77)	65 (50, 80)	66 (52, 80)	67 (53, 80)	< 0.01
Male, n (%)	8489 (56.09%)	2330 (61.36%)	2098 (55.58%)	2072 (54.81%)	1989 (52.58%)	< 0.01
Ethnicity, n (%)						< 0.01
White	9963 (65.83%)	2178 (57.36%)	2417 (64.03%)	2594 (68.62%)	2774 (73.33%)	
Black	1369 (9.05%)	244 (6.43%)	313 (8.29%)	361 (9.55%)	451 (11.92%)	
Asian	455 (3.01%)	76 (2.00%)	112 (2.97%)	136 (3.6%)	131 (3.46%)	
Hispanic/Latino	566 (3.74%)	128 (3.37%)	134 (3.55%)	152 (4.02%)	152 (4.02%)	
Other	734 (4.85%)	216 (5.69%)	182 (4.82%)	180 (4.76%)	156 (4.12%)	
Unknown	2048 (13.53%)	955 (25.15%)	617 (16.34%)	357 (9.45%)	119 (3.15%)	
Admission type, n (%)						< 0.01
EW EMER	13343 (88.16%)	3439 (90.57%)	3445 (91.26%)	3355 (88.76%)	3104 (82.05%)	
OBSERVATION ADMIT	1630 (10.77%)	311 (8.19%)	291 (7.71%)	387 (10.24%)	641 (16.94%)	
EU OBSERVATION	157 (1.04%)	45 (1.19%)	38 (1.01%)	38 (1.01%)	36 (0.95%)	
DIRECT OBSERVATION	4 (0.03%)	2 (0.05%)	0 (0%)	0 (0%)	2 (0.05%)	
URGENT	1 (0.01%)	0 (0%)	1 (0.03%)	0 (0%)	0 (0%)	
First care unit, n (%)						< 0.01
MICU	3815 (25.21%)	807 (21.25%)	990 (26.23%)	1094 (28.94%)	924 (24.43%)	
SICU	3038 (20.07%)	697 (18.36%)	873 (23.13%)	762 (20.16%)	706 (18.66%)	
TSICU	3014 (19.91%)	1078 (28.39%)	802 (21.25%)	645 (17.06%)	489 (12.93%)	

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MICU/SICU	2541 (16.79%)	442 (11.64%)	578 (15.31%)	795 (21.07%)	726 (19.19%)	
CCU	1354 (8.95%)	472 (12.43%)	285 (7.55%)	220 (5.83%)	377 (9.97%)	
CVICU	798 (5.27%)	183 (4.82%)	111 (2.94%)	106 (2.81%)	398 (10.52%)	
Neuro SICU	575 (3.80%)	118 (3.11%)	136 (3.60%)	158 (4.19%)	163 (4.31%)	
Chronic conditions, n (%)						
Diabetes	3577 (23.63%)	798 (21.02%)	887 (23.50%)	938 (24.85%)	954 (25.22%)	< 0.01
Congestive heart failure	3053 (20.17%)	663 (17.46%)	735 (19.47%)	734 (19.44%)	921 (24.35%)	< 0.01
Renal disease	2217 (14.65%)	435 (11.46%)	524 (13.88%)	572 (15.15%)	686 (18.13%)	< 0.01
Malignant cancer	1494 (9.87%)	254 (6.69%)	303 (8.03%)	366 (9.69%)	571 (15.09%)	< 0.01
Severity scores						
CCI, mean (SD)	5 (3, 7)	4 (2, 6)	5 (3, 7)	5 (3, 7)	5 (3, 8)	< 0.01
APSIII, mean (SD)	40 (30, 57)	40 (30, 60)	41 (30, 57)	41 (30, 57)	40 (30, 56)	0.26
Intervention during ICU, n (%)						
Invasive mechanical ventilation	4927 (32.55%)	1582 (41.66%)	1316 (34.86%)	1038 (27.46%)	991 (26.20%)	< 0.01
Dialysis	300 (1.98%)	84 (2.21%)	67 (1.77%)	62 (1.64%)	87 (2.30%)	0.11
Vasopressors	2905 (19.19%)	835 (21.99%)	696 (18.44%)	676 (17.88%)	698 (18.45%)	< 0.01
Outcome						
Hospital death, n (%)	2367 (15.64%)	661 (17.41%)	641 (16.98%)	535 (14.15%)	530 (14.01%)	< 0.01
ICU death, n (%)	1374 (9.08%)	427 (11.25%)	379 (10.04%)	306 (8.10%)	262 (6.93%)	< 0.01
28-days death, n (%)	2947 (19.47%)	777 (20.46%)	784 (20.77%)	719 (19.02%)	667 (17.63%)	< 0.01
Hosptial-LOS, days, median (IQR)	5.7 (3.1, 10.4)	5 (2.8, 9.7)	4.9 (2.8, 9.1)	5.1 (2.9, 9.4)	7.7 (4.6, 12.8)	< 0.01
ICU-LOS, median (IQR)	1.81 (0.98, 3.56)	1.94 (1.1, 4.12)	1.76 (0.95, 3.64)	1.76 (0.97, 3.28)	1.78 (1.03, 3.19)	< 0.01

ED to ICUs time (hours): Q1 (≤ 3.03 hr), Q2 (3.03-4.48 hr), Q3 (4.48-7.54 hr), Q4 (> 7.54 hr)

ED emergency department, ICU intensive care unit, IQR interquartile range, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit, CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, CCI Charlson Comorbidity Index, APSIII acute physiology score III, LOS length of stay

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Supplementary Table 6 Covariate Balance Diagnostics Before and After Generalized Boosted Model Adjustment in Sensitivity Analysis

Metrics	Maximum SMD	Maximum KS	Minimum KS *p-value*
Before GBM	0.64	0.22	< 0.01
After GBM	0.06	0.03	0.07

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Supplementary Table 7 Sample Sizes and Effective Sample Sizes Before and After GBM Adjustment in Sensitivity Analysis

Quartile[a]	Original Sample Size (N)	Effective Sample Size (N)
Q1	3797	3202.7
Q2	3775	3640.1
Q3	3780	3398.9
Q4	3783	2622.8

[a] ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

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Supplementary Table 8 Odds Ratios for All-Cause Mortality in-hospital and ICU mortality from Logistic Regression after Generalized Boosted Models in Sensitivity Analysis

Categories	Model 1		Model 2		Model 3	
	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value
Hospital death						
Quartilea[a]						
Q1	Ref.		Ref.		Ref.	
Q2	1.02 (0.96, 1.09)	0.54	1.02 (0.95, 1.10)	0.55	1.02 (0.95, 1.09)	0.58
Q3	0.88 (0.83, 0.94)	< 0.01	0.88 (0.82, 0.95)	< 0.01	0.80 (0.88, 0.92)	< 0.01
Q4	0.86 (0.81, 0.92)	< 0.01	0.84 (0.78, 0.91)	< 0.01	0.83 (0.77, 0.89)	< 0.01
ICU death						
Quartilea[a]						
Q1	Ref.		Ref.		Ref.	
Q2	0.97 (0.89, 1.04)	0.38	0.95 (0.87, 1.04)	0.28	0.95 (0.87, 1.04)	0.28
Q3	0.84 (0.78, 0.91)	< 0.01	0.86 (0.78, 0.94)	< 0.01	0.82 (0.75, 0.90)	< 0.01
Q4	0.72 (0.67, 079)	< 0.01	0.70 (0.63, 0.77)	< 0.01	0.69 (0.63, 0.75)	< 0.01

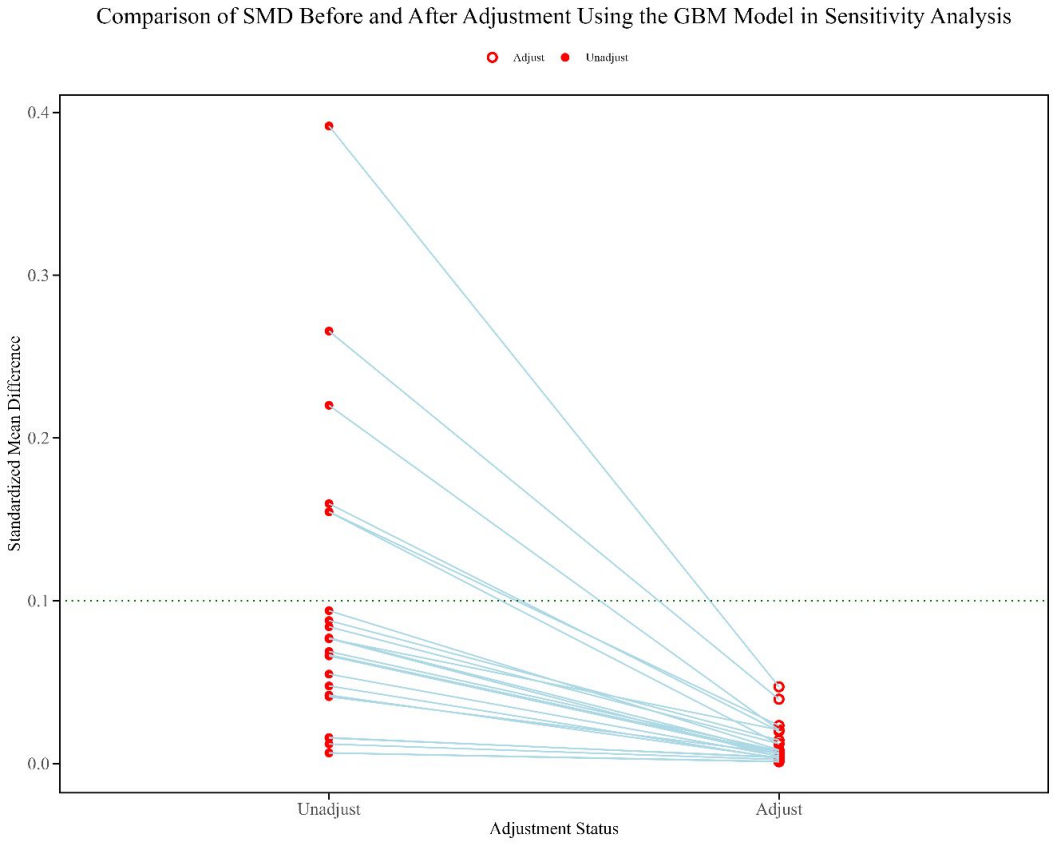
Model 1: unadjusted; Model 2: adjusted for APSIII score, CCI score; Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, invasive mechanical ventilation, analysis and vasopressors. OR: Odds Ratio

[a] ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

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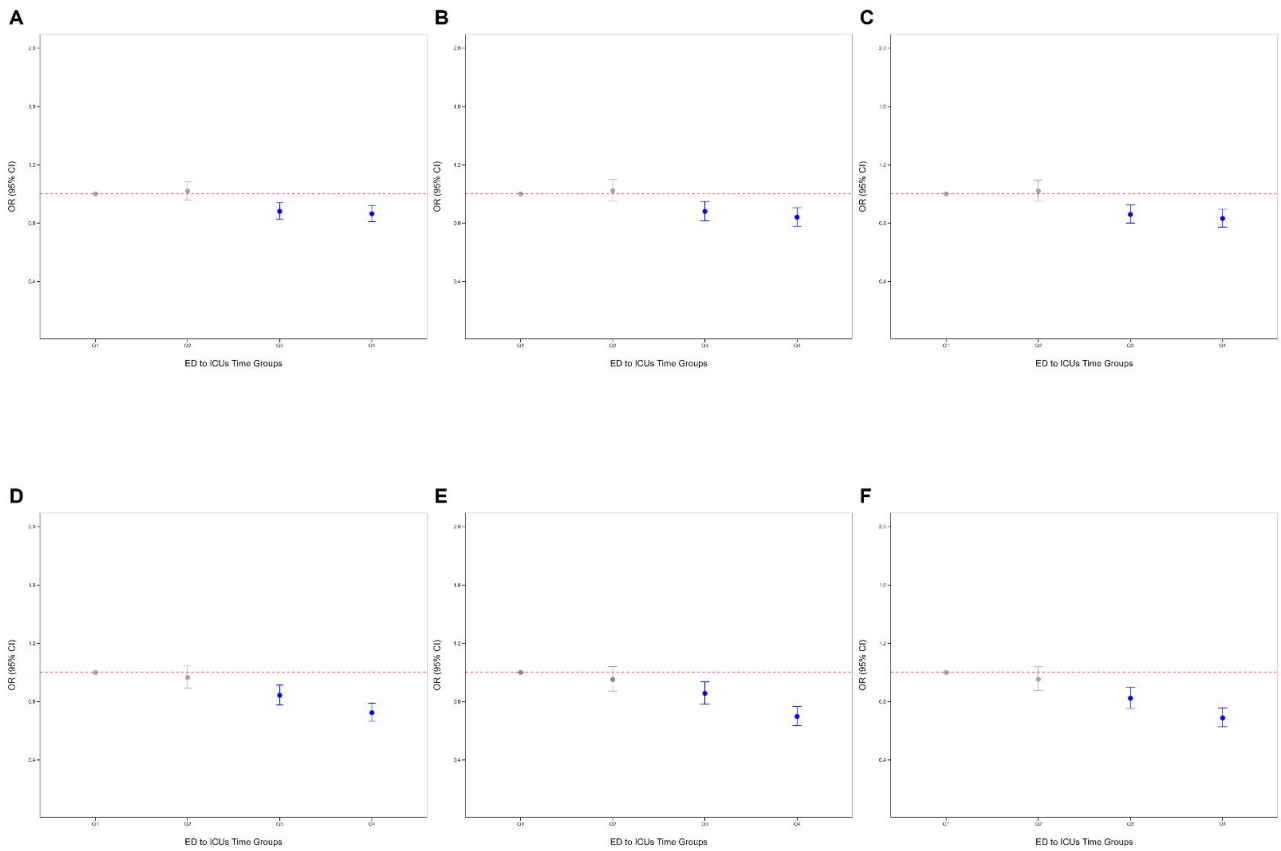
Supplementary Fig. 5 Comparison of Standardized Mean Differences Before and After GBM-Based Propensity Score Weighting in Sensitivity Analysis

Red circles represent the unadjusted SMD values, and open circles represent the adjusted SMD values after weighting. The horizontal dashed line at SMD = 0.1 indicates the threshold for acceptable covariate balance.
Covariates include age, gender, race, first care unit, CCI scores, APSIII scores, diabetes, congestive heart failure, renal disease, malignancy, invasive mechanical ventilation, dialysis, and vasopressor use.

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Supplementary Fig. 6 Odds Ratios for all-cause in-hospital and ICU mortality after generalized boosted model-based propensity score weighting in Sensitivity Analysis

ED to ICUs time (hours): Q1 (≤ 3.03 hr), Q2 (3.03-4.48 hr), Q3 (4.48-7.54 hr), Q4 (> 7.54 hr)

A. Association between ED to ICUs time and in-hospital mortality, unadjusted (weighted).

B. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score and CCI score (weighted).

C. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

D. Association between ED to ICUs time and ICU mortality, unadjusted (weighted).

E. Association between ED to ICUs time and ICU mortality, adjusted for OASIS score and CCI score (weighted).

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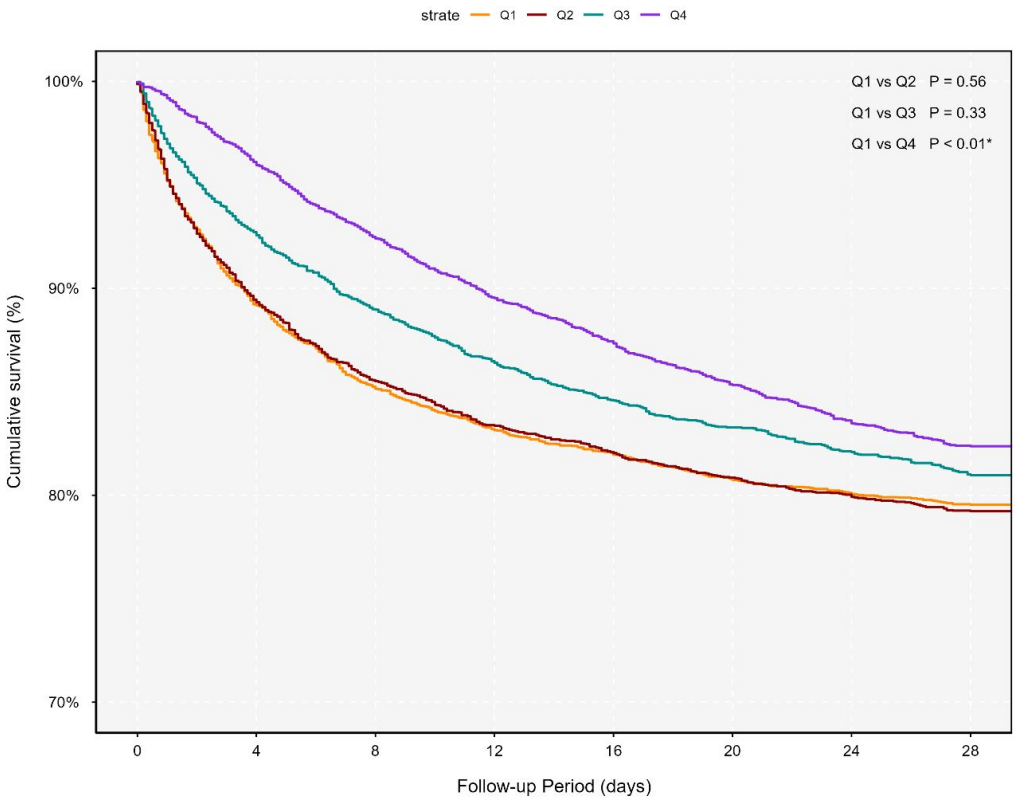
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F. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

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Supplementary Fig. 7 Kaplan-Meier Survival Curves for 28-day All-Cause Mortality in Sensitivity Analysis

ED to ICUs time (hours): Q1 (≤ 3.03 hr), Q2 (3.03-4.48 hr), Q3 (4.48-7.54 hr), Q4 (> 7.54 hr)

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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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Objectives: The association between the duration from the emergency department (ED) to the intensive care units (ICUs) and in-hospital mortality among patients admitted directly to the ICUs from the ED remains controversial. This study aimed to utilize data from the Medical Information Mart for Intensive Care IV (MIMIC-IV) database to explore the relationship between the ED to ICUs time and patient outcomes.

Design: Retrospective observational study.

Setting: Admissions to the Beth Israel Deaconess Medical Center intensive care from 2008 to 2019.

Participants: A total of 15,246 adult patients were identified as admitted directly from the ED to the ICU during their first hospitalization. After excluding those without recorded ED registration times and those with a hospital-to-ICU admission interval exceeding 6 hours (n = 2,432), the final analysis cohort comprised 12,703 patients.

Primary and secondary outcome measures: The primary outcome was in-hospital all-cause mortality. Secondary outcomes included 28-day all-cause mortality and length of stay in ICU and hospital.

Results: The median ED to ICUs time was 3.98 hours. Longer ED to ICUs times were associated with lower in-hospital mortality, decreasing from 17.6% in the shortest to 12.2% in the longest interval group, and shorter ICU stays. After propensity score weighting, adjusted logistic regression models confirmed the inverse association between longer ED to ICUs time and in-hospital mortality (OR: 0.75, 95% CI: 0.69-0.82, $p < 0.01$). Restricted cubic spline analysis showed a nonlinear decline in mortality risk with increasing ED to ICUs time, with a sharper reduction after 5.65 hours. Kaplan-Meier curves indicated consistently better survival in the longest interval group ($p < 0.01$). Sensitivity analysis, reintroducing patients with hospital to ICUs times over 6 hours, confirmed the robustness of these results.

Conclusions: Longer ED to ICUs time was linked to lower mortality and shorter ICU length of stay, suggesting that appropriately extending ED stays may benefit critically ill patients.

Keywords: Emergency Department; Intensive Care Units; Mortality; Generalized Boosted Model; Patient

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Care

STRENGTHS AND LIMITATIONS OF THIS STYDY

- The study utilizes a comprehensive and high-quality dataset from the MIMIC-IV database, offering detailed clinical and demographic data on ICU admissions.
- The use of robust statistical methods, including propensity score weighting based on generalized boosted models, sensitivity analysis, and subgroup analysis, ensures comprehensive covariate balance and enhances the validity and reliability of the results.
- Restricted cubic spline analysis was used to explore non-linear associations, offering a more nuanced understanding of ED to ICUs time effects.
- Lack of data on specific ED interventions may introduce information bias, as we could not assess their potential impact on the outcomes.
- Findings are based on data from a single healthcare system in the United States, which may limit the generalizability of results to other international settings.

Introduction

The emergency department (ED) often serves as the primary entry point for critically ill patients requiring immediate care. In recent years, there has been a marked increase in the capacity of ED clinicians to manage these patients over extended periods [1]. Following initial stabilization in the ED, patients are typically transported to appropriate departments for further diagnostic tests or therapeutic procedures. Most critically ill patients will be transported to specialized intensive care units (ICUs), which provide the highest level of monitoring and care [2]. During the last two decades, escalating acuity and volume of ED visits have resulted in greater demand for critical care services in the ED and ICUs [3]. ED crowding has become an international issue and has long been recognized as posing significant patient safety threats [4]. The impact of ED crowding and ED length of stay (ED-LOS) on inpatient mortality remains contentious and appears to vary by region [5,

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6].

ED-LOS is a critical ED performance indicator and is closely related to crowding, quality of care and patient outcomes [7]. However, the relationship between ED-LOS and in-hospital mortality, particularly for patients admitted directly to ICUs from the ED, remains a subject of debate. Groenland et al. [8] reported that a longer ED to ICUs time is associated with increased hospital mortality in the most severely ill patients. Conversely, Wessman et al. [9] found that patients with the highest triage priority had a decreasing mortality risk with increasing quintiles of ED-LOS for 7-day mortality, and the two highest quintile of ED-LOS had lower 30-day mortality. Saukkonen et al. [10] observed that the length of ED stay was not associated with the outcome of critically ill medical patients.

In light of these conflicting findings and the need for a large cohort to discern subtle outcome differences, our study leverages the extensive Medical Information Mart for Intensive Care IV (MIMIC-IV) database to examine the association between ED to ICUs time and mortality among critically ill patients. This study offers a unique perspective on the potential benefits of extended ED time, addressing gaps in the current understanding of optimal ED management duration before ICU transfer.

Methods

Study design

This retrospective observational study used data from the MIMIC-IV (version 2.2) database, which contains high-quality information on ICU admissions at Beth Israel Deaconess Medical Center from 2008 to 2019. Access to MIMIC-IV is limited to certified individuals; one author (JWQ) with access permission (Record ID 60311526) extracted the data. As all patient records in MIMIC-IV were fully de-identified, the institutional review board at Beth Israel Deaconess Medical Center waived the need for individual patient consent.

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Cohort selection

We initially identified patients with recorded ICU admissions in the MIMIC-IV 2.2 database. From this population, we selected those admitted to the ICU via the emergency department. Patients were excluded if they lacked recorded ED registration time (***edregtime***) or if there was a time difference exceeding six hours between hospital admission (***admittime***) and ICU admission (***intime***), as these cases were not directly transferred from the ED to the ICUs and may have been admitted to a general ward before ICU transfer. For patients with multiple hospital admissions, only the first hospitalization with an ICU stay was analyzed. After applying these criteria, the final study cohort was divided into four quartiles based on the ED to ICUs time.

Data extraction and definitions

All baseline characteristics were extracted within the first 24h of ICU admission from the MIMIC-IV database, including age, gender, ethnicity, admission type, first care unit, chronic conditions, severity scores (Charlson Comorbidity Index (CCI) score, Acute physiology score III (APSIII)), and intervention during ICU. The ED to ICUs time was defined as the interval between *edregtime* (the time of registration in the ED) and *intime* (the time of transfer into the ICU) recorded in the MIMIC-IV database. *Admittime* provides the date and time the patient was admitted to the hospital. Data extraction was performed using pgAdmin4 PostgreSQL 16.1.

Primary outcomes and secondary outcomes

The primary outcome of this study was in hospital all-cause mortality. Of these, hospital death included ICU death and general ward inpatient death. The secondary outcomes included 28-day all-cause mortality, LOS in ICU and LOS in hospital. Patient mortality information for discharged patients was accessed from the US Social Security Death Index.

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Statistical analysis

Our study divided patients into four quartiles based on their ED to ICUs admission times. We assessed the distribution of continuous data using the Kolmogorov-Smirnov test and presented them as means with standard deviations or medians with interquartile ranges. Categorical data were shown in counts and percentages. The Mann-Whitney U test and the Pearson chi-square or Fisher's exact tests were used for comparisons.

We used logistic regression models to estimate odds ratios (ORs) and 95% confidence intervals (CIs) for hospital and ICU mortality. Analyses were conducted using three models: an unadjusted model, a model adjusted for illness severity alone, and a fully adjusted model addressing potential multicollinearity. After fitting the logistic models, pairwise comparisons of the four quartiles were performed, and we applied Tukey's method to adjust for multiple comparisons. The fully adjusted model included a comprehensive set of covariates to assess the relationship between ED to ICUs time and in-hospital mortality, while controlling for key confounders. Age and sex were included to account for demographic differences, while the APSIII and CCI scores provided a detailed assessment of acute illness severity and chronic comorbidity burden, respectively. Additional covariates included diabetes, congestive heart failure, renal disease, and malignancy, given their known impact on critically ill patients' outcomes. Adjustments were also made for invasive mechanical ventilation, dialysis, and vasopressor use, as these reflect the severity of respiratory failure, renal impairment, and hemodynamic instability. This selection of covariates aimed to minimize residual confounding and precise effect estimation of ED to ICUs time on in-hospital mortality. Secondary outcomes, including 28-day mortality, were analyzed using similar models and Kaplan-Meier survival analysis, with additional tests for non-linear relationships. Restricted cubic spline (RCS) models were applied to further explore the non-linear association between ED to ICUs time and both in-hospital and ICU mortality.

To examine potential effect modification, we conducted subgroup analyses based on age, gender, APSIII

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score, CCI score, and the presence or absence of key comorbidities. Interaction terms between ED to ICUs time and each subgroup variable were incorporated into the logistic regression model to assess whether the association between ED to ICUs time and mortality varied across subgroups. Additionally, we performed stratified analyses, fitting separate logistic regression models within each subgroup to estimate the adjusted ORs and 95% CIs. The significance of effect modification was determined using the p-value for interaction from the fully adjusted model.

Generalized Boosted Models (GBM) were utilized to estimate generalized propensity scores (GPS) to achieve covariate balance among the four quartiles of ED to ICU admission times (Q1, Q2, Q3, Q4). The GBM model was implemented using the *twang* R package with the following hyperparameters: 10,000 boosting iterations, an interaction depth of 3, and a shrinkage parameter of 0.01. The GBM model estimated the average treatment effect (ATE) for all four quartiles, ensuring comparability between groups. Covariates included age, gender, race, first care unit, CCI scores, APSIII scores, diabetes, congestive heart failure, renal disease, malignancy, invasive mechanical ventilation, dialysis, and vasopressor use. The balancing process was guided by minimizing the maximum standardized effect size across all covariates. Balance diagnostics were conducted using standardized mean differences (SMD) and Kolmogorov-Smirnov statistics to evaluate the effectiveness of the model in achieving covariate balance before and after adjustment. Sample weights from the GBM model were used to estimate effective sample sizes (ESS) for quartiles.

A sensitivity analysis was also performed by reintroducing patients with a time difference greater than six hours between hospital admission and ICU admission. This analysis aimed to assess the robustness of the study findings by evaluating whether the association between ED to ICUs time and in-hospital mortality persisted when including patients potentially admitted to non-ICU wards before ICU transfer.

We performed statistical analyses with R software (version 4.3.3) and considered two-sided P values below 0.05 as significant.

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Result

Baseline characteristics

From 50,920 unique patients with recorded ICU admissions in the MIMIC-IV database, we identified 15,246 adults admitted directly from the ED to the ICU during their first hospitalization. After excluding those without recorded ED registration times and those with a hospital-to-ICUs admission interval exceeding 6 hours (n=2,432), the final analysis cohort comprised 12,703 patients. The selection process is detailed in **Figure 1**. **Supplementary Table 1** summarizes characteristics of 12,703 study participants categorized by ED to ICUs time quartiles. The median patient age was 64 years, and 56.51% were male. There were 1,185 ICU deaths, 1,988 in-hospital deaths, and 2,490 deaths within 28 days. The median ED to ICUs time was 3.98 hours. Among the chronic conditions in the cohort, diabetes was present in 2,938 patients (23.13%), congestive heart failure in 2,401 (18.90%), renal disease in 1,753 (13.80%), and malignant cancer in 1,090 (8.58%). The median age increased from 61 years in Q1 to 66 years in Q3 and Q4, with an overall median of 64 years ($p < 0.01$). Male patients comprised 56.51% of the study population, with their proportion decreasing from 61.86% in Q1 to 51.43% in Q4 ($p < 0.01$). The CCI score exhibited a statistically significant difference across quartiles, with a median (IQR) ranging from 4 (2, 6) in Q1 to 5 (3, 7) in Q4 ($p < 0.01$). Similarly, the APSIII score varied significantly among groups, with a median (IQR) decreasing from 40 (30, 60) in Q1 to 39 (30, 54) in Q4 ($p < 0.01$).

In terms of outcomes, hospital mortality occurred in 1,988 patients (15.65%), with rates ranging from 17.60% in Q1 to 12.21% in Q4 ($p < 0.01$). ICU mortality was observed in 1,185 patients (9.33%), with the highest rate in Q1 (11.64%) and the lowest in Q4 (6.37%) ($p < 0.01$). The 28-day mortality rate was 19.60% overall, varying from 20.64% in Q1 to 16.85% in Q4 ($p < 0.01$). The median hospital LOS was 5.0 days (IQR: 2.8-9.4) across all patients, with a slight difference among quartiles, from 5.0 days in Q1 to 5.3 days in Q4 (p

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= 0.05). The median ICU LOS was 1.80 days, ranging from 1.96 days in Q1 to 1.70 days in Q4 ($p < 0.01$), indicating significant differences between groups.

Association of the ED to ICUs time with all-cause mortality

Following the application of a GBM and propensity score weighting, we achieved significant enhancement in baseline comparability across the ED to ICUs time quartiles, as shown in **Supplementary Table 2**, where demographic characteristics, first care unit distribution, severity scores, and other covariates demonstrated improved balance across groups. As demonstrated in **Supplementary Fig. 1** and **Supplementary Table 3**, the maximum standardized mean difference was reduced from 0.56 before weighting to 0.04 after weighting, and the maximum Kolmogorov-Smirnov statistic improved from 0.20 to 0.02, with the minimum Kolmogorov-Smirnov p-value increasing from < 0.01 to 0.22. These improvements in covariate balance were achieved through iterative optimization, as illustrated in **Supplementary Fig. 2**. Effective sample sizes remained robust post-weighting: for instance, Q1 retained an effective sample size of 2651.1, and Q4 achieved 2475.5, ensuring sufficient analytical power for subsequent modeling (**Supplementary Table 4**).

Under these more balanced conditions, three weighted logistic regression models were constructed to assess the association between ED to ICUs time and all-cause mortality (**Table 1**). In Model 1, which incorporated weighting without additional adjustments, patients in the longest ED to ICUs time quartile (Q4) already exhibited significantly lower odds of in-hospital mortality compared to those in Q1 (OR 0.80, 95% CI: 0.74-0.86, $p < 0.01$) (**Fig. 2**). Further adjustment for illness severity (Model 2) and comprehensive adjustment for severity scores, comorbidities, demographics, first care unit type, and ICU-level interventions (Model 3) reinforced this inverse relationship. In the fully adjusted Model 3, the odds of in-hospital mortality in Q4 relative to Q1 were 0.75 (95% CI: 0.69-0.82, $p < 0.01$), while ICU mortality odds were 0.63 (95% CI: 0.61-0.74, $p < 0.01$). Variance inflation factor analyses demonstrated no significant collinearity issues within

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Model 3, supporting the stability and interpretability of the final model (**Supplementary Fig. 3**).

Building on these findings, restricted cubic spline analyses revealed a gradual, nonlinear decline in both in-hospital and ICU mortality risk as ED to ICUs time increased, with a more pronounced reduction observed after 5.65 hours (**Supplementary Fig. 4**).

The 28-day survival was further evaluated as a secondary outcome using Kaplan-Meier estimates (**Fig. 3**). Throughout the follow-up period, Q4 consistently exhibited a higher cumulative survival rate compared to Q1, as well as Q2 and Q3. While the survival differences between Q1 and the intermediate quartiles were not statistically significant (P = 0.60 and P = 0.95, respectively), Q4 demonstrated a significantly improved 28-day cumulative survival compared to Q1 (P = 0.03).

Table 1 Odds Ratios (ORs) for All-Cause Mortality in-hospital and ICU mortality from Logistic Regression after Generalized Boosted Models

Categories	Model 1		Model 2		Model 3	
	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value
Hospital death						
Quartilea[a]						
Q1	Ref.		Ref.		Ref.	
Q2	1.01 (0.93, 1.08)	0.86	1.01 (0.93, 1.09)	0.78	1.01 (0.94, 1.09)	0.75
Q3	0.98 (0.92, 1.05)	0.66	0.96 (0.89, 1.04)	0.33	0.95 (0.88, 1.03)	0.23
Q4	0.80 (0.74, 0.86)	< 0.01	0.76 (0.70, 0.82)	< 0.01	0.75 (0.69, 0.82)	< 0.01
ICU death						
Quartilea[a]						
Q1	Ref.		Ref.		Ref.	
Q2	0.89 (0.81, 0.97)	< 0.01	0.87 (0.79, 0.95)	< 0.01	0.82 (0.79, 0.96)	< 0.01
Q3	0.95 (0.87, 1.03)	0.02	0.92 (0.83, 1.01)	0.08	0.86 (0.83, 0.99)	0.04
Q4	0.71 (0.65, 0.78)	< 0.01	0.67 (0.60, 0.74)	< 0.01	0.63 (0.61, 0.74)	< 0.01

Model 1: unadjusted; Model 2: adjusted for APSIII score, CCI score; Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, invasive mechanical ventilation, analysis and vasopressors.

[a] ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

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Relationship between ED to ICUs time and length of stay

Supplementary Table 5 shows the relationship between ED to ICUs time and both hospital and ICU LOS. For hospital LOS, the unadjusted model indicated a significant negative association with ED to ICUs time (Coef. = -0.099, P = 0.002); however, this association lost significance after adjustment in Models 2 and 3 ($p > 0.05$). In contrast, for ICU LOS, a longer ED to ICUs time was consistently associated with a shorter duration of stay across all models, with coefficients of -0.157 ($p < 0.01$) in Model 1, -0.097 ($p < 0.01$) in Model 2, and -0.069 ($p < 0.01$) in Model 3. These results indicate a sustained inverse relationship between ED to ICUs time and ICU LOS, even after adjustments.

Subgroup Analysis

The protective association between longer ED to ICUs time and reduced in-hospital mortality risk was broadly consistent across most examined subgroups in the subgroup analyses hospital mortality, with ORs remaining below 1 (**Fig. 4A**). However, the magnitude of this protective effect differed by specific patient characteristics. Significant interactions were observed with gender ($p = 0.03$), CCI score ($p < 0.01$), and malignant cancer ($p = 0.01$), indicating that the strength of the inverse association increased among patients with higher comorbidity burdens or malignancies, and varied by gender. Similar trends were noted for ICU mortality, with ORs generally showing a reduction in risk with prolonged ED stays across most subgroups (**Fig. 4B**). Although the interaction with gender did not reach significance ($p = 0.06$), CCI score ($p < 0.01$) and congestive heart failure ($p = 0.03$) emerged as significant effect modifiers, suggesting potential variability in the association depending on the severity of comorbidities. These findings underscore the robustness of the protective association between longer ED to ICUs time and improved survival, while highlighting certain patient characteristics that may influence this relationship.

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Sensitivity Analysis

In the sensitivity analysis, patients initially excluded due to a time gap of more than 6 hours between hospital admission (*admittime*) and ICU admission (*intime*) were reintroduced to evaluate the impact of this criterion on the study results. **Supplementary Table 6** presents the baseline characteristics of this expanded cohort. After applying GBM and propensity score weighting, baseline balance remained markedly improved in this expanded cohort (**Supplementary Table 7**). As shown in **Supplementary Table 8**, the maximum standardized mean difference decreased from 0.64 before weighting to 0.06 after weighting, and the maximum Kolmogorov-Smirnov statistic declined from 0.22 to 0.03, with the minimum Kolmogorov-Smirnov p-value increasing from < 0.01 to 0.07. Correspondingly, the effective sample sizes remained robust after adjustment, as detailed in **Supplementary Table 9**. **Supplementary Fig. 5** further illustrates the substantial reduction in standardized mean differences following weighting. The odds ratios from the logistic regression models (**Supplementary Table 10** and **Supplementary Fig. 6**) continued to indicate lower odds of in-hospital and ICU mortality for patients in Q4 compared to those in the other groups. Moreover, Kaplan-Meier survival curves (**Supplementary Fig. 7**) showed the highest 28-day survival rates for Q4, with Q1 vs. Q4 comparisons remaining significant ($p < 0.01$). These findings suggest that the exclusion of patients with longer pre-ICU stays did not materially affect the relationship between extended ED stays and better survival outcomes, thereby reinforcing the robustness of the results.

Discussion

Our study, employing GBM-based propensity score weighting to achieve enhanced baseline comparability, provides a more nuanced and reliable insight into the relationship between ED to ICUs time and mortality outcomes. By leveraging GBM's capability to optimize balance across groups through iterative tree-based

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modeling, as detailed by McCaffrey et al. [11], we ensured robust minimization of baseline confounding. After establishing near-equivalence of patient characteristics across quartiles, we unexpectedly found that longer ED to ICUs time (5.65-34.55 hours) were consistently associated with lower in-hospital, ICU, and 28-day mortality. Notably, this inverse relationship persisted through sequential adjustments for severity of illness, comorbidity burden, demographic factors, and critical interventions. Subgroup analysis further demonstrated that the protective association remained significant across most patient groups, but did not achieve statistical significance among those with malignant cancer. This lack of significance may be attributed to the fact that patients with malignancy often present with complex conditions, where acute issues are typically addressed and stabilized in the ED. In more advanced cases, treatment focuses on symptom relief over intensive rescue, as further ICU intervention may not improve prognosis [12, 13]. Altogether, by employing GBM-based weighting to minimize confounding and achieve comprehensive covariate balance, our findings challenge the conventional assumption that prolonged ED stays necessarily worsen outcomes [8, 14, 15], instead suggesting that, under certain conditions, extended ED management may confer survival benefits.

A large retrospective cohort study with data derived from MIMIC-III showed that every delayed hour in the ED increased the in-hospital mortality, and in particular patients with ≥ 6h of transfer from ED to the ICUs had a significantly higher mortality rate than those with < 6h [14]. However, this analysis may have included patients initially transferred from the ED to general wards before ICU admission, which may not fully reflect outcomes for those transferred directly from the ED to the ICU. Other studies have similarly shown that a longer hospital stay before ICU admission was correlated with mortality, because patients with longer hospital stays may have more comorbidities [16, 17]. We limited the time between recorded hospital admission and ICU admission to 6 hours to identify patients transferred directly from the ED to the ICUs. However, this 6-hour threshold may introduce selection bias, as some patients initially transferred from the ED to non-ICU wards may have stayed there for less than six hours before ICU transfer. To address this, we conducted a

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sensitivity analysis reintroducing patients with hospital to ICUs times over six hours, confirming the robustness of our findings.

The ED to ICUs time is multifactorially influenced by multiple factors beyond patient acuity, including ED crowding [18], triage process [19], availability of ICU beds [20, 21], all of which may shape clinical outcome. In our study, patients with longer ED to ICU times had higher CCI scores (Q1: 4 [2, 6] vs. Q4: 5 [3, 7]), indicating a greater burden of chronic comorbidities [22], whereas those with shorter ED to ICU times had higher APSIII scores (Q1: 40 [30, 60] vs. Q4: 39 [30, 54]), reflecting more severe acute illness [23, 24]. This pattern suggests that patients with multiple chronic conditions but without acute decompensation may have undergone more extensive evaluation and management in the ED before ICU transfer. In contrast, those with critical, rapidly deteriorating conditions were prioritized for immediate ICU admission. These findings underscore the complex interplay between clinical urgency, patient characteristics, and resource allocation in determining ED to ICU transfer times. Trzeciak et al. [25] noted that ED overcrowding, staff may struggle to provide the necessary care for critically ill patients, a limitation associated with increased mortality when patients receive care in settings unable to fully support their critical needs [26]. To address these challenges, EDs worldwide are adopting measures to enhance timeliness and throughput, including bedside registration, self-check-in kiosks, immediate bedding, fast-track systems, and physician-in-triage protocols [27]. Abdulai et al. [28] examined trends in ED throughput in the United States from 2006 to 2016 and found after implementing a series of measures, improved ED throughput allowed healthcare providers more time to assess and manage critically ill patients who are stranded in the emergency department, facilitating appropriate treatment decisions and potentially improving outcomes. Additionally, clinical decision support systems based on intelligent technologies, as noted by Fernandes et al. [29] , have been shown to enhance healthcare professionals' decision-making. These systems help ED staff identify, diagnose, and treat critically ill patients, ensuring appropriate ICU transfers while stable patients remain in the ED. This underscores that extended ED

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stays are not delays but a rational use of resources to improve overall outcomes and reduce mortality. Availability of ICU beds is a crucial factor influencing ED to ICUs time and patient outcomes [30]. A previous study found that ICU admission decisions for critically ill ED patients are affected by medical ICU bed availability [31]. To address these limitations, hospitals are adopting innovative strategies to expand ICU capacity. For instance, Mitra et al. [32] found that establishing high-acuity units, which offer close monitoring and acute care comparable to the ICU but with fewer resources, was associated with reduced in-hospital mortality for patients requiring intensive care. Establishing a critical care resuscitation unit can handle more critically ill surgical patients and speed up the transfer of those with time-sensitive conditions to specialized centers. Furthermore, enhancing ICU capacity and improving its operational efficiency can reduce the time needed for patient admission to the ICU.

This association between longer ED to ICUs time and lower hospital mortality can be explained by several possible mechanisms. One potential explanation is that extended waiting time in the ED provides a crucial period for assessing patient conditions and implementing necessary stabilization measures before ICU transfer. Extended waiting time in the ED may serve as a critical period to stabilize patients by initiating critical care interventions, such as goal-directed therapy, which stabilize patients' clinical status and reduce the risk of deterioration before ICU transfer [33]. Alternatively, extended ED to ICU transfer times may help critically ill patients avoid the significant risks associated with premature intrahospital transport (IHT). Although often unavoidable, IHT exposes patients to a mobile environment, increasing the likelihood of adverse events, including equipment failure, device dislodgement, inadequate monitoring, and cardiovascular or respiratory instability [34, 35]. These adverse events, reported in up to 79.8% of ICU-bound IHT cases in a study of 441 patients [36], can exacerbate clinical deterioration and elevate mortality risk. Premature transport of critically ill patients from the ED, while potentially reducing ED to ICUs time, often limits essential preparation time, increasing the risk of complications [37]. By allowing patients to avoid the risks of

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premature transfer and receive appropriate critical care interventions in the ED, extended ED to ICUs time may provide a vital window for stabilization and improved outcomes. Thus, extended ED to ICUs time may represent opportunities for essential stabilization rather than mere delays, supporting the notion that early critical care in the ED can be pivotal in reducing mortality. Another potential explanation is the significant advancements in the ability of emergency department clinicians to manage critically ill patients over extended periods, combined with the development of advanced emergency medical technologies [38]. For example, the study by Kadar et al. [39] demonstrated that patients receiving ICU telemedicine support in the ED had lower mortality rates and reduced ICU resource utilization compared to those receiving standard ED care. This benefit was attributed to telemedicine teams monitoring patient stability over time, enabling safe non-ICU management and optimized resource use.

Our study highlights that managing critically ill patients in the emergency department involves more than just minimizing ED wait times. Considering the patient's clinical condition, ED circumstances, ED capabilities, transport risks, and ICU bed availability to determine the best timing for ICU admission is essential. A key strength of our research is the use of the comprehensive MIMIC-IV database, which spans from 2008 to 2019. This dataset, the most recent before the onset of COVID-19, allows for an accurate analysis of the association between ED to ICUs time and patient outcomes without the confounding effects of pandemic [40]. The use of GBM-based propensity score weighting minimized baseline confounding and ensured covariate balance, enabling a precise analysis of the unexpected finding that longer ED to ICUs time were associated with lower mortality rates.

Limitations

This study has several limitations. First, as a retrospective analysis drawn from the MIMIC database, these findings cannot establish causality and may not be generalizable to other countries. Second, we excluded

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patients who died in the ED to focus on those successfully transferred to the ICUs. Although necessary, this may introduce selection bias by underrepresenting the most critically ill and potentially underestimating mortality associated with shorter ED to ICUs time. Third, the MIMIC database does not record critical interventions performed in the ED and records surgeries with daily resolution rather than hourly granularity. This limited temporal detail prevents determining whether interventions preceded ICU admission, complicating the interpretation of ED to ICUs time while possibly leading to an underestimation of pre-admission mortality. Lastly, the retrospective design prevented a detailed examination of the components influencing ED to ICUs time, including orientation decisions, clinical assessments, diagnostic workups, boarding durations. Future research should employ prospective, multicenter designs with more comprehensive and temporally precise data collection to better elucidate the impact of ED to ICUs time on patient outcomes.

Conclusions

Patients transferred from the ED to the ICUs within 5.65-34.55 hours experienced lower in-hospital, ICU, and 28-day mortality rates compared to those transferred earlier. These findings suggest that appropriately extended ED stays may facilitate critical stabilization, improving outcomes and emphasizing the need for further research to optimize transfer protocols and ED management strategies.

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Author contributions

JWQ, YNY and ZMS designed the study, conducted the formal analysis, and drafted the manuscript. JWQ extracted the data from the MIMIC-IV database. YNY, ZMS and QXL reviewed and analyzed the data. KSZ, LYZ and WZZ guided the manuscript review and editing. XFJ and MQC instructed the conceptualization. MQC is the guarantor of this study. All authors read and approved the final manuscript.

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Data availability statement

Data generated and analyzed can be available from the corresponding author on reasonable request.

Declarations

Ethics approval

Given that this study analyzed an anonymous public database with prior Institutional Review Board approval, no ethical review was necessary.

Patient consent for publication

Not required.

Patient and public involvement

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Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Competing interests

The authors declare that they have no competing interests.

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Figure Legends

Fig. 1 Flow chart of patients selection for analytic

MIMIC-IV Medical Information Mart for Intensive Care-IV, ICU intensive care unit

Fig. 2 Odds Ratios for all-cause in-hospital and ICU mortality after GBM-based propensity score weighting

ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

A. Association between ED to ICUs time and in-hospital mortality, unadjusted (weighted).

B. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score and CCI score (weighted).

C. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

D. Association between ED to ICUs time and ICU mortality, unadjusted (weighted).

E. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score and CCI score (weighted).

F. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

Fig. 3 Kaplan-Meier survival analysis curves for 28-day all-cause mortality

Kaplan-Meier survival curves for patients stratified into four quartiles based on ED to ICUs time (hours): Q1

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(≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

Fig. 4 Forest plots of odd ratios for the primary outcome in different subgroups

A. Association between ED to ICUs time and in-hospital mortality; B. Association between ED to ICUs time and ICU mortality. ORs with 95% CIs are displayed for each subgroup.

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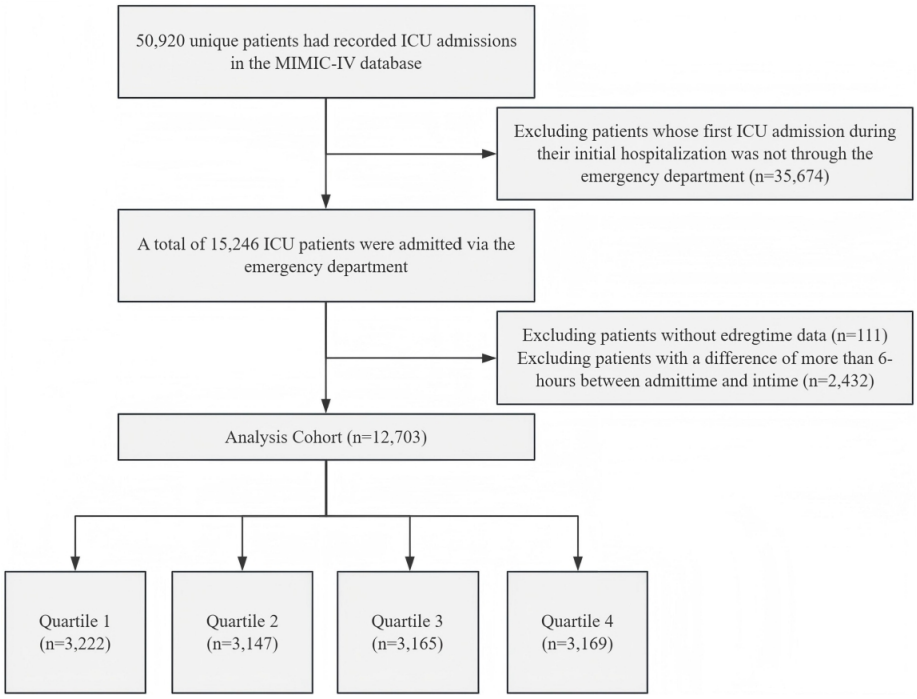


Fig. 1 Flow chart of patients selection for analytic
MIMIC-IV Medical Information Mart for Intensive Care-IV, ICU intensive care unit

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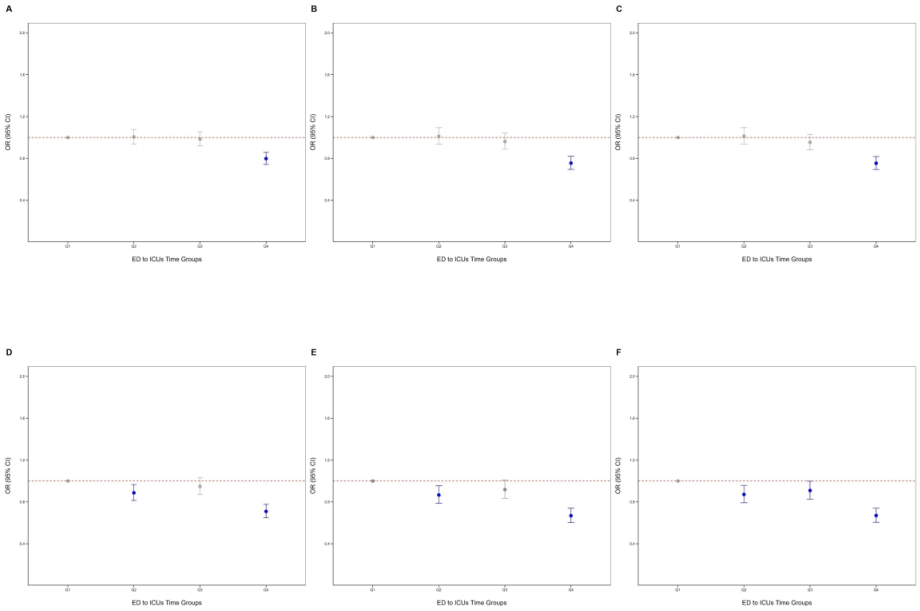


Fig. 2 Odds Ratios for all-cause in-hospital and ICU mortality after GBM-based propensity score weighting
ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).
A. Association between ED to ICUs time and in-hospital mortality, unadjusted (weighted).
B. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score and CCI score (weighted).
C. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).
D. Association between ED to ICUs time and ICU mortality, unadjusted (weighted).
E. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score and CCI score (weighted).
F. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

338x254mm (300 x 300 DPI)

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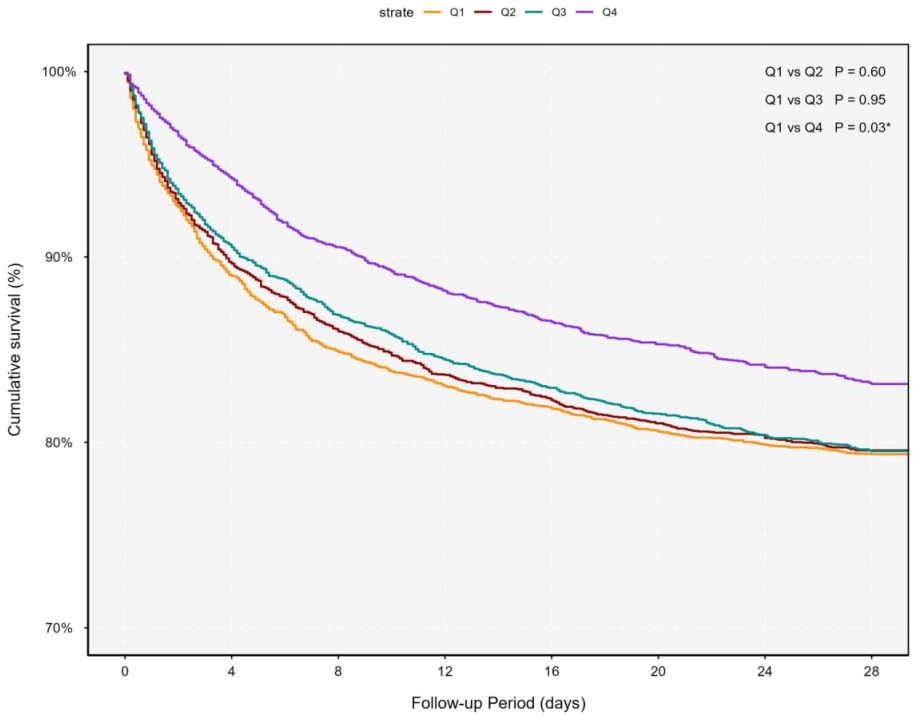


Fig. 3 Kaplan-Meier survival analysis curves for 28-day all-cause mortality
Kaplan-Meier survival curves for patients stratified into four quartiles based on ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

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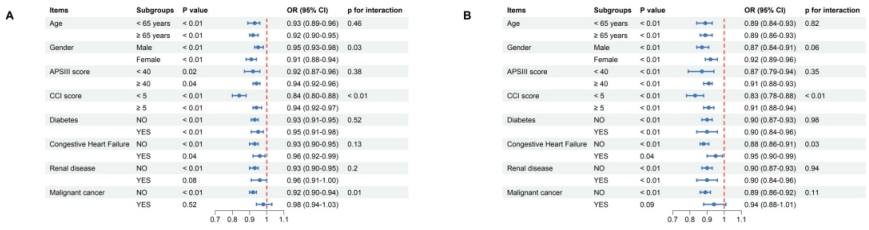


Fig. 4 Forest plots of odd ratios for the primary outcome in different subgroups
A. Association between ED to ICUs time and in-hospital mortality; B. Association between ED to ICUs time and ICU mortality. ORs with 95% CIs are displayed for each subgroup.

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Association between emergency department to intensive care units time and in-hospital mortality: an analysis of the MIMIC-IV database

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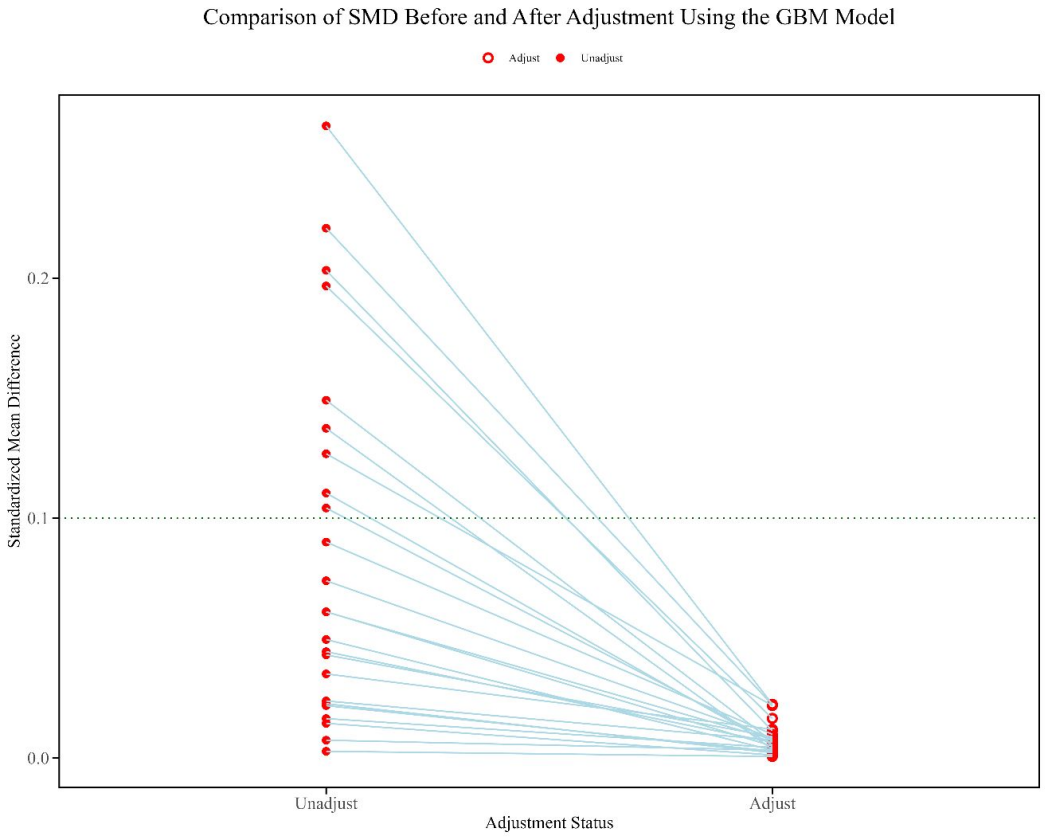
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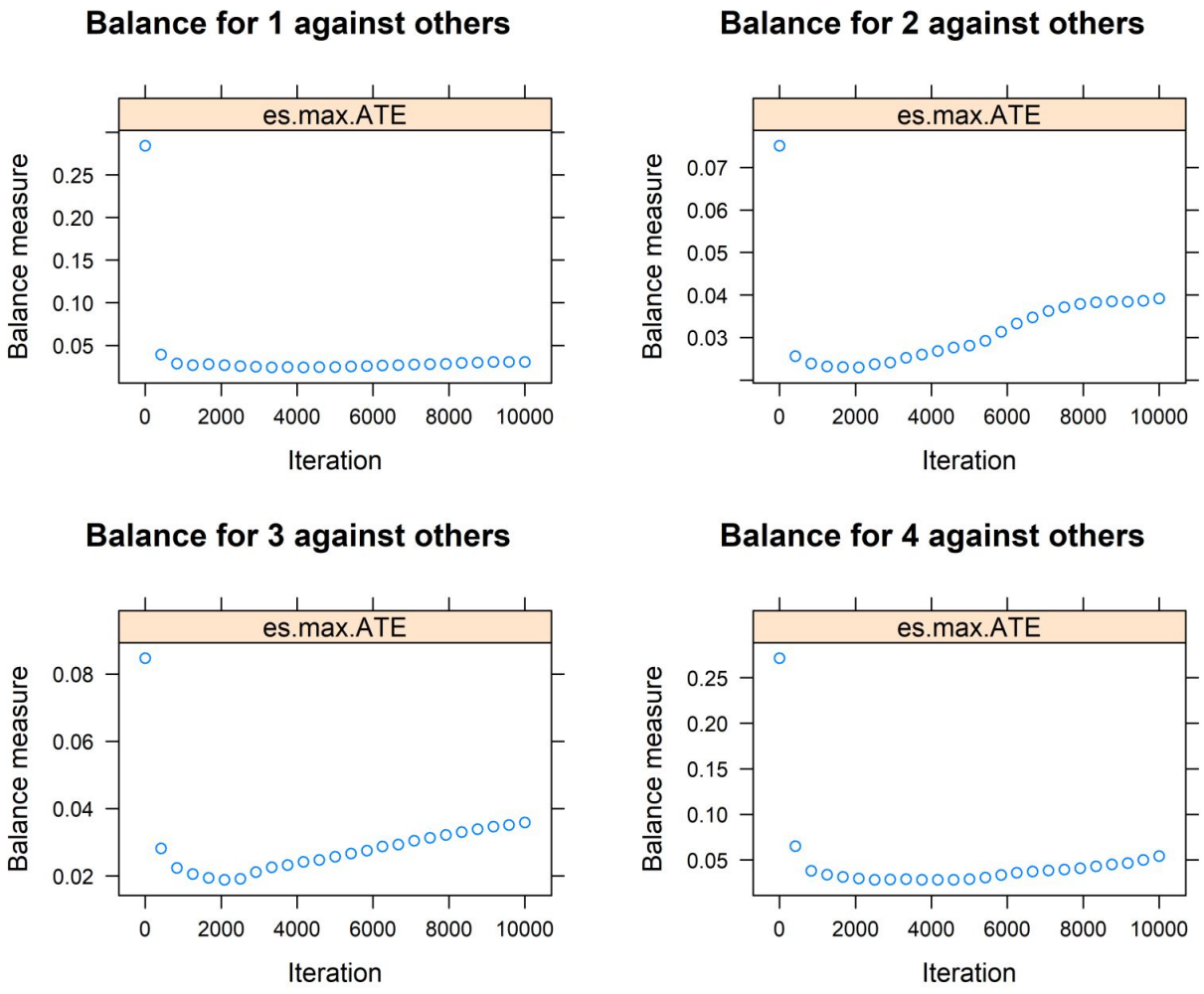
Supplementary Fig. 1 Comparison of Standardized Mean Differences Before and After GBM-Based Propensity Score Weighting

GBM: Generalized Boosted Model, SMD: Standardized Mean Differences. Red circles represent the unadjusted SMD values, and open circles represent the adjusted SMD values after weighting. The horizontal dashed line at SMD = 0.1 indicates the threshold for acceptable covariate balance.

Covariates include age, gender, race, first care unit, CCI scores, APSIII scores, diabetes, congestive heart failure, renal disease, malignancy, invasive mechanical ventilation, dialysis, and vasopressor use.

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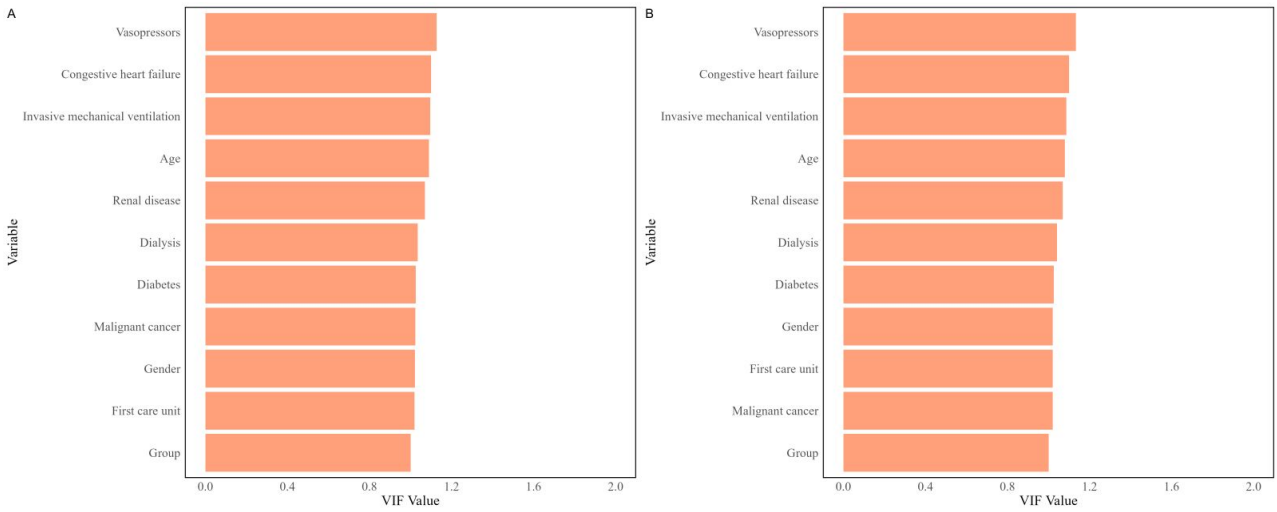


Supplementary Fig. 2 Balance Measures Across Iterations for Propensity Score Weighting Using GBM

ATE: Average Treatment Effect. The figure shows the balance measure (es.max.ATE) across iterations during propensity score weighting using Generalized Boosted Model for each comparison group: Q1 vs others, Q2 vs others, Q3 vs others, and Q4 vs others. Each panel represents the balance assessment for a specific group, with the x-axis indicating the number of iterations and the y-axis showing the balance measure.

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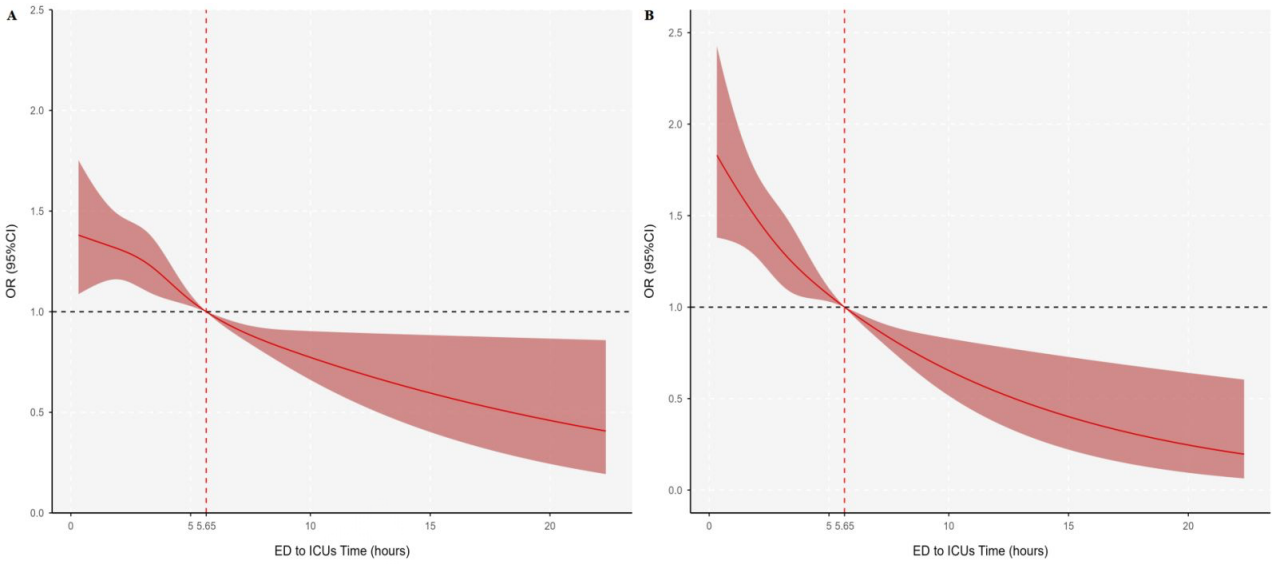
Supplementary Fig. 3 Variance Inflation Factor Analysis for GBM-Weighted Logistic Models: Hospital and ICU Mortality

A. Model analyzing predictors of hospital mortality. B. Model analyzing predictors of ICU mortality. Variables include vasopressors, congestive heart failure, invasive mechanical ventilation, age, renal disease, dialysis, diabetes, malignant cancer, gender, first care unit, and group. All variables have VIF values less than 5, indicating no significant multicollinearity.

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Supplementary Fig. 4 Restricted cubic spline regression analysis of ED to ICUs time with in hospital all-cause mortality

Red heavy central lines represent the estimated adjusted hazard ratios, with shaded ribbons denoting 95% confidence intervals. ED to ICUs time 5.65hr was selected as the reference level represented by the vertical dotted lines. The horizontal dotted lines represent the hazard ratio of 1.0. HR hazard ratio, CI confidence interval

A. Restricted cubic spline for in-hospital death. B. Restricted cubic spline for ICU death.

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Supplementary Table

Supplementary Table 1 Baseline characteristics of ICU patients grouped according to ED to ICUs time quartiles

Categories	Overall (N=12703)	Q1 (N=3222)	Q2 (N=3147)	Q3 (N=3165)	Q4 (N=3169)	P-value
ED to ICUs time, hr, median (IQR)	3.98 (2.83, 5.65)	2.16 (1.66, 2.53)	3.41 (3.12, 3.68)	4.70 (4.2[illegible] 5.12)	7.22 (6.33, 8.72)	< 0.01
Demographic						
Age, years, median (IQR)	64 (49, 79)	61 (46, 77)	65 (49, 79)	66 (51, [illegible])	66 (51, 80)	< 0.01
Male, n (%)	7178 (56.51)	1993 (61.86)	1771 (56.28)	1784 (56.[illegible]7)	1630 (51.43)	< 0.01
Ethnicity, n (%)						< 0.01
White	8149 (64.15)	1799 (55.83)	1986 (63.11)	2122 (67.[illegible]5)	2242 (70.75)	
Black	1103 (8.68)	249 (7.91)	249 (7.91)	277 (8.[illegible]7)	365 (11.52)	
Asian	375 (2.95)	87 (2.76)	87 (2.76)	105 (3.[illegible]3)	117 (3.69)	
Hispanic/Latino	463 (3.64)	112 (3.48)	115 (3.65)	117 (3.[illegible]7)	119 (3.76)	
Other	636 (5.01)	195 (6.05)	146 (4.64)	149 (4.[illegible]7)	146 (4.61)	
Unknown	1977 (15.56)	838 (26.01)	564 (17.92)	395 (12.[illegible]5)	180 (5.68)	
Admission type, n (%)						< 0.01
EW EMER	11357 (89.40)	2912 (90.38)	2883 (91.61)	2834 (89.[illegible]4)	2728 (86.08)	
OBSERVATION ADMIT	1194 (9.40)	269 (8.35)	236 (7.50)	300 (9.[illegible]8)	389 (12.28)	
EU OBSERVATION	147 (1.16)	40 (1.24)	27 (0.86)	30 (0.9[illegible])	50 (1.58)	
DIRECT OBSERVATION	4 (0.03)	1 (0.03)	1 (0.03)	0 (0.00)	2 (0.06)	
URGENT	1 (0.01)	0 (0.00)	0 (0.00)	1 (0.03)	0 (0.00)	
First care unit, n (%)						< 0.01
MICU	3301 (25.99)	683 (21.20)	798 (25.36)	863 (27.[illegible])	957 (30.20)	
SICU	2569 (20.22)	553 (17.16)	732 (23.26)	712 (22.[illegible])	572 (18.05)	
TSICU	2709 (21.33)	941 (29.21)	707 (22.47)	571 (18.0[illegible])	490 (15.46)	
MICU/SICU	2123 (16.71)	365 (11.33)	447 (14.20)	602 (19.0[illegible])	709 (22.37)	
CCU	1053 (8.29)	417 (12.94)	260 (8.26)	200 (6.32)	176 (5.55)	
CVICU	445 (3.50)	164 (5.09)	93 (2.96)	92 (2.91)	96 (3.03)	

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Neuro SICU	503 (3.96)	99 (3.07)	110 (3.50)	125 (3.8[illegible])	169 (5.33)	
Chronic conditions, n (%)						
Diabetes	2938 (23.13%)	674 (20.92%)	723 (22.97%)	767 (24.[illegible]%)	774 (24.42%)	< 0.01
Congestive heart failure	2401 (18.90%)	567 (17.60%)	605 (19.22%)	596 (18.[illegible]%)	633 (19.97%)	0.10
Renal disease	1753 (13.80%)	375 (11.64%)	388 (12.33%)	485 (15.[illegible]%)	505 (15.94%)	< 0.01
Malignant cancer	1090 (8.58%)	208 (6.46%)	240 (7.63%)	297 (9.[illegible]%)	345 (10.89%)	< 0.01
Severity scores						
CCI, mean (SD)	5 (3, 7)	4 (2, 6)	5 (2, 7)	5 (3, 7)	5 (3, 7)	< 0.01
APSIII, mean (SD)	40 (30, 57)	40 (30, 60)	39 (29, 57)	42 (31, [illegible])	39 (30, 54)	< 0.01
Intervention during ICU, n (%)						
Invasive mechanical ventilation	4196 (33.03%)	1366 (42.4%)	1122 (35.65%)	988 (31.[illegible]%)	720 (22.72%)	< 0.01
Dialysis	235 (1.85%)	73 (2.27%)	48 (1.53%)	60 (1.9[illegible])	54 (1.70%)	0.15
Vasopressors	2425 (19.09%)	734 (22.78%)	562 (17.86%)	600 (18.[illegible]%)	529 (16.69%)	< 0.01
Outcome						
Hospital death, n (%)	1988 (15.65)	567 (17.60)	520 (16.52)	514 (16.2[illegible])	387 (12.21)	< 0.01
ICU death, n (%)	1185 (9.33)	375 (11.64)	297 (9.44)	311 (9.8[illegible])	202 (6.37)	< 0.01
28-day death, n (%)	2490 (19.60)	665 (20.64)	643 (20.43)	648 (20.[illegible])	534 (16.85)	< 0.01
Hosptial-LOS, days, median (IQR)	5.0 (2.8, 9.4)	5.1 (2.7, 9.7)	5.0 (2.8, 9.1)	5.0 (2.8[illegible]3)	5.3 (3.0, 9.3)	0.05
ICU-LOS, median (IQR)	1.80 (0.96, 3.59)	1.96 (1.11, 4.15)	1.80 (0.97, 3.64)	1.78 (0[illegible]3.53)	1.70 (0.91, 2.98)	< 0.01

ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr)

ED emergency department, ICU intensive care unit, IQR interquartile range, *EW EMER* emergency ward emergency, EU OBSERVATION: emergency unit observation, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit, CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, CCI Charlson Comorbidity Index, APSIII acute physiology score III, LOS length of stay

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Supplementary Table 2 Baseline Characteristics of the Study Population After GBM Adjustment

Categories	Q1	Q2	Q3	Q4	*p*-value
Weighted Sample Size	12063.96	12306.15	12259.41	12024.68	
Demographic					
Age, years, median (IQR)	65.00 (50.00, 79.00)	65.00 (49.00, 79.00)	65.00 (50.00, 79.00)	65.00 (50.00, 79.00)	0.88
Male, n (%)	6893 (57.14%)	6946 (56.44%)	6926 (56.49%)	6789 (56.46%)	0.94
Ethnicity, n (%)					1.00
White	7806 (64.71%)	7962 (64.70%)	7973 (65.04%)	7862 (65.39%)	
Black	1002 (8.30%)	1028 (8.35%)	1058 (8.63%)	1036 (8.61%)	
Asian	307 (2.55%)	347 (2.82%)	366 (2.99%)	341 (2.83%)	
Hispanic/Latino	424 (3.52%)	439 (3.57%)	435 (3.55%)	421 (3.50%)	
Other	593 (4.92%)	591 (4.80%)	601 (4.90%)	576 (4.79%)	
Unknown	1931 (16.01%)	1939 (15.76%)	1826 (14.89%)	1788 (14.87%)	
First care unit, n (%)					1.00
MICU	3125 (25.90%)	3190 (25.92%)	3193 (26.05%)	3233 (26.89%)	
SICU	2477 (20.53%)	2532 (20.58%)	2511 (20.48%)	2414 (20.08%)	
TSICU	2589 (21.46%)	2652 (21.55%)	2598 (21.19%)	2559 (21.28%)	
MICU/SICU	1992 (16.51%)	2045 (16.62%)	2079 (16.96%)	2051 (17.06%)	
CCU	1008 (8.35%)	1017 (8.26%)	974 (7.94%)	904 (7.52%)	
CVICU	421 (3.49%)	405 (3.29%)	421 (3.44%)	385 (3.20%)	
Neuro SICU	453 (3.75%)	465 (3.78%)	483 (3.94%)	478 (3.97%)	
Chronic conditions, n (%)					
Diabetes	2696 (22.34%)	2870 (23.32%)	2886 (23.54%)	2820 (23.45%)	0.70
Congestive heart failure	2269 (18.81%)	2359 (19.17%)	2258 (18.42%)	2289 (19.03%)	0.90
Renal disease	1590 (13.18%)	1631 (13.25%)	1708 (13.93%)	1692 (14.07%)	0.69
Malignant cancer	954 (7.91%)	1044 (8.48%)	1049 (8.56%)	1014 (8.43%)	0.81
Severity scores					
CCI, mean (SD)	5 (3, 7)	5 (3, 7)	5 (3, 7)	5 (3, 7)	0.68
APSIII, mean (SD)	40 (30, 57)	40 (30, 56)	40 (30, 56)	40 (30, 56)	0.95
Intervention during ICU, n (%)					

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Invasive mechanical ventilation	4050 (33.57%)	4090 (33.24%)	3976 (32.43%)	3839 (31.92%)	0.56
Dialysis	209 (1.73%)	191 (1.55%)	231 (1.88%)	192 (1.60%)	0.76
Vasopressors	2289 (18.97%)	2239 (18.19%)	2303 (18.78%)	2201 (18.31%)	0.86
Outcome					
Hospital death, n (%)	1911 (15.84%)	1960 (15.92%)	1917 (15.63%)	1571 (13.07%)	< 0.01
ICU death, n (%)	1210 (10.03%)	1108 (9.00%)	1173 (9.56%)	882 (7.33%)	< 0.01
28-day death, n (%)	2338 (19.38%)	2460 (19.99%)	2402 (19.60%)	2082 (17.31%)	0.05
Hosptial-LOS, days, median (IQR)	4.80 (2.70, 8.90)	5.00 (2.80, 9.10)	5.00 (2.80, 9.30)	5.60 (3.10, 9.90)	<0.01
ICU-LOS, median (IQR)	1.87 (1.08, 3.67)	1.80 (0.96, 3.59)	1.78 (0.93, 3.58)	1.79 (0.94, 3.48)	<0.01

ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr)

ED emergency department, ICU intensive care unit, IQR interquartile range, MICU medical intensive care unit, SICU surgical intensive care unit, TSICU trauma surgical intensive care unit, CCU coronary care unit, CVICU cardiac vascular intensive care unit, Neuro SICU neuro surgical intensive care unit, SD standard deviation, CCI Charlson Comorbidity Index, APSIII acute physiology score III, LOS length of stay

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Supplementary Table 3 Covariate Balance Diagnostics Before and After GBM Adjustment

Metrics	Maximum SMD	Maximum KS	Minimum KS *p-value*
Before GBM	0.56	0.20	< 0.01
After GBM	0.04	0.02	0.22

KS: Kolmogorov-Smirnov statistic.

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Supplementary Table 4 Sample Sizes and Effective Sample Sizes Before and After GBM Adjustment

Quartile	Original Sample Size (N)	Effective Sample Size (N)
Q1	3222	2651.1
Q2	3147	3058.7
Q3	3165	3031.7
Q4	3169	2475.5

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Supplementary Table 5 Relationship between ED to ICUs time and length of stay

	Coef.	S.E	t-value	*p*-value
Hospital-LOS[a]				
Model 1	-0.099	0.032	-3.153	0.002
Model 2	-0.032	0.030	-1.070	0.285
Model 3	0.031	0.029	1.069	0.285
ICU-LOS[b]				
Model 1	-0.157	0.017	-9.439	< 0.001
Model 2	-0.097	0.016	-6.287	< 0.001
Model 3	-0.069	0.015	-4.503	< 0.001

Model 1: unadjusted; Model 2: adjusted for APSIII score, Charlson Comorbidity Index; Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, invasive mechanical ventilation, dialysis and vasopressors.

[a] The relationship between ED to ICUs time and length of hospital stay was analyzed in a cohort that survived the hospital stay (n=10701)

[b] The relationship between ED to ICUs time index and length of ICU stay was analyzed in a cohort that survived the ICU stay (n=11518)

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Supplementary Sensitivity Analysis

Supplementary Table 6 Baseline characteristics of ICU patients grouped according to ED to ICUs time quartiles in Sensitivity Analysis

Categories	Overall (N=15135)	Q1 (N=3797)	Q2 (N=3775)	Q3 (N=3780)	Q4 (N=3783)	*p*-value
ED to ICUs time, hr, median (IQR)	4.48 (3.03, 7.54)	2.3 (1.77, 2.7)	3.72 (3.38, 4.08)	5.58 (4.9, 6.4)	22.46 (9.95, 72.78)	< 0.01
Demographic						
Age, years, median (IQR)	65 (50, 79)	62 (46, 77)	65 (50, 80)	66 (52, 80)	67 (53, 80)	< 0.01
Male, n (%)	8489 (56.09%)	2330 (61.36%)	2098 (55.58%)	2072 (54.81%)	1989 (52.58%)	< 0.01
Ethnicity, n (%)						< 0.01
White	9963 (65.83%)	2178 (57.36%)	2417 (64.03%)	2594 (68.62%)	2774 (73.33%)	
Black	1369 (9.05%)	244 (6.43%)	313 (8.29%)	361 (9.55%)	451 (11.92%)	
Asian	455 (3.01%)	76 (2.00%)	112 (2.97%)	136 (3.60%)	131 (3.46%)	
Hispanic/Latino	566 (3.74%)	128 (3.37%)	134 (3.55%)	152 (4.02%)	152 (4.02%)	
Other	734 (4.85%)	216 (5.69%)	182 (4.82%)	180 (4.76%)	156 (4.12%)	
Unknown	2048 (13.53%)	955 (25.15%)	617 (16.34%)	357 (9.44%)	119 (3.15%)	
Admission type, n (%)						< 0.01
EW EMER	13343 (88.16%)	3439 (90.57%)	3445 (91.26%)	3355 (88.76%)	3104 (82.05%)	
OBSERVATION ADMIT	1630 (10.77%)	311 (8.19%)	291 (7.71%)	387 (10.24%)	641 (16.94%)	
EU OBSERVATION	157 (1.04%)	45 (1.19%)	38 (1.01%)	38 (1.01%)	36 (0.95%)	
DIRECT OBSERVATION	4 (0.03%)	2 (0.05%)	0 (0%)	0 (0%)	2 (0.05%)	
URGENT	1 (0.01%)	0 (0%)	1 (0.03%)	0 (0%)	0 (0%)	
First care unit, n (%)						< 0.01
MICU	3815 (25.21%)	807 (21.25%)	990 (26.23%)	1094 (28.94%)	924 (24.43%)	
SICU	3038 (20.07%)	697 (18.36%)	873 (23.13%)	762 (20.16%)	706 (18.66%)	
TSICU	3014 (19.91%)	1078 (28.39%)	802 (21.25%)	645 (17.06%)	489 (12.93%)	
MICU/SICU	2541 (16.79%)	442 (11.64%)	578 (15.31%)	795 (21.03%)	726 (19.19%)	
CCU	1354 (8.95%)	472 (12.43%)	285 (7.55%)	220 (5.82%)	377 (9.97%)	
CVICU	798 (5.27%)	183 (4.82%)	111 (2.94%)	106 (2.80%)	398 (10.52%)	
Neuro SICU	575 (3.80%)	118 (3.11%)	136 (3.60%)	158 (4.18%)	163 (4.31%)	

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Chronic conditions, n (%)						
Diabetes	3577 (23.63%)	798 (21.02%)	887 (23.50%)	938 (24.[illegible]%)	954 (25.22%)	< 0.01
Congestive heart failure	3053 (20.17%)	663 (17.46%)	735 (19.47%)	734 (19.[illegible]%)	921 (24.35%)	< 0.01
Renal disease	2217 (14.65%)	435 (11.46%)	524 (13.88%)	572 (15.[illegible]%)	686 (18.13%)	< 0.01
Malignant cancer	1494 (9.87%)	254 (6.69%)	303 (8.03%)	366 (9.6[illegible]%)	571 (15.09%)	< 0.01
Severity scores						
CCI, mean (SD)	5 (3, 7)	4 (2, 6)	5 (3, 7)	5 (3, 7)	5 (3, 8)	< 0.01
APSIII, mean (SD)	40 (30, 57)	40 (30, 60)	41 (30, 57)	41 (30, [illegible])	40 (30, 56)	0.26
Intervention during ICU, n (%)						
Invasive mechanical ventilation	4927 (32.55%)	1582 (41.66%)	1316 (34.86%)	1038 (27.[illegible]6%)	991 (26.20%)	< 0.01
Dialysis	300 (1.98%)	84 (2.21%)	67 (1.77%)	62 (1.64[illegible])	87 (2.30%)	0.11
Vasopressors	2905 (19.19%)	835 (21.99%)	696 (18.44%)	676 (17.[illegible]%)	698 (18.45%)	< 0.01
Outcome						
Hospital death, n (%)	2367 (15.64%)	661 (17.41%)	641 (16.98%)	535 (14.[illegible]%)	530 (14.01%)	< 0.01
ICU death, n (%)	1374 (9.08%)	427 (11.25%)	379 (10.04%)	306 (8.[illegible]%)	262 (6.93%)	< 0.01
28-days death, n (%)	2947 (19.47%)	777 (20.46%)	784 (20.77%)	719 (19.[illegible]%)	667 (17.63%)	< 0.01
Hosptial-LOS, days, median (IQR)	5.7 (3.1, 10.4)	5 (2.8, 9.7)	4.9 (2.8, 9.1)	5.1 (2.9, [illegible]4)	7.7 (4.6, 12.8)	< 0.01
ICU-LOS, median (IQR)	1.81 (0.98, 3.56)	1.94 (1.1, 4.12)	1.76 (0.95, 3.64)	1.76 (0.[illegible], 3.28)	1.78 (1.03, 3.19)	< 0.01

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Supplementary Table 7 Baseline Characteristics of the Study Population After GBM Adjustment in Sensitivity Analysis

Categories	Q1	Q2	Q3	Q4	*p*-value
Weighted Sample Size	14598.11	14800.32	14153.72	14061.59	
Demographic					
Age, years, median (IQR)	65.00 [51.00, 79.00]	65.00 [50.00, 79.00]	65.00 [51.00, 79.00]	65.00 [52.00, 79.00]	0.62
Male, n (%)	8300 (56.86%)	8299 (56.07%)	7845 (55.43%)	7884 (56.07%)	0.72
Ethnicity, n (%)					0.83
White	9673 (66.26%)	9747 (65.85%)	9486 (67.02%)	9540 (67.84%)	
Black	1277 (8.75%)	1302 (8.80%)	1280 (9.04%)	1296 (9.22%)	
Asian	385 (2.64%)	445 (3.01%)	405 (2.86%)	418 (2.97%)	
Hispanic/Latino	519 (3.56%)	533 (3.60%)	494 (3.49%)	498 (3.54%)	
Other	702 (4.81%)	726 (4.91%)	656 (4.63%)	633 (4.50%)	
Unknown	2042 (13.99%)	2048 (13.84%)	1833 (12.95%)	1677 (11.92%)	
First care unit, n (%)					1.00
MICU	3681 (25.22%)	3750 (25.34%)	3629 (25.64%)	3558 (25.30%)	
SICU	2975 (20.38%)	3004 (20.29%)	2863 (20.23%)	2890 (20.55%)	
TSICU	2892 (19.81%)	3008 (20.32%)	2852 (20.15%)	2689 (19.12%)	
MICU/SICU	2408 (16.50%)	2475 (16.73%)	2407 (17.01%)	2437 (17.33%)	
CCU	1335 (9.14%)	1296 (8.76%)	1213 (8.57%)	1246 (8.86%)	
CVICU	768 (5.26%)	719 (4.86%)	662 (4.68%)	752 (5.35%)	
Neuro SICU	539 (3.69%)	548 (3.71%)	528 (3.73%)	490 (3.49%)	
Chronic conditions, n (%)					
Diabetes	3400 (23.29%)	3499 (23.64%)	3310 (23.38%)	3303 (23.49%)	0.99
Congestive heart failure	2916 (19.97%)	3024 (20.43%)	2780 (19.64%)	2872 (20.43%)	0.83
Renal disease	2048 (14.03%)	2129 (14.39%)	2006 (14.17%)	2066 (14.69%)	0.88
Malignant cancer	1306 (8.95%)	1408 (9.51%)	1345 (9.50%)	1429 (10.16%)	0.43
Severity scores					
CCI, mean (SD)	5.00 [3.00, 7.00]	5.00 [3.00, 7.00]	5.00 [3.00, 7.00]	5.00 [3.00, 7.00]	0.28
APSIII, mean (SD)	40.00 [30.00, 57.00]	40.00 [30.00, 57.00]	40.00 [30.00, 57.00]	40.00 [30.00, 56.00]	1.00
Intervention during ICU, n (%)					

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Invasive mechanical ventilation	4808 (32.94%)	4837 (32.68%)	4440 (31.37%)	4303 (30.60%)	0.17
Dialysis	270 (1.85%)	254 (1.72%)	250 (1.77%)	242 (1.72%)	0.97
Vasopressors	2778 (19.03%)	2730 (18.45%)	2619 (18.50%)	2585 (18.38%)	0.90
Outcome					
Hospital death, n (%)	2313 (15.84%)	2383 (16.10%)	2014 (14.23%)	1967 (13.99%)	0.03
ICU death, n (%)	1414 (9.69%)	1389 (9.38%)	1174 (8.30%)	1014 (7.21%)	<0.001
28-day death, n (%)	2840 (19.46%)	2964 (20.03%)	2654 (18.75%)	2401 (17.08%)	0.02
Hosptial-LOS, days, median (IQR)	4.90 [2.80, 9.00]	5.00 [2.80, 9.10]	5.30 [2.90, 9.66]	7.60 [4.50, 12.80]	<0.01
ICU-LOS, median (IQR)	1.87 [1.08, 3.65]	1.76 [0.95, 3.63]	1.81 [0.93, 3.51]	1.84 [1.03, 3.46]	<0.01

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Supplementary Table 8 Covariate Balance Diagnostics Before and After Generalized Boosted Model Adjustment in Sensitivity Analysis

Metrics	Maximum SMD	Maximum KS	Minimum KS *p-value*
Before GBM	0.64	0.22	< 0.01
After GBM	0.06	0.03	0.07

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Supplementary Table 9 Sample Sizes and Effective Sample Sizes Before and After GBM Adjustment in Sensitivity Analysis

Quartile	Original Sample Size (N)	Effective Sample Size (N)
Q1	3797	3202.7
Q2	3775	3640.1
Q3	3780	3398.9
Q4	3783	2622.8

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Supplementary Table 10 Odds Ratios for All-Cause Mortality in-hospital and ICU mortality from Logistic Regression after Generalized Boosted Models in Sensitivity Analysis

Categories	Model 1		Model 2		Model 3	
	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value	OR (95% CI)	*p*-value
Hospital death						
Quartilea[a]						
Q1	Ref.		Ref.		Ref.	
Q2	1.02 (0.96, 1.09)	0.54	1.02 (0.95, 1.10)	0.55	1.02 (0.95, 1.09)	0.58
Q3	0.88 (0.83, 0.94)	< 0.01	0.88 (0.82, 0.95)	< 0.01	0.80 (0.88, 0.92)	< 0.01
Q4	0.86 (0.81, 0.92)	< 0.01	0.84 (0.78, 0.91)	< 0.01	0.83 (0.77, 0.89)	< 0.01
ICU death						
Quartilea[a]						
Q1	Ref.		Ref.		Ref.	
Q2	0.97 (0.89, 1.04)	0.38	0.95 (0.87, 1.04)	0.28	0.95 (0.87, 1.04)	0.28
Q3	0.84 (0.78, 0.91)	< 0.01	0.86 (0.78, 0.94)	< 0.01	0.82 (0.75, 0.90)	< 0.01
Q4	0.72 (0.67, 079)	< 0.01	0.70 (0.63, 0.77)	< 0.01	0.69 (0.63, 0.75)	< 0.01

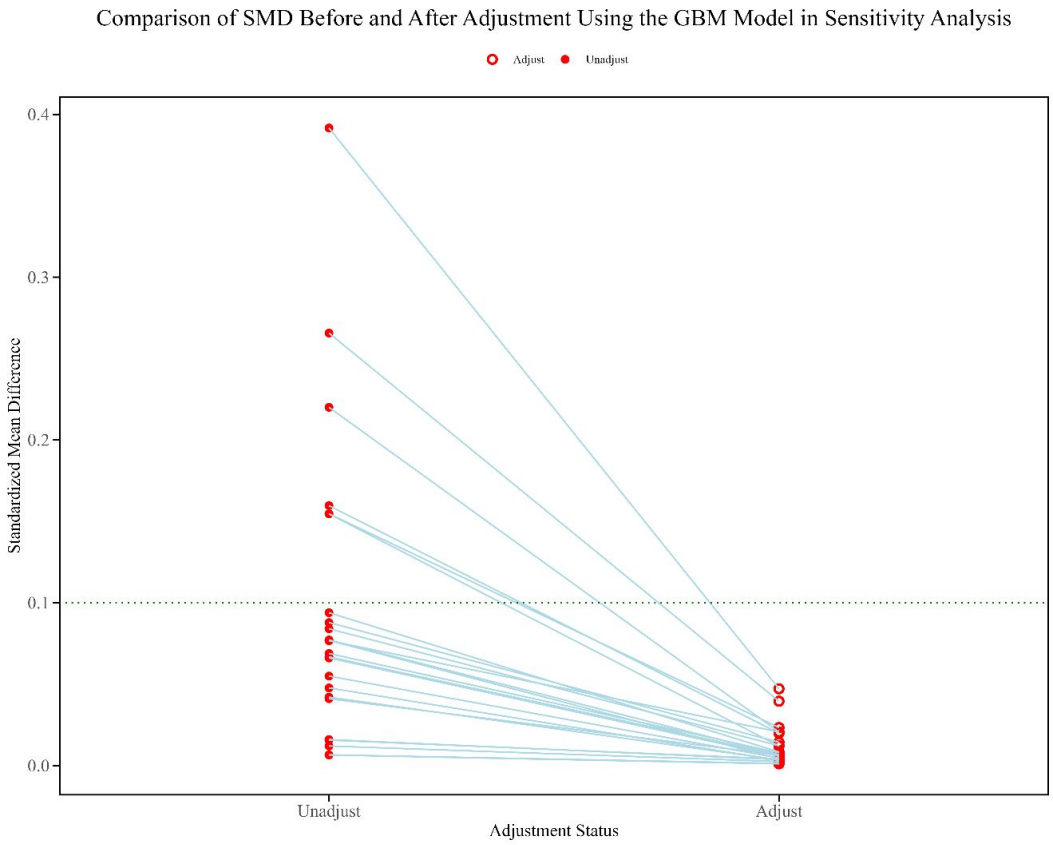
Model 1: unadjusted; Model 2: adjusted for APSIII score, CCI score; Model 3: adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, invasive mechanical ventilation, analysis and vasopressors. OR: Odds Ratio

[a] ED to ICUs time (hours): Q1 (≤ 2.83 hr), Q2 (2.83-3.98 hr), Q3 (3.98-5.65hr), Q4 (5.65-34.55 hr).

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Supplementary Fig. 5 Comparison of Standardized Mean Differences Before and After GBM-Based Propensity Score Weighting in Sensitivity Analysis

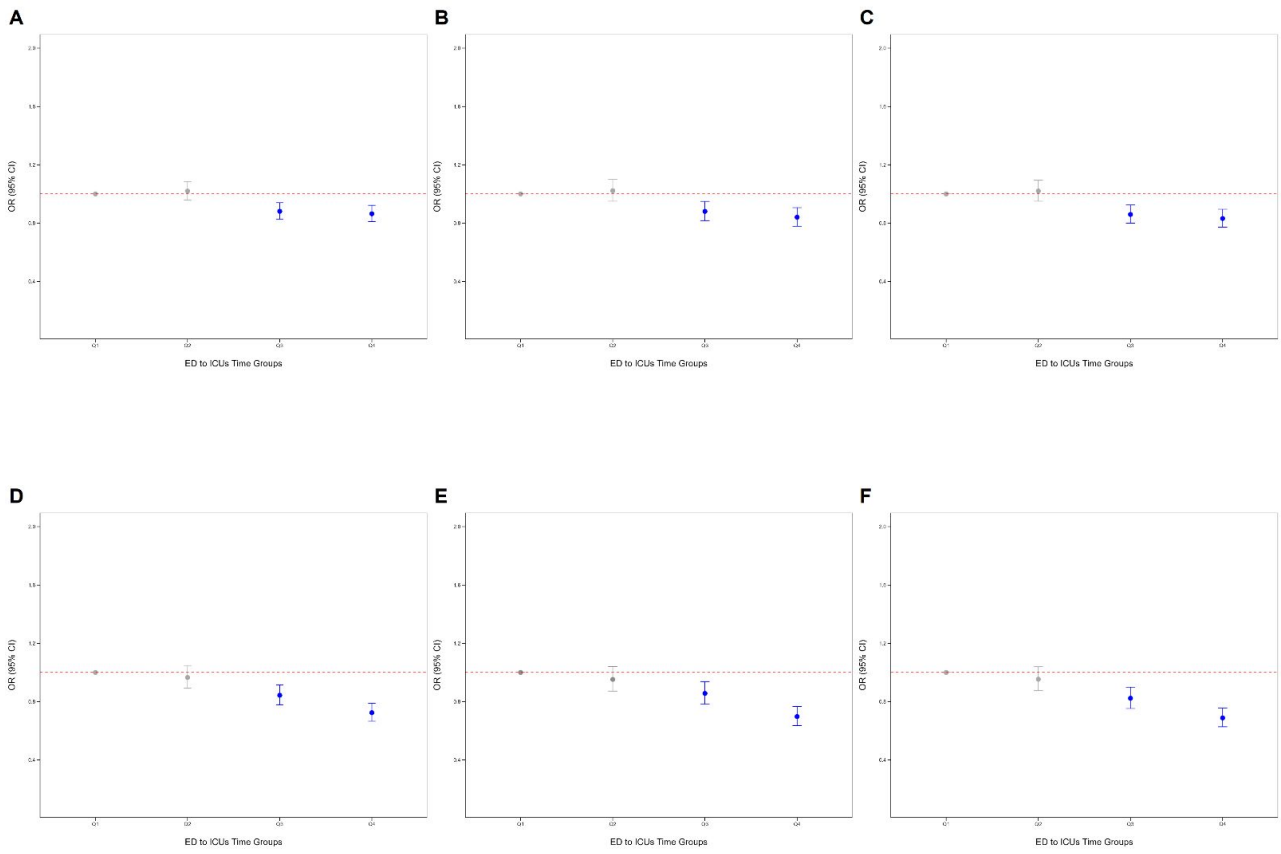
Red circles represent the unadjusted SMD values, and open circles represent the adjusted SMD values after weighting. The horizontal dashed line at SMD = 0.1 indicates the threshold for acceptable covariate balance.

Covariates include age, gender, race, first care unit, CCI scores, APSIII scores, diabetes, congestive heart failure, renal disease, malignancy, invasive mechanical ventilation, dialysis, and vasopressor use.

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Supplementary Fig. 6 Odds Ratios for all-cause in-hospital and ICU mortality after generalized boosted model-based propensity score weighting in Sensitivity Analysis

ED to ICUs time (hours): Q1 (≤ 3.03 hr), Q2 (3.03-4.48 hr), Q3 (4.48-7.54 hr), Q4 (> 7.54 hr)

A. Association between ED to ICUs time and in-hospital mortality, unadjusted (weighted).

B. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score and CCI score (weighted).

C. Association between ED to ICUs time and in-hospital mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

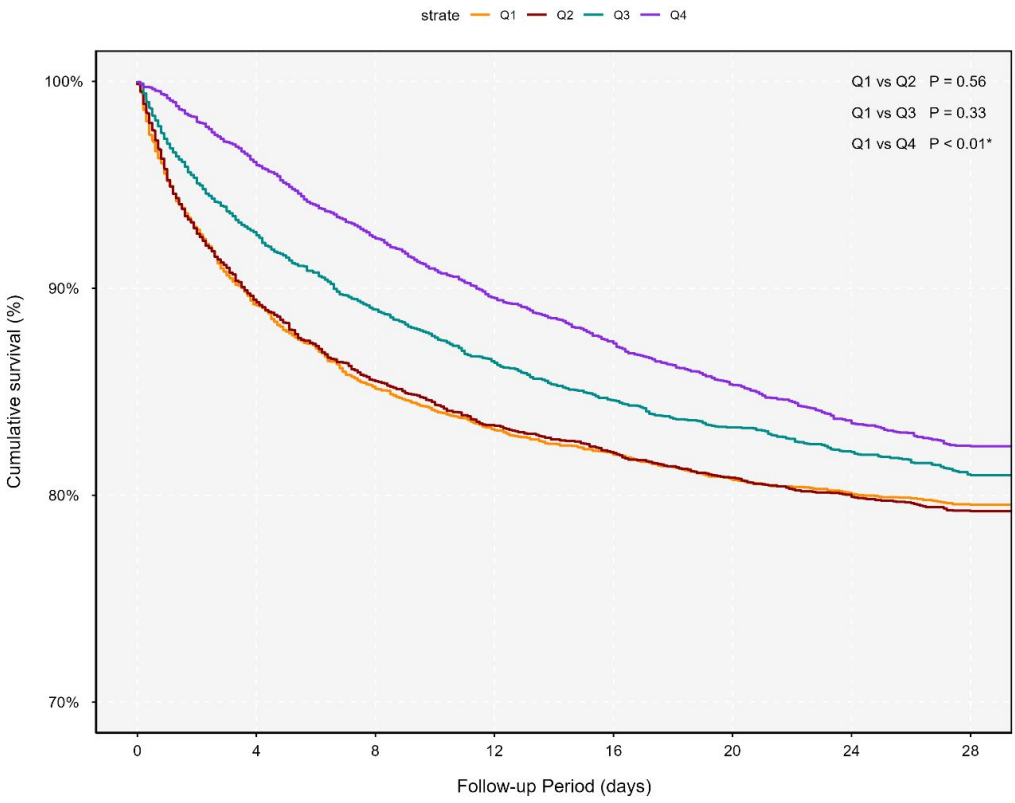
D. Association between ED to ICUs time and ICU mortality, unadjusted (weighted).

E. Association between ED to ICUs time and ICU mortality, adjusted for OASIS score and CCI score (weighted).

F. Association between ED to ICUs time and ICU mortality, adjusted for APSIII score, CCI score, age, gender, first care unit, diabetes, congestive heart failure, renal disease, malignant cancer, mechanical ventilation, dialysis and vasopressors (weighted).

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Supplementary Fig. 7 Kaplan-Meier Survival Curves for 28-day All-Cause Mortality in Sensitivity Analysis

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